

Team 10

Preliminary Design Review

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Table of Acronyms

Abbreviation	Definition
COSPAR	Committee on Space Research
TRL	Technology Readiness Level
STM	Science Traceability Matrix
DSCC	Deep Space Communications Complex
DSN	Deep Space Network
FMEA	Failure Mode and Effect Analysis
VIRTIS	Visible and Infrared Thermal Imaging Spectrometer
IRTLS	Infrared Tunable Laser Spectrometer
LRD	Launch Readiness Date
ESA	European Space Agency
MDR	Mission Design Review
RFA	Request For Action

ADV
MCR
PDR

Advisory
Mission Concept Review
Preliminary Design Review

Preliminary Design Review

1. Mission Overview

1.1. Mission Statement

The purpose of this Aerobot mission to Venus is to collect crucial data that can help scientists understand the planet further and provide insight into Earth's potential climatic future. The mission has been designed to answer the customer-defined goal of "Decadal Question 6.4: How do planetary surfaces and interiors influence and interact with their host atmospheres?" (Origins, Worlds, and Life 2023). The team-specific science objectives of this mission will be to understand the role of volcanic outgassing on atmospheric concentration, the mechanisms of chemical weathering on surficial rocks, and paleoatmospheric conditions. This mission will aid in the scientific understanding of planetary evolution through evidence-based assumptions regarding the planet's geologic history. This mission will be carried out by the deployment of an Aerobot, a robotic aerial vehicle that will utilize buoyancy to navigate through Venus' atmosphere, equipped with advanced spectroscopic instruments of the Virtual and Infrared Thermal Imaging Spectrometry (VIRTIS) and the IR Tunable Laser Spectroscopy (TLS). The Aerobot will measure absorption spectra of sulfur dioxide and carbon dioxide in the infrared range over 5 km², analyze volcanic deposits for evidence of chemical and thermal weathering over 5 km, and measure deuterium-to-hydrogen (D/H) ratios and stable isotopes of carbon-13, oxygen-18, and nitrogen-15 to assess heat fluctuations and gas emissions from found surface fractures. The data will significantly enhance scientific understanding of Venus's geological and atmospheric history, providing critical insights into planetary evolution and climate dynamics across the solar system.

The mission will adhere to specific customer constraints, including a total mass limit of 50 kilograms, stored dimensions not exceeding 1 m x 1 m x 1 m (expandable post-deployment), a cost cap of \$200M for the Aerobot system, use of Radioisotope Thermoelectric Generators (RTGs) is prohibited, and any radioactive materials must not exceed a cumulative mass of 5 grams. The Aerobot system must be ready for integration with other mission components by October 1st, 2028, at the Goddard Space Flight Facility in Greenbelt, Maryland, and achieve launch readiness by March 1st, 2029, from Cape Canaveral, Florida. However, new challenges were introduced due to a scope change in the mission, requiring a 5kg redistribution of the mass limit to the balloon's wall thickness and a mission completion within 36 hours.

The data collected through this mission will provide new, groundbreaking insight into Venusian planetary evolution. Believed to have once been Earth-like, studying the surface and atmosphere of Venus will aid in the prediction of Earth's distant future. This mission will provide invaluable clues into Earth's long-term evolution, including potential planetary responses to climate change and the actions of humanity. This mission also

holds the promise of further understanding Venus' planetary conditions, pointing future studies towards the planet's potential to support life. Through the use of heritage components such as VIRTIS and TLS, which can or have already been adopted for Venusian use, as well as simple supporting systems with low redundancy, the Aerobot will achieve a low-cost and high-efficiency design whilst taking on slightly higher risk than usual for a NASA mission.

1.2. Science Traceability Matrix

Venus's surface and atmosphere are strongly correlated, providing insight into the interrelation of a planet's atmosphere to its surface, geology, and interior ("Venus's Surface" 2015). As part of the decadal question, "*How do planetary surfaces and interiors influence and interact with their host atmospheres?*", this mission investigates how atmospheric composition and evolution shape the geology of Venus. Derived from this overarching goal, the scientific objectives explored in this study examine interactions between the Venusian atmosphere to volcanism, surficial weathering, and plate tectonics. As the planet with the most volcanoes in the solar system, Venus experiences extensive volcanism, releasing large amounts of gases and contributing to its dense atmosphere and high surface temperatures (Wenz 2024). Venusian magma can also be basaltic, similar to Earth's, but with notable differences. Unlike Earth, Venus lacks plate tectonics, which results in a more randomized distribution of volcanic activity dependent on mantle plumes (Sulcanese et al. 2024). In addition, Venus's extreme dryness, having lost most of its water, increases the viscosity of its magma. This characteristic, combined with the low viscosity of basaltic lava, leads to predominantly effusive eruptions rather than explosive, shaping the planet's volcanic landscape and influencing its surface evolution and atmospheric composition (Sulcanese et al. 2024).

Volcanism and outgassing have shaped and driven the composition of the planet's atmosphere ("Origins, Worlds, and Life" 2022, 6.4), making the quantification of gas concentrations from volcanic centers and mantle plumes crucial for understanding the influence of planetary interiors and atmospheric dynamics. Recent radar observations provide strong evidence of ongoing volcanic activity, with detected changes in the surface morphology of volcanic vents, likely due to the new lava flows (Sulcanese et al. 2024). This suggests that Venus remains geologically active, with continuous emissions of gases such as CO₂, SO₂, and H₂O, which play a role in atmospheric evolution and surface weathering. Using high-resolution spectroscopy (1870 - 4300 nm) over a 5 km range (Gilmore et al., 2015; Piccioni et al., n.d.), concentrations of these gases in volcanic emissions shall be measured and analyzed to investigate the relationship between the planet's atmosphere and interior processes. Since these gases are released through volcanic activity, their concentrations can provide insight into Venus's interior composition, the circulation of molten rock, and the potential for ongoing geochemical interactions between the mantle and atmosphere.

Science Goals	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
Planetary Atmospheres Question 6.4: "How do planetary surfaces influence and interact with their host environment?" - Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023-2032	Determine the concentration of sulfur dioxide and carbon dioxide in volcanoes and mantle plumes	Identify gas concentrations of CO ₂ and SO ₂	Collect absorption spectra of CO ₂ and SO ₂ in the 1870 to 4300 nm range over 5 km ²	Wavelength range:	1870 - 4300 nm	1840 - 4990 nm	Visible and Infrared Thermal Imaging Spectrometer High Resolution Subsystem (VIRTIS-H)	Aerobot shall collect absorption spectra
				Integration time:	1 sec	1 sec		
				Spectral Resolution:	1300 - 3000 $\lambda / \Delta\lambda$	1300 - 3000 $\lambda / \Delta\lambda$		
				Sensitivity:	$1.2 \times 10^{-4} \text{ Wm}^{-2}\text{sr}^{-1}\mu\text{m}^{-1}$	$1.2 \times 10^{-4} \text{ Wm}^{-2}\text{sr}^{-1}\mu\text{m}^{-1}$		
	Determine the impact of atmospheric chemistry on surface weathering	Identify volcanic deposits for atmospheric weathering products such as Fe oxides and sulfates	Capture spectral imaging in the 290 - 4300 nm range over 5 km	Wavelength range:	290 - 4300 nm	280 - 5130 nm	Visible and Infrared Thermal Imaging Spectrometer Mapping Subsystem (VIRTIS-M)	Aerobot shall capture spectral imaging for identifying evidence of chemical and thermal weathering
				Integration time:	1 sec	1 sec		
				Spectral Resolution:	100-500 $\lambda / \Delta\lambda$	100-500 $\lambda / \Delta\lambda$		
				Spatial Resolution:	0.25 mrad (250 m at 1000 km)	0.25 mrad (250 m at 1000 km)		
	Determine how Venus's atmosphere has changed over time to investigate ancient tectonic activity	Identify the isotopes fractionation of H, C, N, and O on faulted geology	Measure stable isotope ratios of D/H of carbon-21, oxygen-18, and nitrogen-15 over 1-2 km	Wavelength range:	1000-25000 nm	1000-30,000 nm	IR Tunable Laser Spectroscopy	Mission must enable spectroscopy-based assessment of heat flux and isotopic composition
				Integration time:	1-100 ms	1-100 ms		
				Spectral Resolution:	0.0001 cm ⁻¹	0.0001 cm ⁻¹		
				Sensitivity:	1 ppm	1 ppm		
	Determine how Venus's atmosphere has changed over time to investigate ancient tectonic activity	Identify the isotopes fractionation of H, C, N, and O on faulted geology	Measure stable isotope ratios of D/H of carbon-21, oxygen-18, and nitrogen-15 over 1-2 km	Wavelength range:	1000-25000 nm	1000-30,000 nm	IR Tunable Laser Spectroscopy	Aerobot must use a tunable laser emitting infrared light to analyze the composition of high-precision gas abundance and stable isotope ratios
				Integration time:	1-100 ms	1-100 ms		
				Spectral Resolution:	0.0001 cm ⁻¹	0.0001 cm ⁻¹		
				Sensitivity:	1 ppm	1 ppm		

Table 1: Science Traceability Matrix

Venus's atmosphere also contains thick sulfuric clouds that contribute to a runaway greenhouse effect, trapping heat and reflecting solar radiation (NASA 2023), which may play a role in surface weathering. Additionally, its extreme heat results from a dense carbon dioxide-rich atmosphere intensifying this greenhouse effect. Intense volcanic activity reshapes the surface and releases additional heat and greenhouse gases, further thickening the insulating atmosphere. Venus's slow, retrograde rotation leads to long days and nights, causing additional significant heat buildup on the sunlit side and influencing atmospheric circulation and heat distribution (NASA 2023). Such processes may influence the planet's surface through chemical reactions, changes in mineralogy, and thermal weathering, providing insight into the relationship between the atmosphere and the planetary surface. Capturing spectral imaging (290 - 4300 nm range) in weathered surfaces over a 5 km area shall aid in identifying secondary minerals, analyzing thermal expansion, and determining how atmospheric chemistry plays a role in surficial weathering (Gilmore et al., 2015; Piccioni et al., n.d.).

Venus is hypothesized to have experienced tectonic plate movements in its early history. However, theoretical models supported by current atmospheric data, as indicated in a recent study led by researchers at Brown University, suggest inconsistencies with this possibility (Weller et al., 2023). These findings support the theory that the planet's atmospheric evolution into a hotter and denser state contributed to the loss of conditions necessary for sustained tectonic activity, leading to its transition to a stagnant lid regime ("Venus Had Earth-like Plate Tectonics..." 2023). Evidence from regions with tectonic-like features, such as crustal thickening, fault activity, and deformation, suggests that records of isotope ratios and stable isotopes can provide insights into Venus' tectonic history through atmospheric changes.

The study further proposes that Venus underwent an early phase of plate-tectonic-like activity lasting at least one billion years, during which significant volcanic outgassing released nitrogen and carbon dioxide into the atmosphere. As the planet transitioned to a stagnant lid regime, its outer shell became immobile, causing outgassing rates to decline and ultimately shaping Venus' current atmospheric composition (Weller et al., 2023). Gas emissions from surface fractures can be accessed through spectroscopic measurements of the deuterium-to-hydrogen (D/H) ratio and stable isotopes carbon-21, oxygen-18, and nitrogen-15 over a 1-2 km range to gain insight into Venus' atmospheric evolution and tectonic history. These isotopes record atmospheric changes over time, including temperature and composition, which shall be useful in interpreting paleo-atmospheres ("Climate Science Investigations..." 2023). The objectives of this mission are designed to maximize science return within the constraints of mass, dimension, prohibited materials, cost, and dates. By focusing on the localized ranges to minimize costly measures, these findings shall aid in understanding Venusian surface-atmosphere interactions and inform the development of future planetary missions to Venus.

1.3. Summary of Mission Location

Venus has been of great interest to the scientific community due to its vastly different modern environment, despite having Earth-like characteristics of similar radius, internal structure, and mass. The planet rotates extremely slowly, with an average day of 243 Earth days—longer than its orbit around the Sun of 225 Earth days (NASA 2015). The day and night sides of Venus experience extreme surface temperatures, averaging between 460°C to 470°C, with a surface pressure of 95.6 bar (NASA 2025), due to the planet's intense runaway greenhouse effect. Venus' atmosphere consists of 96.5% carbon dioxide (CO₂), 3.5% nitrogen (N₂), and less than 0.1% water vapor (H₂O) and other compounds (Johnson et al. 2019). Sulfuric acid droplets in the middle atmosphere form a ~20 km-thick layer of clouds and haze, which traps CO₂ in the lower atmosphere, contributing to the extreme surface temperatures and preventing the droplets from reaching the surface (OpenStax 2017). While temperature and pressure increase with decreasing altitude, winds reaching speeds of about 210 - 360 km/h dominate Venus' upper atmosphere at lower latitudes, whereas conditions near the hot, dry surface remain relatively calm (OpenStax 2017). The Venusian surface is primarily composed of volcanic and basaltic flows, which effectively recycled the entire crust approximately 0.5 billion years ago (Alden n.d.). The topography consists mainly of broad highlands and volcanic uplands, punctuated by thousands of mountains and volcanoes (ESO n.d.). This volcanic world is believed to have once existed within the Sun's habitable zone, much like Earth today, potentially supporting liquid water oceans and plate tectonics, both of which have since been lost to the planet's extreme environment (NASA 2015). Understanding the conditions and timeline of Venus' major climatic shifts is crucial for predicting the future of Earth's climate, particularly in the context of anthropogenic CO₂ emissions.

The regions of Ozza and Maat Mons, the Haastse-baad Tessera, and the Ishtar Terra were initially proposed for their potential to address the team's science objectives within the mission requirements. A trade study was conducted to compare these three sites proposed as the mission site in the Mission Concept Review, evaluating their ability to meet the mission's requirements (Table 30). Scoring the highest among all considered locations, the Ozza Mons and Maat Mons are the optimal and final site for the team's mission (Table 30).

As observed on Earth, volcanic processes are major sources of SO₂, CO₂, and H₂O emissions into the atmosphere. Venus hosts over 1,500 major volcanoes and volcanic features exist on Venus (ESO n.d.), prompting investigation into the atmospheric effects of gases emitted from volcanic centers and mantle plumes across the planet's surface. Obtaining absorption spectra data to determine atmospheric concentrations above this region with recent volcanic activity relative to atmospheric

concentrations in different locations on the planet collected during previous and future Venus missions will provide insight on the degree of volcanism as source of greenhouse gasses on Venus. Ozza and Maat Mons are two of the largest shield volcanoes on Venus, with Maat Mons confirmed as an active volcano as recently as 1991 (NASA 2023). Ozza Mons is a dormant shield volcano, thought to have once been positioned above the active mantle plume now beneath Maat Mons (Mason, et al. 2025). Due to their close proximity, the Ozza and Maat Mons region presents a unique opportunity to study multiple volcanic centers at a single site, making it the team's selected location to meet the mission requirements and science objectives. They are located in the Atla Regio, a highland region contains diverse geological features attributed to mantle plume-driven tectonic and volcanic processes, including faulting, rifting, and recently observed changes in volcanic vents (Mason, et al. 2025) (Greicius and Hartono 2023). Ozza and Maat Mons are situated near the equator, approximately 1°N, 195°E (JPL, n.d.). Scientific observations of the atmospheric and surficial characteristics of the Maat Mons region shall provide valuable information for Question 6.4 of the 2023-2032 Decadal Strategy: *"How do planetary surfaces and interiors influence and interact with their host atmosphere?"* To fulfill the science goals, a minimum payload success requires the aerobot to stay within a horizontal radius of 1600 km from the Maat Mons caldera summit. Because the aerobot has limited ability to control the horizontal distance covered during the OpT after parachute deployment, a wide radius surrounding Maat Mons provides minimal payload success. However, as long as meaningful data is collected within the ~500 km of the Maat Mons Crater – within the locations of the targeted features – a horizontal distance greater than 1600 km will not hinder payload success.

Maat Mons is the tallest volcano on Venus, rising 9 km above the surface and containing a 26 x 30 km collapse caldera (Mouginis-Mark, 2016). Measuring the IR absorption spectra of SO₂ and CO₂ in the lower stratosphere and upper troposphere, specifically during the descent within the main cloud layer at 70-50 km elevation and any subsequent measurements, around a 5 km² area over the Maat Mons caldera will allow for the observation of concentration variation of these gasses with decreasing distance from the Maat Mons crater to satisfy Science Objective 1 of the STM. As the aerobot descends out of the main cloud layer at about 50 km elevation (Landis n.d.), visual and infrared mapping imagery will be taken around the center of the Maat Mons, ideally around the center of the feature, to map compositional variations around the highest elevated material around a 5 km horizontal radius around the crater rim. The high elevation of Maat Mons compared to most topography on Venus protects the material from volcanic and tectonic activity from external mantle plumes at Venus' average elevation, while also exposing their surface material to a higher degree of chemical weathering from interactions with the acidic clouds in Venus' lower stratosphere at 50-70 km (Landis n.d.). This isolation of volcanic rock allows for the collection of high resolution scientific data as the Aerobot descends out of the main cloud layer, satisfying the mission's Science Objective 2 in the STM to observe if chemical weathering has occurred to some of the closest geologic material to Venus'

highly acidic atmospheric clouds. Science objective 3 of the STM will be fulfilled as the aerobot is descending within the upper troposphere after exiting the main cloud layer, beginning at about 50 km until the end of Opt at approximately 40 km (Landis n.d.). by measuring the isotopic ratios of H, C, O, and N of the geologic material involved in the hypothesized faulting and graben fissure structures of ancient tectonic events that have been previously mapped on the flanks of Maat Mons and extending into the Alto Region (Mason et al. 2025). These observations will provide insight to answer the mission's science goal of understanding the interactions between Venus' atmosphere with its highly volcanic surface and interior.

1.4. Mission Requirements

The mission objective is to develop a small, low-cost robotic Aerobot System for scientific investigation in Venus' atmosphere. The system must operate within extreme environmental conditions while collecting data related to both atmospheric composition and geological processes. The following mission and system-level requirements have been derived from the given constraints and objectives.

The Aerobot System shall adhere to requirements dictating how it is built to ensure successful operation in the Venus atmosphere. The Aerobot System shall not exceed 50 kilograms in total mass or a 1m x 1m x 1m in stored configuration. Additionally, the system shall be able to withstand Venus' extreme atmospheric conditions. The system shall also not contain more than 5 grams of radioactive material or use a Radioisotope Thermoelectric Generator. In addition, the Aerobot System shall collect data to expand knowledge of how Venus' surface and interior interacts with its atmosphere.

The schedule of Venus Visionaries should ensure that the Aerobot system should be ready for integration with the mission by October 1st, 2028 and be launch ready by March 1st, 2029. The Venus Visionaries' mission should be compliant with NASA's planetary protection policies for Venus. Lastly, the Venus Visionaries' mission shall not cost more than \$200 million.

The mission and system requirements ensure that the Aerobot System aligns with NASA's Discovery Program objectives while adhering to physical, operational, and scientific constraints. These requirements will guide the design and subsystem development to ensure mission success.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
MG-1	Aerobot system shall collect data to expand knowledge of how Venus' surface and interior interacts with its atmosphere	Aligns with the 2023-2032 Decadal Strategy Question 6.4	-	All	Demonstration	Met
PM-1.1	The mission shall follow all accepted given constraints listed by the customer	Customer constraints ensure the design aligns with stakeholder priorities and funding limitations.	MG-1	PM-1.1.1 PM-1.1.2 PM-1.1.3 PM-1.1.4 PM-1.1.5 PM-1.1.6 PM-1.1.7	Inspection	Met
PM-1.1.1	Aerobot system shall not exceed 50 kg in total mass	Makes it compliant to the launch restrictions.	PM-1.1	-	Inspection	Met
PM-1.1.2	Aerobot system shall not exceed 1m x 1m x 1m in stored configuration.	Aligns with deployment and integration constraints.	PM-1.1	-	Inspection	Met
PM-1.1.3	Aerobot system shall be designed to withstand Venus' extreme conditions (475°C, high-pressure, sulfuric acid clouds).	Ensures operational survival in the target environment.	PM-1.1	-	Demonstration	Met
PM-1.1.4	The system shall not contain more than 5 grams of radioactive material or use a Radioisotope Thermoelectric Generator	Customer has maintained that these devices or material will not be necessary.	PM-1.1	-	Inspection	Met
PM-1.1.5	Aerobot system shall be ready for integration with the larger mission by October 1, 2028.	Meets mission schedule constraints.	PM-1.1	-	Demonstration	Met
PM-1.1.6	Aerobot system shall be launch-ready by March 1, 2029	Aligns with the launch window for Venus.	PM-1.1	-	Demonstration	Met
PM-1.1.7	Aerobot system shall be within the budget constraint of \$200 million.	Prevents cost overruns and compliance with customer needs.	PM-1.1	-	Inspection	Met
PM-1.2	Aerobot system shall comply with NASA's planetary protection policies for	Ensures adherence to space exploration ethics and	MG-1	-	Demonstration	Met

	Venus.	contamination guidelines.				
SCI-1.1	The mission shall use VIRTIS-H collect absorption spectra of CO ₂ , H ₂ O, and SO ₂ in the 1870 - 4300 nm range over 5 km	The spectrometer must identify gas concentrations of CO ₂ , SO ₂ , and H ₂ O using spectroscopy around volcanic centers and mantle plumes	MG-1.0	SCI-1.1.1 SCI-1.1.2 SCI-1.1.3 SCI-1.1.4	Demonstration	Met
SCI-1.2	The mission shall use VIRTIS-M to acquire multispectral imaging data across the ultraviolet to near-infrared spectrum (290–4300 nm) and achieve a spatial coverage of 5 km	To determine chemical composition of secondary minerals and thermal expansion on weathered surfaces	MG-1.0	SCI-1.2.1 SCI-1.2.2 SCI-1.2.3 SCI-1.2.4	Demonstration	Met
SC-1.0	The Aerobot shall maintain operational integrity of all subsystems during the designated operation period	A functional probe is required for in situ gathering of data and transmission back to earth	MG-1.0	SC-1.1 SC-1.2 SC-1.3	Demonstration	Met
SC-1.1	Thermal systems shall be able to radiate heat into space to maintain internal temperatures necessary for operation	To allow the instruments and batteries to operate properly and prevent quick mission degrading	SC-1.0	TS-1.0	Demonstration	Met
SC-1.2	Power system shall be capable of providing power as related to power draw for each subsystem	Continuous power is critical for all components on board	SC-1.0	PWR-1.1	Analysis	Met
SC-1.3	Computer Hardware shall have enough power to store the data and have enough processing power to organize data	The mission will collect large amounts of data and needs to be sorted to send back to Earth	SC-1.0	CDH-1.0	Demonstration	Met

Table 2: Mission Requirements

1.5. Concept of Operations (ConOps)

The Concept of Operations outlines the tasks that occur once the aerobot has entered the Venusian atmosphere and the parachute has confirmed to be inflated. Per the mission requirements, the instrumentation within the aerobot system will collect science data during a 36-hour period of cruising altitude within Venus' stratosphere (Landis n.d.). For this mission, the aerobot shall utilize a helium balloon powered by a controlled pump to remain suspended at a specified altitude between 70 and 30 km vertical altitude above Venus' surface to begin data collection after the aerobot has exited the main cloud deck that sits within the upper atmosphere at the 100 - 80 km altitude range (Takagi et al. 2019), but limits risk of aerobot destruction from hellish conditions of the Venusian troposphere during OpT (Landis n.d.) . By using the pump to control balloon volume, the buoyant forces, and thus the altitude, can be precisely controlled. The aerobot system is a system containing an MCA-led/designed payload. The aerobot system has various subsystems, including thermal, and power command (the generator's operation is monitored and managed by a control system), mechanical, data dandling (managing, collecting, and analyzing data in such a manner useful for calculated decisions), and communication required for the mission success.

Once stabilized at the desired altitude, the system will deploy all subsystems and science instrumentation. Because no active propulsion technology will be used, upon deployment, the aerobot shall coast in the direction of wind flow for 36 hours. The planned range of descent from 70 - 30 km over 36 hours leads to an ideal lowering rate of the parachuted aerobot system of 0.23 m/sec throughout the coasting period. During the coasting period, the aerobot shall have periodic communication windows every 90 minutes, during which all gathered data will be shared with the Primary Spacecraft.

Once the parachute has been deployed, OpT begins at 70 km altitude within the Venusian stratosphere. VIRTIS-H will begin measuring absorption spectra starting from the western flank of Maat Mons as the aerobot is traversed across the Maat Mons region by the eastward trending wind. To measure the effect of atmospheric composition surrounding major volcanic centers (STM Science Objective 1), this will ideally begin at an altitude of 70 km and a horizontal distance of the western flank of Maat Mons (~120 km from the edge of the collapse caldera).

After an OpT of 720 minutes, the aerobot descends to a vertical altitude of 60 km and is traversed within a 5 km horizontal radius of Maat Mons' caldera. At this point, VIRTIS-M will begin taking spectral images of the elevated caldera and surrounding topography for Science Objective 2 of the STM. Infrared photography of one of the highest features on Venus, the Maat Mons' crater rim ~9 km above the surface located

at 1°N, 195°E (“Venus - 3-D Perspective View of Maat Mons” 1996), and surrounding elevated geology will provide information on how the planet’s geology interacts with the lower CO₂ troposphere and relatively close sulfuric acid clouds (Landis n.d.).

At an OpT of about 1440 minutes and a vertical altitude between 50 to 40 km within the lower cloud haze of Venus’ stratosphere (Landis n.d.), the aerobot system will traverse past the Maat Mons caldera and IRTLS will begin collecting absorption spectra data of the rifted geology of Maat Mons’ eastern flank and Ozza Mons’ southwestern flank. Science Objective 3 will be fulfilled by collecting infrared spectroscopic data on the geology of Maat Mons that surrounds hypothesized faulting and graben fissure structures of ancient tectonic events that have been mapped in Maat Mons and extending Alta Region (Mason et al. 2025). At this point, VIRTIS-H has fulfilled Science Objective 1 and will stop collecting data, transferring the dataset to the CDH subsystem.

Once the aerobot system has passed the eastern flank of Maat Mons and Ozza Mons, IRTLS and VIRTIS-M will stop taking spectral images of the Venus surface. This will occur at approximately 2160 minutes at an estimated elevation of 40 km.

Any additional measurements until EoL providing a more robust understanding of the movement of volcanic aerosols and their dispersion rate along an atmosphere with varying wind speeds (Colozza and Landis 2004) as the aerobot descends throughout the upper, middle, and lower haze layer of the Venusian atmosphere.

After this period, the harsh atmospheric conditions will degrade the balloon, leading to its failure and subsequent aerobot freefall until it crashes into Maat Mons. As the system descends, a short window will exist for gathering data at lower altitudes. The higher temperatures and pressures of the lower atmosphere and the eventual crash will make such a window brief. To ensure Science Objectives will be fulfilled, the ConOps has prepared for data collection to occur solely within the OpT, with any subsequent data collection during the descent providing further clarity of surficial conditions. If balloon failure occurs outside of the window of communication with the Primary Spacecraft, the final data recorded will likely not be retrievable. In this scenario, it might be wise to consider a preemptive controlled descent during the communication window. On the other hand, because the exact balloon failure time is unknown, maximizing overall data collection by keeping the aerobot afloat until natural descent might be preferred.

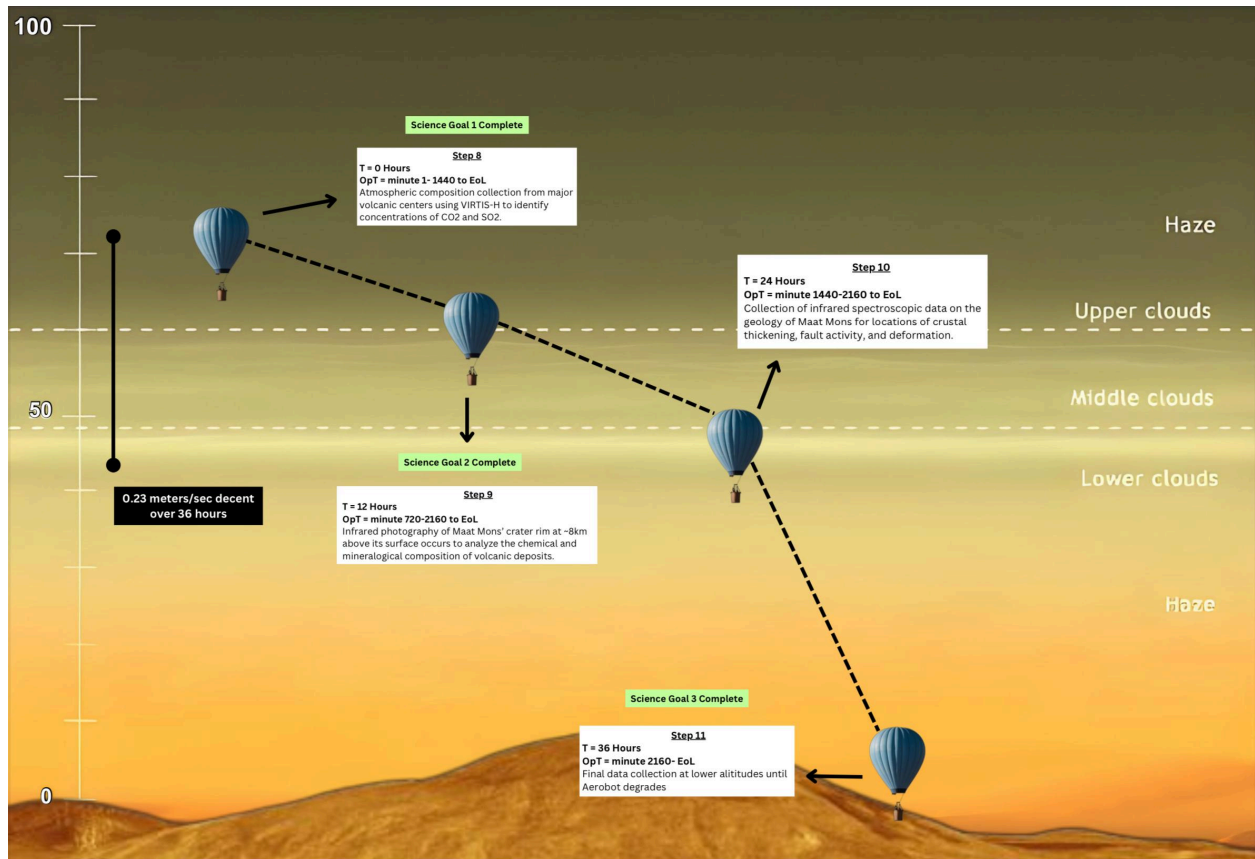


Figure 1: ConOps of Mission throughout 36 hour duration after provided Step 1-7 ConOps.

1.6. Vehicle Design Summary

The spacecraft that shall be capable of descending through the atmosphere of Venus whilst withstanding the extreme environment it is subject to. To accomplish this, the Aerobot System will consist of a 1x1x1 m cube just on the edge of the mass requirements that houses two primary science instruments capable of taking in situ measurements. The descent will be managed by a balloon that the rest is suspended to. Roughly 36 hours of mission time will be allotted for the Aerobot to operate with any extra time being used until the very end of transmission. These operations will commence the moment the system has been inserted into the atmosphere, a task the design team is not responsible for.

System	Mass (kg)	Dimensions LxWxH (cm)	Maximum Power Draw (W)
Aerobot	49.996	100 x 100 x 100	243

Table 3: Aerobot Specifications

A visual example of the Aerobot System assembly has been pictured below. This rendering may not be entirely accurate to the final design of the vehicle and all mass, dimensions, and power draws of the system and subsystems will be subject to change throughout the design process as the concept matures.

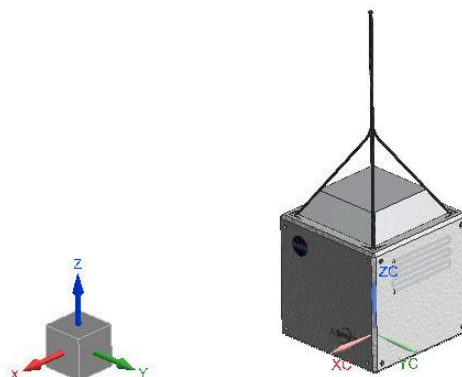


Figure 2: Aerobot Assembly

1.7. Science Instrumentation Summary

The team's aerobot includes the Virtual and Infrared Thermal Imaging Spectrometry (VIRTIS) and the IR Tunable Laser Spectroscopy (IRTLS) to complete objectives that further scientific understanding of Venus's atmosphere, surface, and their interactions. VIRTIS is an advanced imaging spectrometer designed to study the Venusian atmosphere, surface, and volcanic activity through its two main components, the VIRTIS-M mapping spectrometer and the VIRTIS-H high-resolution slit spectrometer (Piccioni et al., n.d.). The VIRTIS-M captures low-resolution two-dimensional spatial images within a spectral range of 280–5130 nm using both visible and infrared light (Piccioni et al., n.d.). Utilizing a Shafer telescope and Offner Relay spectrometer, the VIRTIS-M collects data through scans by a rotating primary mirror within multiple observation modes (Piccioni et al., n.d.). This mapping tool uses two detectors, the Thomson TH7896 CCD for visible light and HgCdTe for infrared (Piccioni et al., n.d.). For the team's science objectives, the VIRTIS-M will be used to capture spectral imaging to examine the impact of atmospheric chemistry on the surface through weathering. Volcanic deposits will be assessed for chemically driven weathering products such as iron oxides and sulfates. The VIRTIS-H provides detailed information in the 2–5 μm range using an off-axis parabolic mirror and an Echelle spectrometer

coupled with the same HgCdTe detector used by VIRTIS-M (Piccioni et al., n.d.). Equipped with a prism-grating cross-disperser, spectral orders are separated and validated with internal calibration lamps and filters (Piccioni et al., n.d.). The VIRTIS-M will be used to measure the gas concentrations, including carbon dioxide and sulfur dioxide, emitted from Ozza Mons and Maat Mons through absorption spectra to understand how volcanic outgassing impacts Venus's atmosphere. The VIRTIS system shall be provided with power through a main bus of 22V to 35V and will draw no more than 36W of power (VISIBLE and INFRARED THERMAL IMAGING SPECTROMETER" n.d.). Both instruments are centered together for simultaneous data collection and are designed with cryogenic cooling to withstand Venus's harsh environment (Piccioni et al., n.d.). The IRTLS is capable of detecting trace gases and isotope ratios with high precision (Tunable Laser Spectrometers | Capabilities" 2022), collecting stable isotopic data of D/H, carbon-21, oxygen-18, and nitrogen-15 to investigate any traces of ancient tectonic activity that shall provide insights into Venusian atmospheric evolution. The IRTLS shoots a tunable infrared laser that measures and quantifies the absorption of gases and isotopes as a closed cell, for rover missions, or an open path, suited for orbit and aerobot use (Tunable Laser Spectrometers | Capabilities" 2022). Equipping the aerobot with such technology will allow for in-depth studies of Venus and its geology.

1.8. Programmatic Summary

1.8.1. Team Introduction

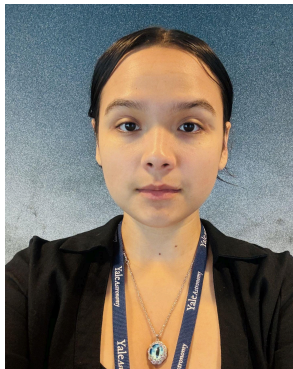
Name	University	University Location	Experience
Nicole Fenger 	University of Miami	Coral Gables, Florida	• Aerostructure and Recovery member of Rocket 'Canes - an on campus team ◦ participated in NASA Student Launch • Member of American Society of Mechanical Engineers ◦ participating in NASA Lunabotics Challenge • Previous Summer Internship with Vertol Systems Co. in Destin, FL • Working towards Level 1 Rocketry Certification through National Association of Rocketry
Cailin Stauffer	Binghamton University	Binghamton, New York	• Peer Mentor/Orientation Leader for first-year students at undergraduate university • Geoscience Intern at Lithocheimia LLC environmental consulting firm • NSF I-Corps Program for innovate technology startups to build customer relations and understanding • Leading geophysics researcher for a master's thesis project: Utilizing Smartphones as Magnetometers for Locating Undocumented Orphaned Oil and Gas Wells

			• Undergraduate geochemistry researcher on Kiamichi River Protection project through undergraduate university's chemistry department • Lead research on the western delta front of Mars' Jezero Crater for university's undergraduate research program • Graduate Teaching Assistant for Geology of the Solar System and Mineralogy courses • President and editor of the Sigma Gamma chapter of Sigma Alpha Iota • Co-creator and President of undergraduate university's Fiber Arts Club and Students for Sustainability Club
Gabriel Paine 	University of Central Florida	Orlando, Florida	• Treasurer and 2 year mentor for First Step, a STEM peer mentorship club on campus for freshmen engineering students • Former Manufacturing Process Analyst for Embraer Executive Aircraft • Member of a drone delivery development project • Former restoration hanger volunteer at aircraft museum Valiant Air Command
Himaja Mallipudi	Indiana University - Bloomington	Bloomington, Indiana	• Peer tutor in an Advanced Excel course that is offered to Kelley School of Business students only ◦ Tutor students in data visualization, data cleansing, and analysis of dashboards such as pivot tables and

			○ pivot charts ○ Instruct proper methods in development of dashboards for businesses to use ● Assistant Director of Education committee for Social Engagement Enterprise at Kelley ○ Author and publish newsletters with a team of 6 to deliver articles about socially-pressing topics ● Member in Diversity, Equity, & Inclusion committee of Women and Company ○ Develop and host events across IU campus to teach about diverse cultures that are found on campus
Eduardo Castro 	Union College	Schenectady, New York	● Published research on main-belt and trojan asteroids as a member of the Union College Scholars Astrophysics Research Team ○ “Composite Lightcurves of Three Main-Belt Asteroids,” Minor Planet Bulletin 2025 ○ “Observations of (617) Patroclus Mutual Events in Support of the Lucy Mission to the Trojan Asteroids,” Minor Planet Bulletin 2025 ○ “Additional Observations of (617) Patroclus-Menoetius Mutual Events,” Minor Planet Bulletin 2025 ○ “Lightcurves for Three Koronis Family Asteroids,” Minor Planet Bulletin 2024 ● Vice-President of Union College Rocket Club ○ Lead improvements in new member

			experience by advising projects and teaching rocketry principles and software • Physics Supplemental Instruction Leader at Union College ◦ Prepare and host supplemental problem-solving sessions and office hours in support of PHY-120 students
Angy El-Dakkak 	New Jersey Institute of Technology	Newark, New Jersey	• NCE STEM Academy at NJIT ◦ Covered fundamentals and interdisciplinary topics in engineering with hands-on essentials of prototyping sessions in the NJIT Makerspace. • Authored and released a novel titled ◦ CLASS OF COVID-19: Second Wave, belonging to the real-life genre, selling 2000 copies. Earned acclaim from the Wall Street Journal, NBC, and CNN for groundbreaking, unprecedented real-life stories ◦ Wrote three poems in the Pegasus Literary Magazine • Guest Ambassador at Liberty Science Center Jersey City, ◦ Maintained seamless operations and guest experience for Planetarium shows through effective communication skills and interpersonal interactions • Assistant Teacher at Meadowlands Environmental Center Lyndhurst, NJ ◦ Researched and observed

			amphipods, turtles, shrimp, and other wildlife within the marshes ○ Engaged in educational interactive sessions to analyze animal structures and appearances deeply through activities like dip-netting and utilizing microscopes, leading to the identification of key findings
Veronica Mok	Georgia Institute of Technology	Atlanta, Georgia	● Engine Test Stand Avionics Lead @ the Yellow Jacket Space Program and led the electronics development of the Helluva Engine Test Stand (HETS) to characterize liquid rocket engines at high testing cadence and achieved 10 hotfires in 1 month with precise sensor data utilizing thermocouples, load cells, pressure transducers, and differential pressure sensors. ● Opportunity Research Scholar @ The Propagation Group and used Keysight ADS to design the RF energy harvester and impedance matching network for the control board of a Reconfigurable Antenna Array crown utilizing RFID tags to alter radiation patterns. ● Director of Operations and Master Peer Instructor in electronics benchtop equipment @ The Hive Electrical and Computer Engineering Makerspace
Madeline Maldonado Gutierrez	Barnard College of Columbia	New York, New York	● Worked at the American Museum of Natural History in research projects involving computational astrophysics and exoplanetary science for the past two

	University		consecutive summers • Involved in Columbia Astronomy Outreach, a group of undergraduate and graduate students teaching astronomy education for K-12 students • Conducting an independent research project in stellar astrophysics at Columbia University • General Assistant at the Columbia University Libraries
Laiba Mahmood	CUNY Queens College	Queens, New York	• Participating in departmental research projects spanning from environmental microbiology, modeling the evolution of landscape and life, to understanding magmatic processes of historical eruptions through crystal imaging and mineralogy. • Completed the L'SPACE NASA Proposal Writing Academy in Spring 2024 as an outreach officer, addressing a gap in cryogenic storage tank usage. • Partook in a summer research program at the University of Michigan studying the stable isotopes $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of a microbial ecosystem mirroring Precambrian life during the summer of 2024. • Serving as Vice President of Knights of Science, a newly established STEM student organization focused on supporting student success in the sciences. • Mentor first-generation underclassmen pursuing STEM degrees.

			• Awarded the Barry M. Goldwater Scholarship (2025) and recognized as an outstanding undergraduate interested in a research career. • Incoming NASA ICONS intern for the summer of 2025, exploring the organic geochemistry and habitability of Europa's oceans.
Karthik Nagaraju 	University of Central Florida	Orlando, Florida	• American Institute of Aeronautics and Astronautics (AIAA) Team Lead, Assembly Lead, Aviation Vehicle Design Competition ◦ Led a team to design, fabricate, and analyze an original aviation vehicle, improving deliverables efficiency by 35%. • Knights Experimental Rocketry (KXR) – Aerodynamics Engineer, Intercollegiate Rocket Engineering Competition (IREC) ◦ Designed and analyzed rocket components for IREC, focusing on nose cone geometry, aerodynamics, and materials. • IDEAS Showcase Research Competition ◦ Engineered origami-inspired foldable structures and performed comparative load analysis against traditional methods. • Autonomous Boat Race Competition – Co-Lead Systems Engineer ◦ Designed and built an autonomous boat using Onshape and MATLAB, managing development with Gantt

			charts and cost analysis. ● FPP Coalition, Bright Minds Fresh Ideas Think Tank, Business Management Intern ○ Led a business proposal to reduce food insecurity by 20%, conducting research on logistics, materials, and sustainability.
Christian Odish	Washtenaw Community College (Transfer Program)	Ann Arbor, Michigan	● PHP, MySQL, Java, and web development, including work on database-integrated applications and weather API interfaces. ● Working towards finishing a Computer Science degree.
Akilesh Yelchuru	Florida Institute of Technology	Melbourne, Florida	Vex U (Documentation Lead, Build Lead, and Strategy Lead) - Designed a competitive competition robot with CAD with iterative prototyping to integrate mechanical subsystems to build an excelling high-performance robot.. - Performed detailed documentation of the team's engineering process through CAD sketches, performance analysis, and work log. - Developed a comprehensive analysis system for match optimization data derived from opposing team robot analysis. AIAA - Designed and built a high-power rocket for a space shot by defining mission goals, using simulation software for aerodynamic and propulsion analysis, constructing the airframe and recovery systems, conducting rigorous ground testing, and successfully

			launching and recovering the rocket to achieve mission objectives and gather post-flight data. Battlebots (CAD Lead and Mechanical Design Lead) - Designed and built a competitive 3 lb BattleBot by modeling in CAD, integrating a weapon and drivetrain system, testing for maneuverability and durability, and successfully preparing for battle through iterative refinements and strategic repairs. NASA Lunabotics - Member of the club and assisting in the recruitment and setup of the program.
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1.8.2. Team Management Overview

Delegation of Workload & Team Organization

The delegation of Team 10's workload revolves around the three subteams: Science, Engineering, and Programmatic. Based on each deliverable, Team 10 will carefully read over the instructions and criteria and understand which sections each subteam needs to be in charge of. From there, each subteam will have meetings and conversations to decide how the workload should be split amongst everyone. For instance, the Chief Scientist will communicate with the Planetary Geologists, Astrobiologists, and Astrophysicists regarding deliverable sections that are related to the Science team. Likewise, the Deputy Project Management of Resources will communicate with the Program Analyst and Mission Assurance Specialist about specific Programmatic sections. Lastly, the Lead Engineer will communicate with Mechanical, Thermal, Computer Hardware, and Electrical Engineers regarding tasks that need to be delegated to the Engineering team. Team leads will have active communication through Discord for decision making purposes, task delegation, and clarifications. This will result in fairness and an organized structure for the overall deliverable.

Decision Making

Decision-making in a group of twelve members will need to be organized as it is important to value each member's thought process. In a situation of split opinion, the approach will be to facilitate discussions and find commonalities to reach a solution that will benefit the team. That being said, it will follow a majority voting system because this allows each individual on the team to present why this is the best path to follow. Additionally, it enhances the team's decision making and will ensure better outcomes in the long-run. In a situation of time-sensitivity, an individual in a leadership role will make the executive decision on behalf of the group. A team lead making decisions in a quick manner will ensure a smoother workflow, less risks, and confirms that everyone will work in the same direction. Furthermore, a team lead will inspire confidence, leadership, and motivation to the team. It will be crucial to work in a timely manner as the team will need to decide on internal deadlines that will pave a smooth path for the deliverable. Relying on past experiences, situations, and judgments may help prioritize speed while also being able to submit a quality deliverable.

Conflict Resolution

During situations of conflict resolution, it will be crucial to early on understand the importance of communication and what channels will be used for it (Discord). Having

communication at least once every one to two days will allow the team to be on the same page. Additionally, creating an environment where each team member can speak freely will be crucial as this will increase activity and accountability within the team. Also, assigning roles to the deliverables will provide smooth transitions because the workload will be evenly distributed and makes it easier for everyone to check in with each other. Lastly, utilizing the resources and support systems of the mentors and L'Space leaders will be valuable due to everyone's experience in understanding the nature of the Mission Concept Academy.

Task Management & Accountability

In order to successfully manage all the tasks for each deliverable, it is important to stay organized and be on top of the assignments. For tracking each task in a manageable form, Team 10 has a Google Spreadsheet dedicated to this and it is found in the shared Google Drive. In this spreadsheet, there are tabs for each deliverable so that each team member can see all the tasks in an organized format. On each deliverable sheet, there are headings such as Category (subteam), Task ("Section X"), Assigned To, Pages Expected, Status, Date Assigned, and Due Date. This chart will provide easy access for all of the team members to see how much time needs to be dedicated and how they should communicate with the rest of the team regarding new ideas, decisions, and time sensitive tasks. As of right now, Team 10 is working well with having this shared Google Spreadsheet across all of the subteams. With this spreadsheet, it is important that each team member references this as it proves a clear idea of the workload and assignments. Everyone will be held accountable for completing tasks on time, however, the Team Leads will be having individual meetings with each member to check how the work is progressing and if any support is needed.

Current Team Organizational Chart

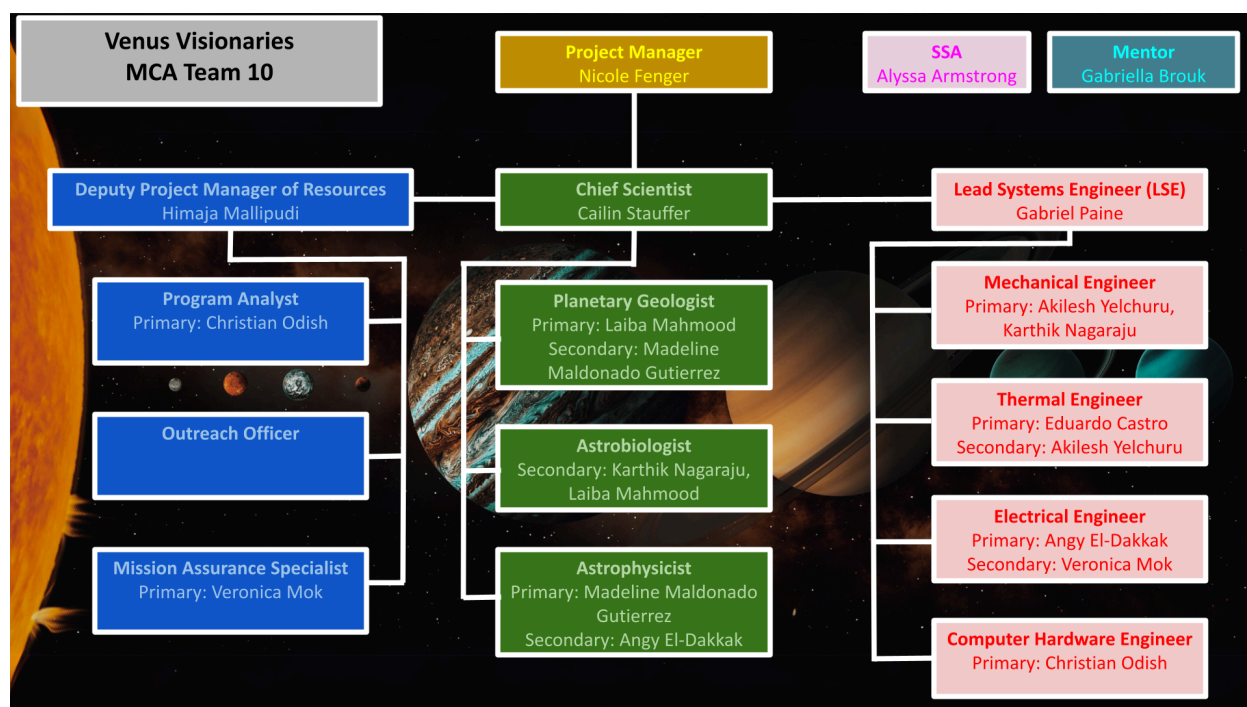


Figure 3: Current Organizational Chart

1.8.3. Schedule Overview

The Venus mission schedule is structured to support the efficient completion of all mission phases, from preliminary design through launch and operations. The timeline below provides an overview of the major milestones and key activities spanning Phases C through F, following the completion of Pre-Phase A, Phase A, and Phase B. These earlier phases represent pre-sunk work and will therefore not be included in the Gantt chart. The schedule focuses on Phase C (Preliminary Design) and continues through Phase D (Final Design and Fabrication), Phase E (Launch, Operations, and Data Analysis), and Phase F (Mission Closeout).

Each phase is aligned with NASA's standard mission lifecycle framework. The schedule ensures adequate time for critical design reviews, integration and testing, and risk mitigation, while maintaining flexibility for iteration and contingency planning.

Phase	Description	Est. Date of Completion
Phase B	Preliminary Design and Technology completion	April 21, 2025
	PDR Standing Review Board completed by SRB	April 21, 2025
Phase C	Key Decision Point C	May 1, 2025

	Final Design and Fabrication, completing engineering validation	May 2025 - Aug 1, 2025
	Critical Design Review (CDR) completed by SRB	Aug 1, 2025
	System Integration Review (SIR) Completed by SRB	Sep 2025
Phase D	Key Decision Point D	Oct 1, 2025
	System Assembly, Integration, and Test	Oct 2025 - Oct 1, 2028
	Operational Readiness Review (ORR) completed by SRB	Dec 1, 2028
	Mission Readiness Review (MRR)	Feb 1, 2029
	Key Decision Point E	Feb 2029
	Launch Date	Mar 1, 2029
Phase E /Phase F	Post-Launch Assessment Review	Apr 1, 2029
	Prime Mission Operation and Data Collection	March 1 - Sep 18, 2029
	Assessment of mission readiness for sustainability	March 2029
	Decommissioning Review (DR)	Oct 1, 2029
	Key Decision Point F	Nov 1, 2029
	Mission Closeout	Sep 17 - Dec 31, 2029
	Disposal Readiness Review (DRR)	Dec 1, 2029

1.8.4. Cost Overview

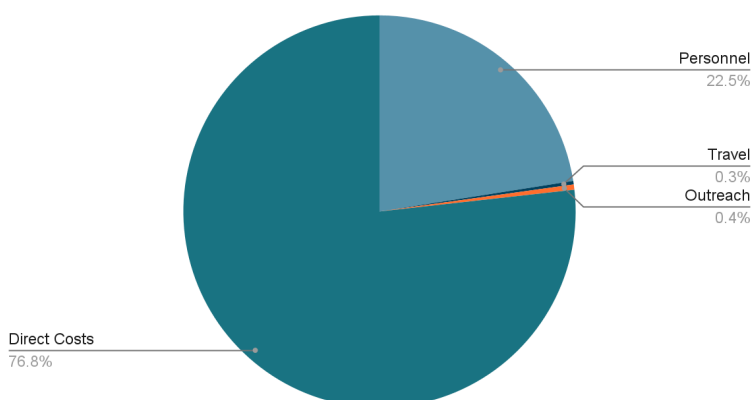
The mission's overall budget is limited to \$200 million. Based on past estimates that include system development, testing and integration, launch preparation, missions operations, personnel, outreach and travel expenses our total cost can be expected to be very close to the limit of the budget of around \$140 million. This gives the team 30% margin under the \$200 million limit and offers a small flexibility for unknown expenses that can occur in the future of the mission's development. This margin is defined by NASA's total cost margin from the Project Management Skill Module. If unexpected cost growth arises or extra programmatic needs show up the team will have to pursue a slight descoping to remain within the budget of the mission.

Parameter	Amount
Budget Cap	\$200 million

Total Estimate Cost	\$140 million
Remaining Margin	\$60 million
Margin percentage	30%

Phase by phase details will be provided. This will include fluctuations that can potentially arise in personnel, and allocations for travel and outreach. This can make sure the team can properly account for all expenses while staying in the budget cap.

Mission Overview Costs



2. Overall Vehicle and System Design

2.1. Spacecraft Overview

There are four top-level subsystem requirements, one for each. The payload system has three separate requirements based on the needs of the science traceability matrix. The table below rehashes the requirements from section 1.4 that are specifically applicable to the subsystems.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
SC-1.0	The Aerobot shall maintain operational integrity of all subsystems during the designated operation period	A functional probe is required for in situ gathering of data and transmission back to earth	MG-1.0	SC-1.1 SC-1.2 SC-1.3	Demonstration	Met
SC-1.1	Thermal systems shall be able to radiate heat into space to maintain internal temperatures necessary for operation	To allow the instruments and batteries to operate properly and prevent quick mission degrading	SC-1.0	TS-1.0	Demonstration	Met

SC-1.2	Power system shall be capable of providing power as related to power draw for each subsystem	Continuous power is critical for all components on board	SC-1.0	PWR-1.1	Analysis	Met
SC-1.3	Computer Hardware shall have enough power to store the data and have enough processing power to organize data	The mission will collect large amounts of data and needs to be sorted to send back to Earth	SC-1.0	CDH-1.0	Demonstration	Met

Table 4: Top-Level Subsystem Requirements

The engineering system is designed for a low-cost, short-duration mission to Venus, focusing on surface exploration and atmospheric analysis. The aerobot operates in Venus's extreme environment, with high temperatures, high pressure, and a harsh atmosphere. There are five primary subsystems within the aerobot; mechanical, power, command and data handling, thermal, and payload.

The power system will be responsible for supplying the needs of the thermal, CDH, and payload subsystems through a battery storage format. The thermal system will play a similar role, keeping all other systems in a safe operating range as the environment of Venus heats them up. It will also play a secondary role of protecting the interior from a relative amount of the environment outside. The command and data handling system will provide inputs to all other systems to allow for operation of the vehicle. Most importantly, it will maintain a connection to Earth to send the data collected and stored onboard. The mechanical system will provide mounting for all other systems while also providing a connection to the balloon pump subsystem to allow for safe passage. Lastly, the payload system will consist of all instruments required to gather data in situ to fulfill the goals of the mission.

The final design of the aerobot resembles a cubesat hot air balloon fusion in which the mechanical assembly is wrapped in thermal protective coating with openings allowing for observations to be made as the in atmosphere time progresses. The overall dimensions have been compiled into a drawing while omitting exact specifications of the balloon system.

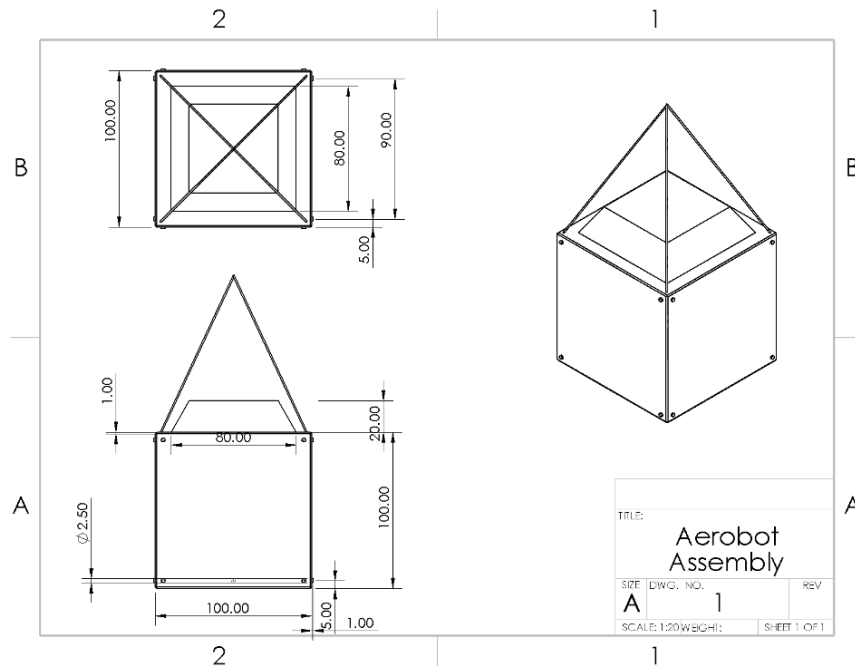


Figure 4: Aerobot Assembly Drawing

Below the mass, dimensions, and power draw of each subsystem has been specified. Individual component specifications have been summed while the dimensions represent the overall size of each subsystem assembly. It should be noted that the dimensions of the mechanical system represent the overall size of the spacecraft since it must house every other subsystem. Additionally, the mechanical and power systems have no listed power draw as the mechanical system does not use any and the power system generates.

Subsystem	Mass (kg)	Dimensions LxWxH (cm)	Maximum Power Draw (W)
Mechanical	5	100 x 100 x 100	0
Power	4.2	40 x 30 x 20	0
Command and Data Handling	3	30 x 30 x 40	80
Thermal	5.796	14.68 x 7.62 x 18.8	88
Payload	32	60 x 50 x 40	75

Table 5: Subsystem Specifications

2.1.1. Mechanical Subsystem Overview

The mechanical subsystem is responsible for the integration of various subsystems for this mission. As such, multiple factors such as the ability to regulate temperature, pressure, and altitude in the corrosive and gassy atmosphere of Venus. The mechanical subsystem consists of the structure, sensors, mobility, and a warm electronics box. The structure is the chassis, which is a Blue Canyon Technologies - X-Sat Saturn-Class Bus. The sensors are an IMU, lidar, receiver, and transmitter. The mobility consists of the pump. Since the Venus atmosphere is so intense in terms of corrosion and temperature, the following requirements were set. This would ensure for the mechanical subsystem that the other subsystems could be managed effectively and allow for the aerobot to operate at maximum efficiency.

2.1.1.1. Mechanical Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req Met?
1	The structure subsystem should support 50 kg of mass	Makes it compliant with the launch restrictions.	Customer Constraint		Demonstration and Test	Structure	TBD [1]
2	The Aerobot system shall not exceed 1m x 1m x 1m in stored configuration.	Aligns with deployment and integration constraints.	Customer Constraint		Analysis and Test	Aerobot System	TBD [1]
3	The GNC subsystem shall, in relation to the x and y axes within 1mm and the z-axis within 0.1 mm, have its center of gravity.	Aligns with deployment and integration constraints.	Customer Constraint		Analysis	GNC	TBD [3]
4	The GNC subsystem should survive the transportation from Earth to Venus as well as the atmosphere of Venus	Aligns with deployment and integration constraints.	Customer Constraint		Analysis	GNC	TBD [4]

	by itself, about 50 to 70 feet above the aerobot system itself.						
5	The structure should hold the electronics with the ability to thermally manage the electronics of the mechanical subsystem.	Makes it compliant with the launch restrictions.	Customer Constraint		Analysis and Test	Warm Electronics Box	TBD [5]
6	The pump should be able to alter aerobot buoyancy to allow the aerobot to remain about 50 to 70 km above Venus's ground.	Makes it compliant with the launch restrictions	Customer Constraint		Analysis	Pump	TBD [6]

Table 6: Mechanical Subsystem Requirements

2.1.1.2. Mechanical Sub-Assembly Overview

2.1.1.3. Mechanical Subsystem Recovery and Redundancy Plans

There are four major mechanical subsystems: primary spacecraft, EDI system, aerobot system, and descent system. However, the aerobot system is the part of interest to the MCA team. The aerobot system is composed of the payload, mechanical, CDG, communication, and thermal subsystems. The aerobot has various mechanical subsystems, including mobility, sensors, structure, warm electronics box, deployment, and payload.

For this mission, the aerobot shall utilize a helium balloon powered by a controlled pump to remain suspended at a specified altitude between 50 and 70 km. By using the pump to control balloon volume, the buoyant forces, and thus the altitude, can be precisely controlled. Since this design has been conceptualized and no other steps have occurred yet, the deployment system is TRL 2.

The aerobot will utilize various sensors to identify and convey logistics. These sensors include IMU, lidar, transmitters, and receivers. The IMU tracks positioning in the x,y, and z axes as well as tracks the aerobot's velocity. The lidar sensor uses lights and lasers to gauge distance and terrain. Since the use of these sensors can work in a laboratory and

are functioning, the TRL level for the sensor subsystem is TRL 4.

The structure of the aerobot should be lightweight yet durable; to achieve this, it will have a mixture of materials, including aluminum and composites. The actual design of this structure will be similar to Blue Canyon Technologies - X-Sat Saturn-Class Bus. Since this design has been utilized in space and has worked on multiple occasions, the structure subsystem would have a TRL level of TRL 3. This is because while it has been utilized in space, it has not been used in the conditions of Venus, only space, meaning it has only experienced half or less of the conditions that the aerobot will experience.

The lowest TRL subsystem under the mechanical subsystem rating applies to the whole mechanical subsystem, meaning the overall mechanical subsystem TRL level is TRL 2.

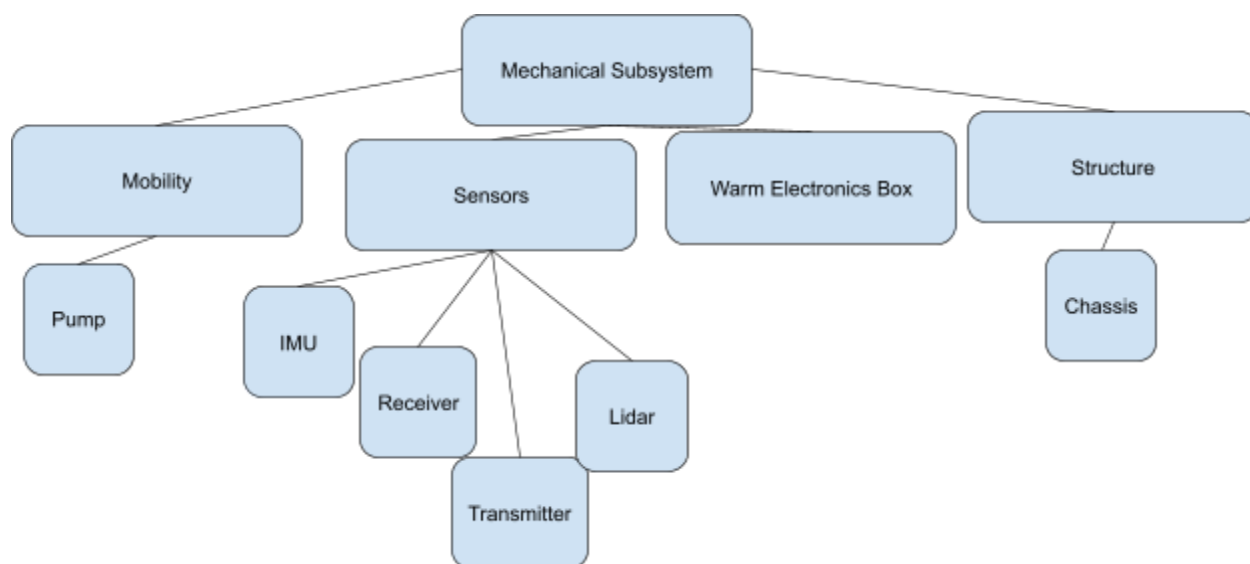


Figure 5: Diagram of the various subsystems that compose the mechanical subsystem.

2.1.1.4. Mechanical Subsystem Manufacturing and Procurement Plans

Per NASA policy, the Aerobot System will be designed and constructed through an industry-wide effort that leverages domestic and international partnerships. For each component, an ideal supplier was identified alongside a backup supplier. Suppliers were chosen based on experience, reliability, and capability of producing the desired product. Specialization in the aerospace industry and American origin were also considered. Beyond supplier credibility, another crucial consideration for the timely construction of the Aerobot is the lead time of each component. Expected lead times were thus

estimated for each component. For the component lead times, it is important to note that publicly available information was extremely scarce and that individual suppliers should be contacted for more precise estimates. Due to lack of available information, lead time estimates were generally not derived from the chosen suppliers. Thus, it was decided that offering separate expected lead times for primary and backup suppliers was not beneficial.

Mechanical Subsystem

The LiDAR instrument that makes up the mechanical subsystem will be provided by Honeywell. As a long established aerospace company, Honeywell has continually worked with NASA in various missions including the Artemis program. Most importantly, Honeywell has direct experience with Lidar systems. One of Honeywell's products, the High Altitude LiDAR Atmospheric Sensing (HALAS) system, is capable of incredibly detailed high altitude atmospheric mapping ("High Altitude LiDAR Atmospheric Sensing (HALAS) | Honeywell" 2025). This is almost identical to the use case on Venus. With the collaboration of Honeywell, HALAS can serve as the basis for the Aerobot's lidar instrument. As a backup supplier, BAE Systems was chosen for their aerospace and LiDAR systems experience ("LiDAR Services" 2025). Although existing products such as HALAS are available, significant changes in design will need to be made to apply LiDAR in the Venusian environment. Because of this, lead times are estimated to be around 50 weeks for the LiDAR instrumentation.

Due to their extensive history in aerospace and communications technology, including GPS technology, Collins Aerospace will be contracted to design and manufacture the communications instruments ("GPS Antennas and Receivers" 2025). As the backup supplier, General dynamics was chosen for their experience in the space sector and their existing GPS products ("Satellite GPS Receivers - General Dynamics Mission Systems" 2025). Due to the unique nature of this application which differs from the typical satellite-based products in the market, the communications instruments must be contracted as a unique system. This increases expected lead times to 75 weeks.

Collins Aerospace will also provide the inertial measurement unit (IMU). Collins Aerospace is the ideal supplier as they already develop several high precision and reliable IMUs, such as their SiIMU02 and LITIS products ("Inertial Measurement Units" 2025). Honeywell will act as the backup supplier, with their selection of over a dozen IMUs of several application types ("Inertial Measurement Units | Honeywell Aerospace" 2025). Due to the independent nature of the IMU and its location within the temperature controlled zone of the aerobot, parts can largely be off the shelf and thus have a modest lead time. Using Honeywell's HG4930CA51 as basis, a lead time of 28 weeks can be expected ("HG4930CA51 | DigiKey Electronics" 2025).

2.1.1.5. Mechanical Subsystem Verification Plans

Verification and validation, also known as V&V, tend to be defined for a mission simultaneously, but are essentially very different topics. This procedure is used to define the testing and calculations required for the success of each subsystem. Verification testing occurs in a controlled environment to corroborate that the object of interest, whether it be a system as a whole, subsystem, or even a component, operates as intended to achieve the requirement defined ahead of time. Validation is the testing of the vehicle under conditions simulating the final destination and required trajectory to reach there. The methods of verification and validation are the same; the only difference between the two processes is that validation deals with not just a part of interest, but rather, subsystems and systems are tested.

There are four methods of verification and validation: analysis, demonstration, inspection, and test. Analysis is done utilizing modeling software in tandem with analytical techniques. This is done in order to anticipate the constraints of a part of interest using the derived data or calculations from other structures (typically a lower system) or verification from end products. The software can be used to make a mathematically representative model that can be tested in a simulation software. The analysis verification is utilized for an integrated system or product that cannot be tested directly. The second type of verification is demonstration. Demonstration is proving the requirement is met through testing the part of interest directly. This method requires the least amount of data gathering. The third verification method is inspection. Inspection involves visual verification and examination to identify the various design elements and identification from manufacturers. Inspections include documentation, various records, and drawings. The fourth form of verification is a test. Testing occurs towards or during the final interactions of a particular item of interest. Typically, detailed analysis occurs as most of the product is tested once to verify that everything works. This tends to be the most intense technique in terms of utilization of resources under specific conditions and at discrete points.

In order to properly identify and decide on aerobot subsystems, a list of criteria must be developed to aid in the decision-making process. The aerobot will be tasked with fulfilling a variety of mission requirements, and this will necessitate different criteria for each subsystem. With regards to mission requirement 1, the weight of every subsystem must add up to no more than 50kg, and therefore, weight must be a primary consideration. Requirement 1 utilizes a mixture of demonstration and testing verification methods. This is because the aerobot can only have 50 kg of mass. As the structure subsystem is responsible for supporting various systems, adding up to 50 kg, the structure should be able to hold 50 kg of mass. The structure weight is included in this calculation, but having tolerance in the various elements of the design allows for a more robust and functional vehicle when completed. Requirement 1 will be demonstrated by

placing a mock weight on top of the structure so that it can support 50 kg, and through final iterations, this will be tested from the construction of the aerobot. Requirement 2 defines a maximum size for the stored configuration of the aerobot, necessitating a size criterion. If this requirement is not closely monitored, a “Tetris-like” process may have to be used. Requirement 2 employs a mixture of analysis and testing verification methods. Through analysis, the system size requirements are created and tested, while through testing, the size requirements are measured and ensured that the size requirements are maintained. Requirement 3 generates a variety of criteria for all systems onboard the aerobot to be able to survive the environment they are subject to. The extreme heat and pressures that can be found as a vehicle descends through the atmosphere necessitate robust materials and electronics to be chosen. Therefore, heat tolerance and pressure tolerance shall be the primary criteria in all component considerations. Further, all electronic systems will draw a certain amount of power that the power generator must compensate for. Power generation for the aerobot shall be decided upon specifically based on longevity, maximum output, and other criteria specific to energy-generating and storing systems. Requirement 3 only uses the analysis verification method as it is easier to model and simulate the actual design before building to adjust the center of mass, rather than create the physical model and modify the design afterward. Requirement 4 is related to the science suite on board the aerobot. Science subsystems such as sensor suites will be responsible for gathering and relaying data to the command suite. Some criteria for decision-making related to these sensors will include alignment, i.e., where that sensor needs to be able to point, if at all. Data generation will need to be closely considered along with onboard data storage and communication bandwidth, as there will be a limited amount of time to transfer data back to Earth before the aerobot fails. If a sensor generates too much data, it may overload communications and lead to failed science goals. Power draw, price, TRL, and availability will be further criteria that must be looked at for the science subsystem. Requirement 4 also uses only analysis verification methods as emulating space travel as well as Venus conditions, to verify and test separation distance from the primary spacecraft is a major task. The most efficient way to verify the functionality of this system is through modeling and simulation. Requirement 5 states: “the structure should hold the electronics with the ability to thermally manage the electronics of the mechanical subsystem,” which means that although multiple verification and validation methods can be utilized, analysis and test work best in this scenario. This is because the analysis would be able to use software such as CFD to analyze heat transfer in the warm electronics box. Then, using this data, the warm electronics box can analyze and identify issues before the test V&V method. The test method would then verify or deny the analysis method’s findings to further allow the tuning of the warm electronics box to allow for the fine-tuning of its performance. Requirement 6 states, “the pump should be able to alter aerobot buoyancy to allow the aerobot to remain about 50 to 70 km above Venus’s ground”.

Since the environment of Venus and the resources required to emulate this situation would be far too extreme, the best V&V method to utilize would be analysis. This would allow the comparison of the whole aerobot and, specifically, the aerobot pump's performance in such a situation to be analyzed over time for as long as required.

2.1.2. Power Subsystem Overview

2.1.2.1. Power Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
PWR - 1.1	The power system shall have enough power to supply all components on board with the necessary power for the duration of 48 hours	System need power to function, thrive, and survive	SC-1.2	PWR - 1.2 - 1.9	Analysis	Met
PWR - 1.2	The power system shall successfully expended during descent and lower aerobot	The power system must handle peak power demands for control systems to ensure a smooth and controlled descent.	PWR - 1.1	PWR - 1.3	Demonstration	Met
PWR - 1.3	The power system shall withstand a wide temperature molten-salt based Lithium battery module that can meet the required temperature range of Ozza and Maat Mons	The Lithium battery, with its high-temperature stability and robust design, is ideal for Venus	PWR - 1.3 SYS - 3.0	-	Test	Met
PWR - 1.4	The power system shall include high-temperature supercapacitors to handle peak power demands during descent and instrument operation.	Power demands exceed the continuous power output of the primary battery and fuel cells, and supercapacitors provide the necessary high-power bursts.	PWR - 1.1	-	Test	Met
PWR - 1.5	The power system shall include a Power Management and Distribution (PMAD) system to optimize energy use and prevent overloads.	Efficient power management is essential to ensure stable voltage levels, prevent overloads, and	PWR - 1.1	-	Analysis	Met

		optimize energy use across all subsystems.				
PWR - 1.6	The power system shall include a thermal management system to maintain operational temperatures for all components.	Venus's extreme temperatures require a robust thermal management system to protect sensitive electronics and ensure reliable operation.	TS - 1.0	-	Test	Met
PWR - 1.7	The power system shall supply enough power for a dependable avionics and control systems with an onboard computer, flight control systems, and sensors	Avionics and control systems are critical for navigation, stability, and mission success and require a stable and reliable power supply to function effectively.	PWR - 1.1	-	Test	Met
PWR - 1.8	The power system shall provide power effective and reliable communication system	Communication systems are essential for transmitting scientific data and maintaining contact with Earth.	PWR - 1.1	-	Analysis	Met
PWR - 1.9	The power system shall provide high precision to power the following instrumentation components: VIRTIS, analysers, and weathering sensors	Scientific instruments require precise and stable power for accurate data collection and analysis.	PWR - 1.1	SCI - 1.1 SCI - 1.2	Test	Met

Table 7: Power Subsystem Requirements

2.1.2.2. Power Sub-Assembly Overview

The foundation of the Aerobot power system is high-temperature Lithium-Sulfur (Li-S) primary batteries. The batteries are chosen due to their high energy density of approximately 500 Wh/kg and high-temperature operation capability. Two redundant stacks of 15–20 series-connected Li-S cells provide a total capacity of around 12.4 kWh, enough to power the Aerobot throughout a full 48-hour mission cycle. The batteries provide baseline power to the major systems, supplying around 259 watts: 125 W to science equipment, 80 W to command and data handling systems, and 54 W to thermal control. Aerobot's power system relies on wired connections resistant to the rugged environment of Venus. All of the critical devices—lithium-sulfur battery, supercapacitors, solid oxide fuel cells, and Power Management and Distribution (PMAD) unit—use supra-temperature, radiation-resistant cables like platinum or nickel alloys. They ensure stable conductance and structure under high-radiation, high-temperature conditions. To protect these high-strength cables, the system also incorporates dedicated connectors like ceramic or high-temperature metal connectors that have been specially built to be capable of Venus's high temperatures and pressure. A prime example of one such offering is Omnetics Space-Native Series, which is specifically known to meet aerospace and planetary application standards. The primary power source, the lithium-sulfur (Li-S) battery, is a non-rechargeable system, designed to provide a single discharge cycle during the mission. This design is made simpler by reducing system design complexity and maximizing reliability for the short-duration mission profile. During normal flight, the Li-S battery supplies a baseline level of approximately 50–100 watts of power, which should be sufficient for sustained operation of avionics, communications devices, and thermal management units. This constant level of energy assures that the Aerobot is active and responsive at all times throughout its mission duration, even during the most austere atmospheric conditions.

Each stack is operated safely within a voltage range of 31.5–42 V by a specially designed Battery Management System (BMS) that safeguards individual cell health, prevents overcharge or deep discharge, and manages power distribution in cooperation with the larger PMAD system. To augment the Li-S batteries and manage variable power needs during mission-critical stages, such as, atmospheric descent and high-instrument load operation, the Aerobot uses high-temperature supercapacitors. They provide 200 to 500 watts of instantaneous power for brief periods, buffering sudden spikes in demand that would destabilize the system otherwise. The supercapacitors are replenished by the main batteries or by the Aerobot's secondary power source—solid oxide fuel cells. The SOFCs form an essential part of the Aerobot's hybrid power approach. The SOFCs function within a temperature range of 600°C to 1,000°C and are hence very well adapted to the Venusian environment. The SOFCs produce steady electrical power ranging from 50 to 100 watts by electrochemically

oxidizing hydrogen and oxygen with 50–60% conversion efficiency. This sustained power is required for missions of long duration, particularly during descent when navigation, propulsion, control, and communication systems are all operating simultaneously. Not only do the SOFCs generate electricity, but they also assist with thermal management; the SOFCs' waste heat is utilized to help keep the batteries and other subsystems at their optimal operating temperatures, minimizing the requirement for special heaters. The Power Management and Distribution (PMAD) system stabilizes voltage supply to all onboard systems, regulates the levels as per demand and also prioritizes the essential components during high activity periods. PMAD incorporates DC-DC converters that step down the Li-S battery output to avionics levels (28 V) and sensor levels (5 V), voltage regulators that stabilize variable outputs, and power distribution units (PDUs) that allocate power based on mission phase and system priority.

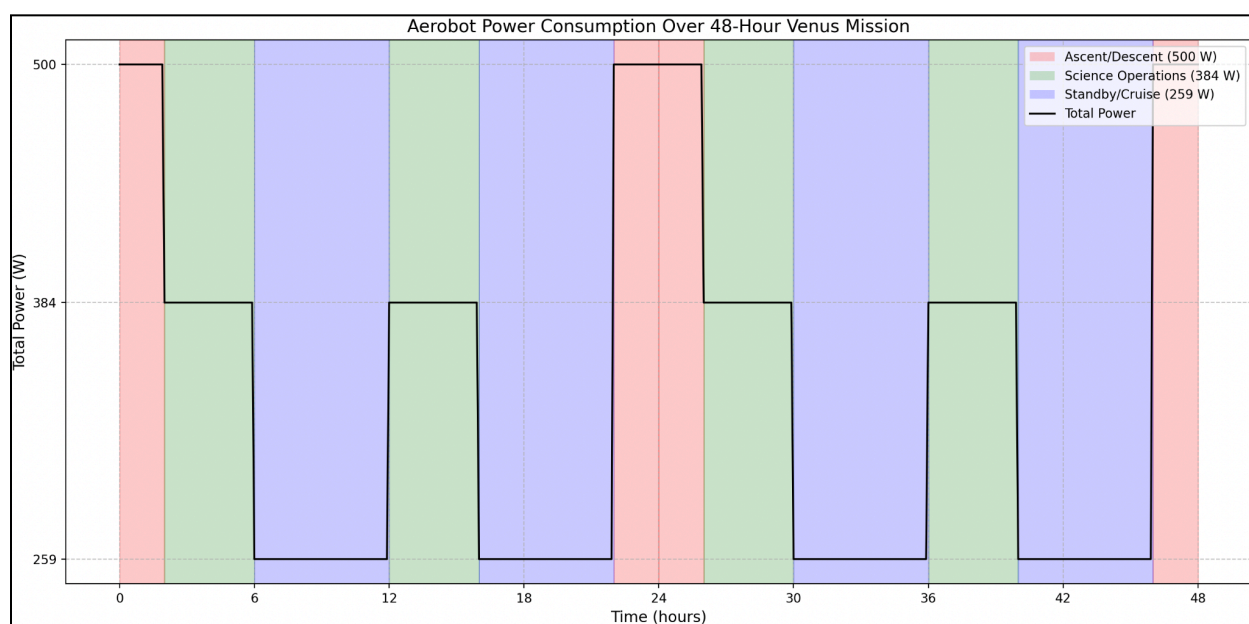
The power subsystem is endowed with many levels of thermal and environmental ruggedness. The heat output from the SOFCs is wisely directed to ensure thermal balance of the system so that core electronics and energy storage devices are preserved within safe levels. Safety and redundancy are built into the system. Dual battery stacks eliminate single-point failure of the primary power supply, and supercapacitors dampen random load spikes. The BMS provides continuous battery condition monitoring and communicates with the PMAD system for balanced load management. Fail-safe circuitry and protection schemes also eliminate electrical fault conditions, allowing the Aerobot to respond in real-time to mission condition changes without loss of performance or reliability.

Lithium-Sulfur Primary Battery technology is at a Technology Readiness Level, TRL, of 5–6. Lab-tested cells have demonstrated operation in high-temperature environments by surviving 26 days at 475°C. However, little to no data exists regarding conditions that mirror the surface of Venus, such as its corrosive environment and the combined stress of long-term heat and electrical exposure. To advance the TRL to 7–8, planned work involves testing in a Venus simulation chamber, such as NASA Glenn's GEER facility, to validate performance under conditions of 96% CO₂, 90 atm pressure, and sulfur-cycle corrosion. Long-duration cycling tests for more than 100 days also need to be conducted to assess degradation rates and overall durability.

Solid Oxide Fuel Cells (SOFCs) are at present at a current TRL of 5. They have undergone testing under conditions similar to those in space including vacuum and thermal cycling, but not yet those on the surface or in the atmosphere of Venus. While capable of providing 50–100 W of steady power, they have not been interfaced yet with mission units like the Venus Aerobot, for example, to provide buoyancy control and heat coupling. Raising the TRL to 7 will entail prototype testing under high-pressure CO₂ conditions that mimic Venusian atmospheric chemistry. The system must also be shown to operate autonomously with a mission-specific balance-of-plant, e.g., fuel storage and waste heat rejection elements.

Power Management and Distribution (PMAD) systems are at TRL 7 and already flight-proven in numerous space missions, such as technologies such as voltage regulators, DC-DC converters, and Power Distribution Units (PDUs). Despite not yet discovering Venus-specific risks, derating high temperatures might take only minor changes to design. For increasing TRL to levels of 8–9, components need qualification into Venusan-like thermal regimes, particularly with operational margins for temperatures of 475°C. Moreover, the features of redundancy must be examined with conditions resembling prolonged mission lives in order to check system dependability.

Thermal Control Systems are currently at TRL 6. Key components like high-temperature insulation, heat pipes, and coolers have been demonstrated to operate in harsh conditions, including lunar and Martian space missions. However, Venus also presents unique requirements like corrosion resistance under sulfuric acid clouds and thermal stability for long-term periods in harsh environments. To advance the TRL to level 8, materials will be exposed to supercritical CO₂ and sulfuric acid vapor in proprietary test chambers. Additionally, a ground demonstrator—a sample being a high-altitude balloon—may be employed to demonstrate thermal control algorithms in a relevant environment.



Subsystem	Mass (kg)	Max Power Draw (W)	Dimensions (cm)
Baseline Power System	12	100	40 x 30 x 20
Instrumentation	13	125	30 x 30 x 40
Communications	6	30	10 x 10 x 10
Descent Operations	5	200	10 x 15 x 15

Total	36	230	90 x 85 x 85
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Table 8: Power Sub-Assembly Power Draw and Dimensions

Solar panels function very well where sunlight is present in abundant profusion, but Venus's vast cloud cover severely limits their operation. Only 1–2% of sunlight can reach the surface because of the atmosphere on the planet, which makes the use of solar panels for electricity generation impossible. Solar panels, though economical for operations in sunlight-dominated atmospheres, inefficient to use on Venus, would need back-up power arrangements, thus increasing overall cost, under whose limitation we have to operate. Li-ion batteries are lightweight, high energy density, and are capable of delivering sufficient power for a short-duration mission. But they cannot operate at the extreme surface temperature on Venus. Li-ion batteries are capable of meeting power requirements for a short mission if operated within their temperature envelope, but the need for further thermal management systems increases complexity and cost. Li-ion batteries are durable in Earth-like environments but would degrade rapidly under Venusian conditions with significant alteration. Li-S batteries (molten-salt electrolytes with Li-Si or Li-Al anodes) are specifically designed for high-temperature use because they will operate at Venus's high surface temperature and provide sufficient power for the mission. Batteries are very reliable under extreme conditions, as observed under laboratory testing. Saltwater Li batteries are adequate to supply power for all the mission phases, including descent, surface maneuvers, and data gathering, without needing additional thermal management systems. Though Li-S batteries are more expensive than Li-ion batteries, they are cheaper due to the fact that they are able to operate without incorporating thermal management systems since they are corrosive-resistant in the environment of Venus by the nature of their robust structure and high-temperature materials.

Criteria	Explanation	Grade	Weight	Solar Panel	Li Ion Battery	Li-Sulfur Battery
Performance	The power system's ability to constantly and efficiently provide the required energy output throughout the 48 hours of operation for the mission, including descent, surface exploration, and data collection.	10 = high, 5 = medium 1 = low 0 = Fail	30%	2	5	9
Reliability	The power system's ability to operate consistently without failure under the conditions of Venus. including withstanding high temperatures (>400°C), corrosive atmospheric gases, during descent and surface operations.	10 = high, 5 = medium 1 = low 0 = Fail	25%	3	6	9

Fulfill Science Requirement	The power system must enable the aerobot to study Venus's atmospheric composition, surface geology, and thermal properties. This requires powering high-precision instruments and ensuring constant operation for the full 48-hour mission duration.	10 = high, 5 = medium 1 = low 0 = Fail	25%	2	5	9
Durability	The power system's ability to withstand Venus's harsh environment without degradation over 48 hours.	10 = high, 5 = medium 1 = low 0 = Fail	20%	3	4	9
		TOTALS :	100%	24.50%	50.50%	90.00%

Table 9: Power Subsystem Trade Studies

2.1.2.3. Power Subsystem Recovery and Redundancy Plans

The extreme conditions on Venus demand a power system that is not only robust but also fault-tolerant, with redundant layers and adaptive recovery mechanisms. For its 48-hour mission, the power design of the Aerobot is constructed to ensure seamless operation with an integrated combination of redundant power supplies, dynamic load handling, and fail-safe operation. At the center of the system are two Li-S battery stacks, each made up of 15–20 series-connected cells delivering a nominal voltage range of 31.5–42 V. The two stacks offer fault-tolerant failover capability; if one stack fails due to cell aging or thermal stress, the other can assume the full load. The (BMS) monitors and controls cell dynamics in real-time, isolating faults and re-routing power accordingly. The combined capacity of the system of approximately 12.4 kWh has a buffer to accommodate unexpected demands of energy. Backing up the batteries are supercapacitors and SOFCs. Supercapacitors provide 200–500 W of instantaneous power during peak loads, while SOFCs provide steady 50–100 W and recharge the capacitors on demand. SOFCs are particularly desirable on Venus with its elevated operation temperature range of 600–1,000°C because this harmonizes well with available heat, reducing thermal management complexity. Power management is controlled by the PMAD system, distributing power according to mission phase and system criticality.

Essential subsystems like avionics and communications receive uninterrupted power, and non-critical loads are throttled during power shortages. DC-DC converters provide stability against voltage fluctuations, and excess energy from SOFCs or supercapacitors is redirected to support core functions. Fail-safe circuitry prevents overcurrent and short-circuiting, isolating faults in milliseconds to prevent cascading failure. Deep discharge prevention is also ensured by the BMS to protect battery integrity. In addition, SOFC waste heat is utilized to maintain Li-S batteries and electronics within their desired thermal operating range (300–400°C), reducing the need for targeted heaters. Platinum/nickel alloys and radiation-resistant cabling is employed using high-temperature, corrosion-resistant ceramic connectors to guarantee performance under thermal cycling and corrosive conditions. The system adapts to mission phases: in ascent and descent, supercapacitors and SOFCs share the management of navigation and propulsion up to 500 W of power, and for science operation modes, power is dynamically allocated to prioritize critical instruments, with non-essential systems temporarily powered off if necessary. Contingency provisions include graceful degradation, whereby SOFCs sustain critical avionics and communication systems at reduced capacity when both battery stacks are lost.

Also, PMAD facilitates autonomous subsystem resets through reserved supercapacitor power, avoiding full reboots that could cause data loss. Overall, the Aerobot power system is a systematic design of resiliency that combines dual-redundant batteries, hybrid power augmentation, adaptive power routing, and environment hardening for the maintenance of operational integrity throughout its 48-hour Venus flight. Even through the experience of partial system malfunction, the Aerobot is able to recuperate and adapt to ensure continued uninterrupted continuation of its mission.

2.1.2.4. Power Subsystem Manufacturing and Procurement Plans

The power system manufacturing strategy for Aerobot emphasizes in-house development at strategic NASA centers to maintain control of proprietary technologies and ensure mission-critical reliability. NASA's Jet Propulsion Laboratory (JPL) is where it will be assembled, tested, and finally put together, while NASA Glenn Research Center produces high-temperature components such as Lithium-Sulfur (Li-S) batteries, Solid Oxide Fuel Cells, and specialty cabling. Structural and thermal testing, as well as environmental simulation, are conducted at NASA Langley Research Center (LaRC). The justification for in-situ fabrication lies in the need to control proprietary materials such as platinum cables, ceramic connectors, and sulfur-composite electrodes essential to Li-S and SOFC system performance and survival on Venusian environments. Additionally, NASA's exclusive access to test facilities such as the Glenn Extreme Environments Rig (GEER) with 460°C thermal and 92-bar pressure with sulfuric acid

exposure simulation makes end-to-end test and verification easy. Security features also play a major role; systems like the Power Management and Distribution (PMAD) and Battery Management System (BMS) must comply with ITAR rules and have controlled access environments.

Some components, though, are sourced from good external suppliers. Supercapacitors come from Maxwell Technologies (a subsidiary of General Atomics), which has contributed to NASA missions such as Mars rovers and Europa Clipper. Radiation-hardened and high-temperature connectors are supplied by Omnetics Connector Corporation, and nickel/platinum alloy cabling is supplied by TE Connectivity, a NASA-approved vendor with past collaboration on Venus lander missions. The purchasing plan is blended in-house manufacture and outsourcing based on component sophistication, manufacturing capability, and heritage in spaceflight. Critical items and corresponding suppliers are SOFC stacks (fabricated at NASA GRC) and Li-S battery cells, supercapacitors manufactured by Maxwell Technologies, PMAD system in-house developed at JPL, and cables and connectors for high temperatures from TE Connectivity and Omnetics, respectively. The lead times vary between two and eight months and estimated component costs total approximately \$3.3 million.

The acquisition process is such that long-lead items such as SOFCs and Li-S batteries are ordered first, which begin fabrication upon design completion. At the same time, components such as cables, connectors, and supercapacitors are ordered to fulfill integration schedules. Supplier audits are performed by JPL's acquisition team, including on-site visits to Maxwell and Omnetics facilities to ensure compliance with AS9100 aerospace standards. System integration and assembly occur in JPL's Class 1000 cleanroom (Clean Room 10-100) and are accomplished using a disciplined, four-phase process. Subsystem construction occurs within the initial 12 months: GRC constructs the battery stacks and SOFCs and conducts performance qualification in GEER, and JPL assembles the PMAD units and adds fail-safe circuitry. Maxwell provides supercapacitors and tests them under simulated Venus thermal conditions. Months 13-18 are devoted to structural integration with mounting of the Li-S batteries and SOFCs onto radiation-shielded titanium framework. The PMAD is linked to all subsystems through high-performance connectors and cables, and supercapacitors are mounted proximally to facilitate rapid energy transfers.

Thermal-vacuum tests at LaRC, vibration tests at JPL, and corrosion exposure tests in GEER are scheduled for months 19-21. Final system assembly occurs in month 22, when the integrated power system is combined with Aerobot avionics, propulsion modules, and scientific instruments. Pre-launch system-level testing (months 23-24) is the final phase. The eight-month SOFC fabrication window and three-month GEER test window are the critical path, and the overall timeline spans more than 24 months. Critical milestones include design freeze in month 1, component acquisition between months 2 and 4, subsystem completion by month 12, and flight-readiness by month 24.

There is a dedicated team from NASA centers, such as 10 engineers at GRC for the high-temperature systems (8,000 people-hours) and 8 engineers at JPL for the PMAD, BMS, and integration (6,000 people-hours). Thermal analysis is supplemented by four external contractors contributing an additional 2,000 people-hours. Major equipment includes GEER at GRC, vibration tables at JPL, and cleanrooms at Kennedy Space Center. Non-labor expenses are projected at around \$4 million, leveraging NASA's in-house cost savings and existing infrastructure.

Most development is done in-house because of the highly specialized nature of Venus-translated technologies. SOFCs and Li-S batteries use materials developed on NASA funding including zirconia-based electrolytes and sulfur-composite cathodes, which are not yet available commercially. Both PMAD and BMS likewise rely on the expertise of JPL in designing radiation-hardened FPGAs, essential for mission-critical performance. All that is allowable is outsourced hardware where flight-proven high-reliability hardware is being provided by commercial collaborators—such as Maxwell supercapacitors and Omnetics' connectors, high-temp—excluding R&D overhead wherever it is unnecessary and reducing risk.

A good risk mitigation plan supports the technical and project schedule success. Redundancy in the system is achieved by dual-supplying the Maxwell supercapacitors and maintaining an in-reserve batch at GRC. Risk is controlled against the schedule by adding a buffer of three months in the SOFC testing. For minimizing integration errors, mockups 3D printed of the wiring harnesses and chassis are being prototyped and tested prior to final assembly at JPL.

To conclude, this manufacturing and integration strategy favors indigenous development of the most critical Venus-specific technologies at NASA GRC and JPL with balanced selective commercial outsourcing of established spaceflight components. This blended approach allows the Aerobot power system to achieve its 24-month development target with continued environmental resilience, system redundancy, and mission success for the 48-hour Venus mission. Last assembly and validation at JPL, along with Kennedy Space Center launch operations, closed a tightly controlled and strategically executed production loop.

2.1.2.5. Power Subsystem Verification Plans

There is a stated method of verification for every primary requirement. For example, minimum power capability (PWR-1.1) will be validated through energy modeling and a 48-hour thermal-vacuum test to provide the system with a minimum of 12.4 kWh without falling below 259 W. Maximum power capability (PWR-1.2) will be illustrated by inducing a 500 W load during descent with supercapacitors. Li-S batteries (PWR-1.3) will undergo 48-hour high-temperature endurance test in NASA's GEER chamber, passing through 460°C and 92-bar pressure to maintain them at voltage ranges of 31.5–42 V. Supercapacitors (PWR-1.9) will be exposed to shock bursts of

power to make sure that they can deliver and recharge accordingly. For thermal control (TS-1.0), high-temperature testing and thermal simulation will be conducted to confirm that all components are at a safe temperature of operation, -20°C to 400°C. The instrument power accuracy (SCI-1.1) will be verified with voltage ripple testing and data integrity checks during simulated science operations. The PMAD system (SYS-3.0) will be tested with power distribution simulation and fault injection to confirm efficiency and speed in the recovery from faults.

Verification techniques are properly separated: Analysis is the forecasting of energy capability and thermal performance using software like MATLAB and COMSOL. Demonstration is replication of specific capability in actual though non-destructive environments, like replication of descent power peaks or PMAD reaction to battery failure tests. Testing is verification of real performance under extreme Venus-like environments, such as cycling Li-S batteries and SOFCs on GEER or supercapacitor bursts at 400°C. Instrument testing provides voltage stability and data accuracy. Inspection ensures that material like nickel/platinum wiring and ceramic connectors, is resistant to corrosion and of production quality using facilities like X-ray and visual examination.

Several NASA facilities will be employed for verification. GEER at NASA GRC will operate components under Venus surface conditions. JPL's Venus Simulation Chamber will simulate the cloud-level environment for ascent and descent tests. Thermal vacuum chambers at NASA LaRC will perform thermal cycling, and JPL's vibration tables will qualify components for launch stress.

The checkout schedule stretches across important milestones. Following the month 3 Preliminary Design Review, component-level testing is conducted months 4 through 10. Subsystem testing is conducted months 11 through 14, and an integrated system test of the entire system, such as a 48-hour simulation of the mission, is conducted months 15 through 18. Final inspections and pre-launch verifications bring it to an end in months 19 and 20.

Risk avoidance is the theme that runs through all of them. Fault injection will test the system resilience, such as single-point failure recovery like battery stack loss. GEER tests will monitor for thermal runaway with automatic termination if safe temperatures are reached. Sulfuric acid corrosion testing will guarantee all materials are resistant to exposure to it.

In summary, this test plan incorporates a combination of modeling, physical testing, demonstrations, and inspections to ensure the Aerobot power system is safe, reliable, and fully capable of meeting mission requirements. By utilizing NASA's state-of-the-art test facilities and qualified industry partners like Maxwell Technologies and Omnetics, the system is thoroughly validated to function on one of the most hostile planets in our solar system.

2.1.3. CDH Subsystem Overview

The Command and Data Handling (CDH) subsystem plays a foundational role in maintaining spacecraft operations and scientific productivity as it is the backbone of spacecraft operations. Put simply, this subsystem will tie telecommunications, data computing, and software architecture in one integrated stable environment. Its reliability, resource efficiency, and fault tolerance determine how effectively mission objectives will be met. By applying NASA's life-cycle management practices, focusing on TRL advancement, and addressing risk mitigation at each developmental milestone, the project team can ensure that the CDH subsystem is ready for the challenges of launch, on-orbit operations, and long-duration science collection. To help ensure the mission's goals are achievable, the design must contain data integrity checks, communications protocols, and fault tolerances which must be robust all while maintaining strict mass, power and thermal constraints. The requirements for each subsystem are directly from broader mission objectives and system-level guidelines as stated in detail in the (NASA Systems Engineering Handbook 2016, NASA Project Planning and Control Handbook 2016). These requirements make sure the subsystem will be reliable when merged and reach its performance goals through the development and operation timeline of the mission.

As a whole, the key CDH functions are as listed:

- Receive and validate commands from Earth or relay satellites.
- Coordinate execution of commands among subsystems and instruments.
- Route and buffer telemetry and science data.
- Compress, timestamp, and store data for downlink.
- Initiate autonomous fault management and saving procedures.

More specifically, the CDH subsystem integrates telecommunications, data computing, and software control into a cohesive, interdependent system. Rather than treating these elements separately, this integration supports reduced interface complexity and better fault tolerance. For telecommunications, the CDH subsystem will command all intra and inter-spacecraft communications, merging telecommunications, data computing, and software architecture subassemblies into a single unit (instead of separating each entity). Under ordinary circumstances, telecommunications is treated as a standalone subsystem but for this mission framework, it is more efficient for it to be merged along with on-board data processing and software-based command handling to emphasize their essential dependence.

The CDH subsystem must also ensure that commands from Earth are received and executed correctly and that scientific data gathered is collected, stored and relayed

back to the ground safely and efficiently. This will be achieved by the CDH subsystem executing uplinked commands, collecting telemetry and scientific data, ensuring data integrity, and downlinking collected information to Earth.

CDH Subsystem Specifications

Dimensions:

- Must fit within a maximum envelope of 30 cm × 30 cm × 40 cm. This ensures that the subsystem complies with the spacecraft's allocated volume for the avionics system as well as minimizes the interference with neighboring subsystems

Mass:

- The total allowable mass for this subsystem (including telecommunications hardware, onboard computing units, and embedded flight software) is capped at 19 kilograms. This constraint aligns with the spacecraft's overall dry mass budget and keeps launch vehicle requirements in check.

Maximum Power Draw:

- This subsystem is limited to a peak combined power draw of 80 watts, assuming all components operate at high capacity but without simultaneous full-load operation of the communication and data processing elements. During typical operations, however, the subsystem will consume closer to 60 watts.

2.1.3.1. CDH Subsystem Requirements

Below define the requirements how the CDH subsystem must function to satisfy mission objectives, focusing on constraints like mass, power, data throughput, redundancy and environmental resilience. Below is a table format standardized to capture each requirements rationale and verification method. This is in accordance with NASA systems engineering practices. They will also adhere to required fault tolerances and data protocols reflecting on NASA's emphasis on reliability in spacecraft operations.

Requirement ID	Requirement Statement	Source	Verification Method
CDH-01	The CDH subsystem will provide ongoing data and command handling capability in both nominal and	Makes sure critical inter and intra spacecraft data exchange will remain unbroken under fault	Analysis, Functional Testing

	contingency spacecraft operation modes.	conditions (NASA Systems Engineering Handbook 2016).	
CDH-02	The integrated CDH subsystem, this includes telecommunications and onboard computing, should not exceed a mass of XXkg and will have to abide by the dimension limits of 30 x 30 x 30 cm.	Aligns with overall system-level constraints on mass and volume according to the OfficialMCR_Team10 document. Any overruns could likely compromise other subsystems' allocations (NASA Project Planning and Control Handbook).	Mass measurement Inspection Dimension checks
CDH-03	The CDH subsystem must work within an average power budget of 60 W, and power.	Will preserve level power margins and will make sure operations are stable during high demand throughout the mission (NASA Schedule Management Handbook 2010).	Power draw testing under certain conditions that relate to the mission.
CDH-04	Data throughput capability must meet or exceed the speed of XX Mbps for downlink and YY Kbps for uplink. This will accommodate housekeeping and	Enables timely data return and real time ground control for the instruments onboard (NASA Project Planning and Control Handbook).	Budget analysis, In-lab testing

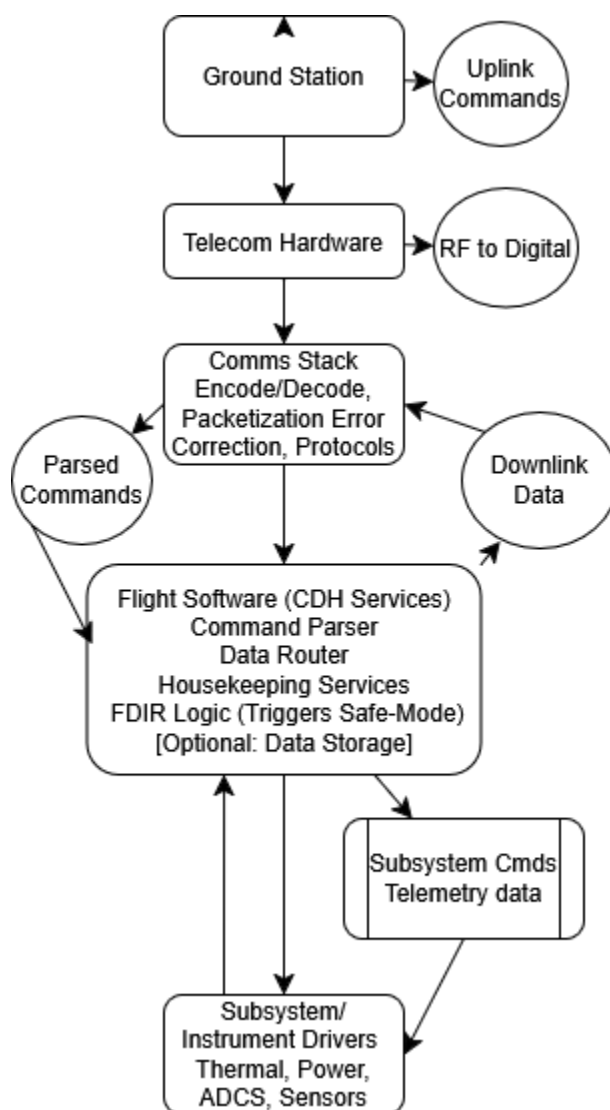
	data volumes during the mission's life cycle.		
CDH-05	The subsystem must merge software and hardware fault detection with an automated failover to protect command and data integrity if a single component ends up failing.	Reduces risk of mission loss from hardware and software inconsistency (NASA Risk Management Handbook)	Fault injection tests, Thorough inspections, analysis
CDH-06	Software protocols for data handling and command execution must adhere with NASA software assurance standards, making sure that data integrity through checksums or CRCs and authorized command access.	Protects against corruption or error, making sure that software behaviors will be fully functional for the mission (NASA Systems Engineering Handbook 2016).	Extensive software review, Simulation tests
CDH-07	Each subassembly within CHD must provide and be able to maintain temperatures that are operable within X-X celsius, this will be coordinated with the thermal subsystem for flight phases.	Make sure that various thermal conditions are reliable and will factor in potential heating during modes that are high power (NASA Systems Engineering Handbook 2016).	Thermal-vacuum testing, Qualified environmental testing.

2.1.3.2. CDH Sub-Assembly Overview

Scope of the CDH Subsystem

The Command and Data Handling (CDH) subsystem of this mission merges three formerly distinct subsystems: telecommunications, on-board computing, and software architecture. As a result, the CDH subsystem is now a tightly integrated system that prioritizes fault tolerance, reduced interface complexity, and operational efficiency. This architecture directly supports NASA's emphasis on modularity and subsystem cohesion in order to reduce risk as well as to increase maintainability across the mission lifecycle.

General Structure of CDH Subsystem



Component Relationship Flowchart

The flowchart illustrates the core data handling and command execution process of the CDH subsystem. It begins with the Ground Station, which transmits uplinked commands to the spacecraft via Telecom Hardware. These signals are converted from RF to digital, then passed into the Comms Stack, where they undergo encoding, decoding, error correction, and protocol handling. Once processed, the Parsed Commands are routed to the Flight Software, which is responsible for parsing, data routing, housekeeping services, and executing FDIR (Fault Detection, Isolation, and Recovery) logic. Commands are then directed to various Subsystem/Instrument Drivers that interface with systems like thermal control, power, ADCS, and onboard sensors. These subsystems return telemetry data and status information, which the flight software processes and packages for downlink. The Comms Stack handles the final formatting and relays the Downlink Data back through the telecom system to the Ground Station. This cycle ensures continuous, secure, and fault-tolerant command and data exchange throughout the mission.

CDH Subassembly Overview

The CDH subsystem can be roughly described as having the following subassemblies:

- **Telecommunications Subassembly:** Handles RF communication (uplink/downlink), encoding/decoding, and error correction. It is evaluated across X-band, S-band, and Ka-band with X-band selected for balance of heritage, data rate, and power demands. To ensure fault tolerance, dual-string redundancy and modular transceivers
- **Data Computing Unit (OBC):** Processes commands, routes data, and monitors subsystems. Combines rad-hard CPU with FPGA module for real-time compression and analysis. Balances speed, resilience, and energy efficiency. Expected TRL: 5–7 depending on final configuration.
- **Software Architecture:** Built on a real-time OS layered with mission applications. Includes data services, FDIR logic, task scheduling, and autonomy routines. Ensures modularity for upgrades and integration ease.
- **Integration and Interfaces:** CDH interfaces with ADCS for attitude data and torque commands, Power for surge-tolerant voltage supply, Thermal for thermal balance, and Payload for instrument telemetry routing.

Telecommunications Subassembly

The telecommunications subassembly handles all uplink and downlink pathways and manages the spacecraft's communication with Earth and potentially with orbiters or other assets. It includes transceivers, encoding/decoding logic, and forward error correction modules all of which are housed within a radiation-shielded environment. It is

also worth mentioning that data flow begins with RF signal reception and then is followed by digitization, decoding, validation, and dispatch to the OBC.

Prior to this report, a critical design tradeoff evaluated S-band, X-band, and Ka-band transmission paths. From the study, the following was determined:

- S-band offers robust reliability but low data throughput
- Ka-band allows high throughput but imposes narrow beam pointing and higher power demand
- X-band provides a middle ground where there is sufficient data rate for scientific and housekeeping data with proven flight heritage and minimal complexity

Since redundancy will be built into the telecom module, dual-string transceivers, separate power buses, and RF switch paths will all be implemented into this subsystem. This enables continued operations if one line fails—an approach encouraged by NASA's Systems Engineering Handbook for deep-space risk resilience.

Data Computing Unit (OBC)

Because the OBC is the core processor of spacecraft operations, it is responsible for parsing and routing commands from Earth, managing real-time task scheduling, processing and compressing payload and telemetry data, and initiating autonomous responses to anomalies. Prior to this report, a trade study was conducted and it was determined that a hybrid approach using a rad-hard CPU alongside a configurable FPGA offered the best balance between reliability and performance. It was determined that in this setup, the CPU ensures software fault tolerance while the FPGA accelerates compute-heavy tasks like image compression and signal analysis. In addition to this, the OBC is able to interface with sensors, the ADCS system, payload instruments, and power systems. However, this bidirectional traffic requires a robust bus. To account for this, SpaceWire has been selected due to its fault detection features and effective latency support.

Thermal and Power Interfaces

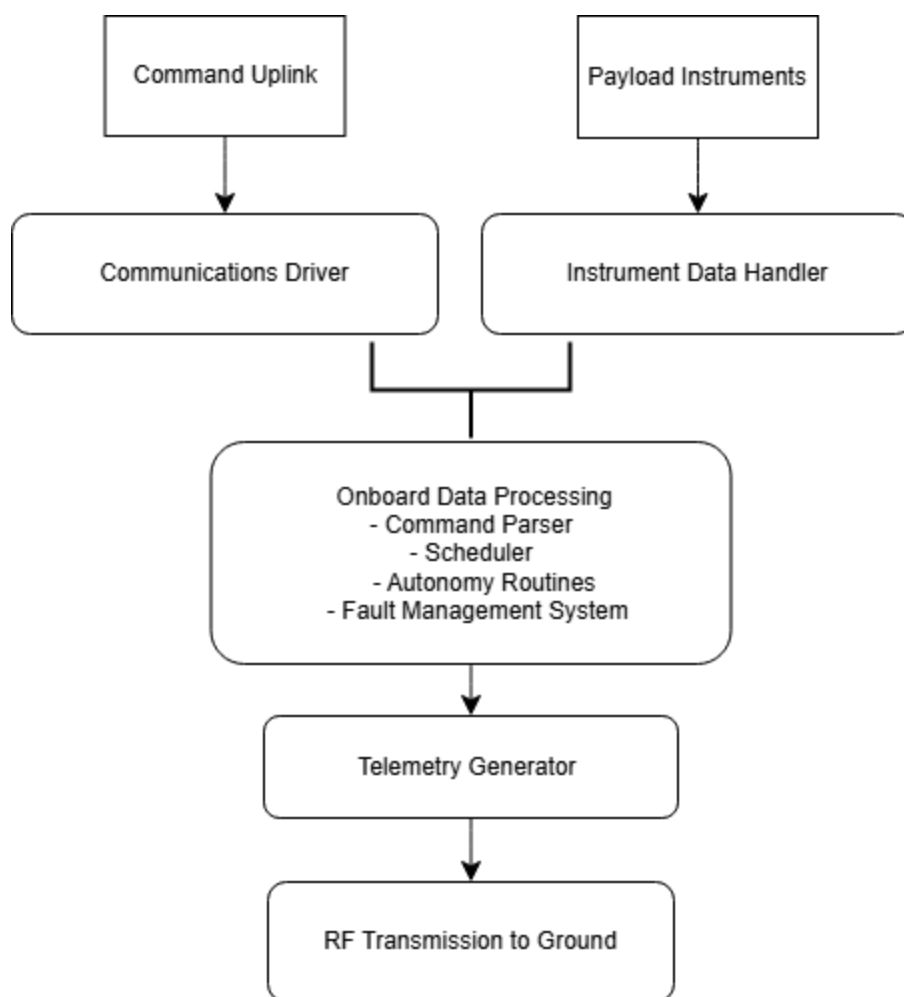
According to the Spacecraft Thermal Control Handbook the best practice for thermal interfaces is to have each subassembly thermally coupled to spacecraft radiators or heat pipes. To ensure compliance with component operating limits typically between -20°C to +55°C, heat-intensive periods (such as high-throughput downlinks) are modeled in the thermal analysis to measure thermal performance. Power draw profiles are also monitored dynamically to avoid overloading buses during high-load events like FDIR-triggered reboots.

Subsystem Integration and TRL Summary

Once integrated, the CDH subsystem's maturity will be governed by the lowest-performing component until it is fully tested across environment and interface parameters.

Subassembly	Initial TRL	Target TRL (Post-Testing)	Integration Notes
Telecommunications	6–7	7	Flight heritage; dual-string
OBC (CPU + FPGA)	5–6	7	Requires validation in thermal-vac
Software Architecture	4–6	7	NASA heritage + new autonomy logic

Software Architecture and Data Flow



Software Architecture Flowchart

The CDH software stack is designed around a clear functional data flow, visualized above. There is an emphasis on prioritizing autonomy, reliability, and efficient command execution and this is seen in the choices made for each module of the flowchart. As per the standard, the chosen architecture aligns closely with NASA heritage systems while also incorporating custom routines to support mission-specific requirements. In addition to this, the software architecture leverages NASA's Core Flight System (cFS) as its foundation. This allows for reuse of validated libraries and simplifies integration. This also allows for additional mission-specific applications to be layered on top of the existing architecture. This is due to the system supporting in-flight software patching via secure command uploads, enabling adaptability throughout the mission lifecycle.

At the first step (the top of the data flow), Command Uplink data received from ground control is passed into the Communications Driver. This will handle decoding,

validation, and initial buffering. In parallel, raw data from the Payload Instruments is managed by the Instrument Data Handler. This in turn formats and queues data for central processing. Next, both data streams converge within the Onboard Data Processing unit. As this module serves as the core of the software system, it integrates several critical services:

- Command Parser: Will interpret and route commands to the correct subsystems or application
- Scheduler: Will manage the execution timing of commands and software tasks, ensuring proper sequencing and resource use
- Autonomy Routines: Allows the spacecraft to operate independently during communication blackouts, anomaly events, or science collection windows

Fault Management System: Will execute continuous fault detection such as thermal, voltage, and timing anomalies and will execute recovery procedures

After this, processed data is sent to the Telemetry Generator. This system packages housekeeping, science, and system health data into telemetry frames according to CCSDS standards. Then, the output is relayed through the telecom stack for RF Transmission to Ground during the allotted scheduled downlink windows.

As a whole, the chosen approach ensures that each software function operates as a modular component within a larger pipeline. This helps to simplify testing, fault recovery, and allows for more seamless future upgrades. Underlying the entire architecture, a real-time operating system (RTOS) will be responsible for task management, memory isolation, and inter-process communication. The RTOS supports efficient execution, something that is vital for timing-critical operations such as fault recovery and command sequencing. In addition to this, Fault Detection, Isolation, and Recovery (FDIR) routines are embedded throughout the stack to strengthen redundancy. These routines will monitor heartbeat signals, sensor thresholds, memory integrity (via CRCs), as well as bus communication validity. In the event that a critical fault is detected, the system can autonomously transition to a Safe/Recovery Mode (which will be further discussed later), shutting down nonessential services while preserving power to command and telemetry capabilities.

Payload

In terms of payload, the data needs of each instrument, along with the needs of the other subsystems are also considered. For the different subsystems nested within CDH, instruments require high-throughput interfaces to send science data, ADCS requires real-time command exchange and feedback, power requires stable voltage supply and surge tolerance during high data rates, and thermal needs passive/active

control to keep CDH hardware within operational temperature.

Integration with Other Subsystems

The CDH subsystem's success depends on effective collaboration with other spacecraft systems. The power subsystem must require stable voltage rails, supply redundancies if required, and carefully integrate surge spikes during transmission that have high data rate. The thermal subsystem provides conduction or radiation pathways that will keep the electronics at stable operating temperatures, especially during high computing power or long downlinks. Attitude determination and control (ADCS) interacts with CDH for real-time sensor data and torque commands, and the payload suite may generate large volumes of data that place further strain on the subsystem's storage and throughput capacities. The NASA Work Breakdown Structure (SP-20210023927) highly encourages thorough planning of these interactions, making sure that the tasks for each subsystem are defined and budgeted accurately and correctly.

2.1.3.3. CDH Subsystem Recovery and Redundancy Plans

In a space environment defined by remoteness, radiation, temperature extremes, and non-interventionist constraints, system resilience is not just a desirable feature—it is a mission-critical requirement. As a result of this, the CDH subsystem must remain operational through hardware degradation, software anomalies, and transient environmental faults as the environment the hardware it is supporting in is very harsh and unforgiving. As described in the NASA Risk Management Handbook, NASA's fault management doctrine emphasizes building in systemic resilience through redundancy, autonomy, and recoverability. The design of this subsystem adheres to that philosophy by combining robust engineering with a fail-operational mindset.

Mission-Critical Role of Recovery and Redundancy

The CDH subsystem as a whole serves as the spacecraft's central nervous system—from executing time-tagged commands to maintaining onboard state awareness, coordinating subsystems, and ensuring that mission data reaches Earth. A failure here would impact all mission functions and can even result in total mission failure. With this level of systemic dependence, even a minor failure can cascade into catastrophic mission loss. Because of the grave importance of this subsystem, recovery and redundancy planning and procedures are to be embedded throughout this subsystem's architecture, lifecycle, and verification protocols. The strategy follows the four pillars of fault tolerance: detection, isolation, containment, and recovery. With this in mind, the main recovery and safety architecture is to be implemented in layered safeguards across hardware, software, and system architecture. For recovery to work properly, it should ensure that general state is preserved, telemetry is recovered, and operations resume in a controlled, validated manner.

Redundancy Architecture

The CDH subsystem will incorporate cold redundancy in all critical components. For this, dual-string telecommunications hardware should be mounted on separate boards, with electrically isolated power buses and independent interface lines. In addition to this, the onboard computer (OBC) exists in two fully independent units: a primary OBC that runs continuously and a cold-redundant backup that can be powered up and initialized by the flight software upon fault detection. Each unit includes its own memory bank and separate software stack to prevent shared point-of-failure modes. This will all help there are multiple modes of recovery in case multiple systems are impacted. For the system's memory components (including mission data buffers, configuration tables, and housekeeping logs), ECC-protected flash and RAM should be utilized as they are highly redundant. For Radiation-induced errors, they will be automatically corrected in-line through software. In addition to this, periodic scrubbing routines scan the memory for persistent soft faults which can help resolve any unintentional bit flips or radiation-induced errors. These routines will be triggered by flight software and validated using checksum comparators. All these measures will help reduce chances of catastrophic failure as there are multiple modes of recovery in case multiple systems are impacted.

Software Fault Management and Autonomous Recovery

Due to the drastic nature of software anomalies ranging from minor memory leaks to major command-handling stalls, the CDH software stack must include multiple layers of built-in defense to manage these risks and increase reliability. This can be achieved if task execution times are monitored against expected timing models. In this setup, if a service or task exceeds its window, it is forcibly terminated and restarted. Inter-process communication can also be validated using handshakes and acknowledgments and heartbeat signals from the OBC to peripheral devices are monitored continuously to further redundancy. Flight software will be built using a modular architecture where, if one application becomes unstable (e.g., an instrument handler), it can be safely restarted without rebooting the full OBC. This process is similar to "hot-swapping" in high-availability server environments and preserves continuity in spacecraft operations.

If the unlikely event that all else fails, a watchdog timer can be configured to detect systemic software stalls. In this case, upon expiration, the watchdog will reset the OBC and subsequently reboot the system into recovery mode. This mode will allow power retention by stopping all other power usage except for essential functions such as telemetry transmission, minimal thermal regulation, and beaconing.

Safe/Recovery Mode and Controlled Reinitialization

The greatest layer of protection is the system's safe/recovery mode. When a major anomaly which cannot be resolved locally (such as a sustained power anomaly, OBC memory fault, or bus timeout) is detected, the system will be engineered to automatically transition to a predefined safe mode. This mode will inherently be simple and conservatively engineered. All extraneous systems including all instruments will be powered down, science data collection is paused, and all power is diverted to thermal control, RF beaconing, and command reception to maintain overall system health. Safe mode telemetry packets are minimal but will be sufficient enough to help assess spacecraft health as well as inform ground-based recovery actions.

After the errors have been resolved, the craft will need to exit safe/recovery mode through the reinitialization process. This process is staged to ensure proper reactivation can be operated by ground operators that have the ability to initiate the reactivation sequences once telemetry confirms nominal conditions. These work by first restoring telecom parameters, then ADCS control, and then finally reactivating command to payloads and instruments. This recovery process will be extensively tested during integration and system validation to ensure successful operation.

End-to-End Testing and Verification

For end-to-end testing, recovery and redundancy logic are only as strong as their validation (verification is also discussed later on). This subsystem will undergo several hardware-in-the-loop simulations, testing common failure modes such as single-event latch-ups, ADCS command loop interruptions, corrupted command packets. These tests will also help evaluate system response. Fault injection testing at both the electrical and software layer. To ensure each fault path is both detectable and recoverable under worst-case assumptions, fault injection testing will be conducted at both the electrical and software layer. These tests are accompanied by end-to-end mission scenario simulations using spacecraft testbeds, often running for hundreds of simulated mission days.

With these redundancies, the CDH system is not only designed to withstand faults but to adapt, recover, and continue the mission with minimal intervention—a standard not just of NASA compliance, but of long-term spaceflight survivability.

2.1.3.4. CDH Subsystem Manufacturing and Procurement Plans

As a whole, manufacturing and procurement represent the operational core of implementation. This area also is prone to delays, cost overruns, and schedule deviations. As per NASA's Project Planning and Control Handbook, this plan outlines a

disciplined, integrated supply chain and fabrication strategy that balances cost, risk, schedule, and performance.

Overview of Procurement Philosophy

For procurement, the mission team adheres to a mixed procurement model. The following are combined:

- Select proven, flight-qualified COTS components for high-reliability and low-risk elements
- Select commercial-grade modules (screened and requalified to space standards)
- Fabricate custom parts for unique mechanical and electrical integration needs

The manufacturing and procurement plan for the CDH subsystem includes both commercial procurement (e.g., processors, telecom modules, passive components) and government-furnished equipment (GFE), such as previously flown boards or NASA-developed middleware. Procurement begins in Phase B and follows a gated approval process managed under Configuration Management guidelines. For trackability, each part number has a traceability path from vendor quote to final integration. This satisfies NASA's configuration assurance requirements.

Item Procurement Strategy

The procurement plan emphasizes early identification of long-lead components, such as radiation-hardened processors, memory chips, or space-qualified transceivers. As these items often carry 6–12 month lead times, they are often prioritized during early systems engineering reviews to avoid production delays.

Prior to final assembly, an early prototyping run of engineering models (EMs) is planned to verify interfaces and accelerate software integration ahead of flight unit delivery. This will help address integration errors early on in the process, further reducing delays and schedule deviations. To achieve this, qualification units (QUs) will be used for environmental testing such as TVAC, vibration, EMI for example. This helps in preserving the flight models for final assembly and payload integration.

In terms of vendors, they will be selected not only based on technical compliance but also on supplier reliability, manufacturing capacity, and history of export control compliance. This ensures the most efficient transition between plans and designs to the final product. By also instituting reliable vendor qualification, further steps can be taken to make sure any supplier deviation can be mitigated without delaying the critical path.

Manufacturing and Assembly Workflow

To ensure proper assembly, subsystem and final integration will be performed in an ESD-safe cleanroom at ISO 9001-certified facilities. The simplified workflow is divided into the following steps:

- Board Population and Assembly: Components are placed on PCBs via automated reflow soldering and visual inspection/X-ray scans verify placement
- Enclosure Fabrication: Custom milled or 3D-printed enclosures are built to exact mechanical tolerances and matched to chosen spacecraft frame dimensions
- Thermal Coupling and Shielding: Components are fitted with thermal interface material (TIM), heat spreaders, or heat pipes per the Spacecraft Thermal Control Handbook. EMI gaskets and grounded shields are installed at signal interfaces.
- Software Loading and Burn-In: Firmware and flight software are uploaded and verified; the unit then undergoes a 72-hour burn-in at operational temperature limits.
- Testing: Integration testing harnesses will simulate ADCS, payload, and power interfaces using emulators and flight-like telemetry scenarios

Quality Assurance and Verification Protocols

Per protocol, all manufacturing steps are logged using tracking systems (Digital Traveler Systems for example). Within this system, components and assemblies are tagged with unique serial numbers. This will allow end-to-end traceability to be possible in the case that failure analysis or root cause evaluation will need to be conducted.

Some verification activities include the following:

- Thermal-vac testing across operating range (-30°C to +60°C),
- Vibration and shock testing at launch-load levels,
- EMI/EMC testing per MIL-STD-461 standards,
- Functional simulation of mission phases (traverse, downlink, idle, anomaly response)

In these phases, if there any detected non conformances, they will be documented under Material Review Board (MRB) proceedings. In this case, deviations from design specifications require either rework authorization or formal waivers signed by subsystem leads and mission QA authorities.

Cost, Schedule, and Risk Control

For CDH mission costs, NASA's bottom-up CER-based model as outlined in the Cost Estimating Handbook will be used for cost estimating. WBS cost elements are tagged and integrated with earned value tracking (EVM). This enables active monitoring of burn rate, variance, and contingency levels which helps in monitoring current

spending as well as preventing possible further expenditure. This is done by creating schedule margins which are embedded at every stage (procurement, assembly, software upload, testing). This will help to ensure that one delay does not halt overall progress. In terms of percentages, contingency funds are set to around ~20% for hardware and 15% for labor. Actual values are displayed in the tables below.

Total Estimated Costs

Subsystem	Estimated Cost Range	Basis of Estimate
CDH	\$650,000 – \$950,000	Parametric estimate based on rad-hard avionics, dual-string redundancy, and deep space environmental tolerance

Component	Estimated Cost Range	Notes
Onboard Computer (OBC)	\$200K – \$300K	Radiation-hardened CPU + FPGA; includes redundant logic controller
Telecom Module (X-band)	\$150K – \$250K	Redundant transceivers, encoding, and RF conditioning
Software Development & FDIR	\$150K – \$250K	Includes autonomy logic, middleware, ground command support
Integration, Test, Verification	\$100K – \$150K	TVAC, EMI/EMC, HIL, functional testing and QA
Misc (shielding, harnessing, etc)	\$50K – \$75K	Includes connectors, enclosures, internal bus hardware

Subsystem Manufacturing and Procurement Plans Conclusion.

In this structured, traceable approach, the team aligns procurement and manufacturing execution with NASA's stringent expectations for reliability, affordability, and mission assurance. On this large of a scale, this helps to ensure that the CDH subsystem will be delivered not only on-time and within scope, but ready for the harsh realities of spaceflight.

2.1.3.5. CDH Subsystem Verification Plans

The verification of the Command and Data Handling (CDH) subsystem is an important part of making sure that the system works reliably under all mission conditions. This section will provide a comprehensive strategy to make sure that all performance, interface, environmental, and fault management requirements are met. The verification approach follows the guidelines outlined in the *NASA Systems Engineering Handbook* and other referenced NASA resources.

Requirement ID	Requirement Statement	Verification Methods
CDH-01	Provide ongoing data and command handling in all spacecraft modes.	Functional testing, mission simulation, software command-response testing
CDH-02	Meet mass and dimensional limits.	Physical inspection, subsystem measurement, integration fit test
CDH-03	Maintain operations within the power budget.	Electrical load testing under nominal and peak usage profiles
CDH-04	Ensure sufficient data throughput for mission demands.	RF loopback test, link margin analysis, throughput stress testing
CDH-05	Incorporate automated fault detection and failover.	Hardware-in-the-loop (HIL) tests, injected fault simulations, watchdog timer validation
CDH-06	Adhere to NASA software integrity standards.	Static code analysis, checksum/CRC testing, unit-level simulations
CDH-07	Maintain operational temperatures within	Thermal vacuum testing, temperature cycling, stress

	XX–YY°C.	analysis
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Each of these verification methods is designed to be executed during different phases of system integration and qualification, beginning at the component level and continuing through full system end-to-end testing.

1. Hardware-in-the-Loop (HIL) Testing

- CDH hardware components will be tested in real-time with simulated mission inputs.
- Ensures that hardware reacts appropriately to commands, telemetry cycles, and interrupts.

2. Environmental Qualification

- Vibration, shock, and thermal vacuum testing will validate hardware integrity for launch and space environments.
- Conducted using space rated environmental chambers per mission class requirements.

3. Software Validation and Simulation

- Fault injection testing (bit-flips, memory corruption, command loss) to validate error-handling logic.
- Verification of safe-mode behavior and recovery using watchdog timers.
- CRC and packet validation logic evaluated under operational loads.

4. End-to-End System Tests

- Command sequences sent through actual hardware and software stack.
- Different telemetry generation tested from payload > OBC > telemetry queue > RF transmission.
- Command parsing and actuation response verified using ground controls simulations.

5. Subsystem Interface Testing

- Validate interoperability with:
 - Power subsystem (voltage, start and shutdown sequencing)
 - Thermal subsystem (shared sensor and heating logic)
 - Payload subsystem (data routing and storage capacity)
 - ADCS (command timing, sensor data)

This plan makes sure that the CDH subsystem is not only compliant with mission requirements, but also robust, redundant, and adaptable to the dynamic conditions of space operations.

2.1.4. Thermal Control Subsystem Overview

The thermal management subsystem is made up of six main components: surface finishes, insulation, active cooling, phase change material, temperature monitoring, and heat distribution. Together, these components ensure that the system temperature of the aerobot is maintained at 300 K. Although the operating temperature of the payload is between 65 K to 90 K, the instruments are equipped with their own cryocoolers enabling operation at higher surrounding temperatures. Therefore, the thermal subsystem must only ensure the internal temperature is low enough for the operation of such cooling coolers, while also rejecting the heat dissipated by the cryocoolers (Piccioni et al., 2007). More specifically, the thermal subsystem was designed to maintain an internal temperature of 300 K at an altitude of 50 km in the Venusian atmosphere. Along with this, the system is capable of maintaining this temperature for a short time at 70 km while the aerobot descends to 50 km.

Heat flow begins with the surface finish. This is a high reflectivity, low absorption coating of silverized Teflon which helps to maximize radiative heat transfer to the surrounding while reducing heat gained through radiation. Beneath this coating, the foam insulation layer slows heat transfer to the inside of the aerobot, allowing the surface temperature to reach equilibrium at a higher temperature than the internal operating temperature. Maintaining a higher temperature enables the aerobot surface to radiate more heat to the planet and its atmosphere. Within the aerobot, fans are used to circulate heat, ensuring a uniform temperature gradient throughout. At the same time, fluid pumps operate a cycle which expels heat from the inside of the aerobot. Finally, thermostats placed throughout the aerobot interior communicate with the electronic components of the thermal and command subsystems to adjust component operation and maintain a stable temperature. The thermal subsystem also contains a transitory component. A small amount of phase-change material with a low melting point will undergo phase change and release heat to warm up the aerobot while it transitions from the cold upper atmosphere.

The thermal subsystem was devised through the help of heat flow analyses partially represented by the heat flow maps in Figure 6. Because of the rapid rotation of the Venusian atmosphere, air temperatures are assumed to be constant in both night and day time and are only a function of altitude. Given this, the determining factor between the hot and cold cases is the influence of solar flux which is only present during the day time. For both cases, an active cooling system is needed.

For the hot case, solar flux was estimated as a fraction of the solar flux at a distance from the Sun equal to the semimajor axis of Venus. This fraction is taken from an upper bound of 50 percent solar intensity at 50 km altitude (Landis, Colozza, and LaMarre 2003). Solar flux is diminished at these altitudes by the Venusian cloud layers and the planet's thick atmosphere. The resulting flux was also used to estimate flux from albedo at this altitude by multiplying it with an albedo value of 0.77 (Williams 2018). Planet shine was estimated using the method described by Rickman 2014 which states that maximum planet shine flux is given by:

$$q = \frac{q_{Sun}(albedo)}{4} \left(\frac{R}{R+h} \right)^2$$

Here, q_{Sun} is the solar flux, albedo is the planetary albedo value, R is the planet radius, and h is the altitude of the vessel. While solar flux was assumed to only impact the upper face of the aerobot, albedo and planet shine calculations considered the effects of view factors towards the atmosphere and planet for each face.

Another consideration was radiation between the aerobot surface and the environment. The atmosphere and the planet were modelled as a black bodies with temperatures of 350 k and 226.6 k respectively. Thus, the heat transfer for each face was given by:

$$Q_{rad} = \varepsilon \sigma A_p F (T_{out}^4 - T_{bot}^4)$$

Where ε is the emissivity of the reflective outer coating, σ is the Stefan-Boltzman constant, A_p is the projected area of each surface, F is the view factor, T_{out} is the temperature of the outside body, and T_{bot} is the surface temperature of the aerobot. Planet and atmospheric view factors for the side surfaces considered the cruising altitude. The balloon, view factors related to it, or its shadow were not considered for any heat flow calculations.

Convection estimates assumed an atmosphere of conduction coefficient, dynamic viscosity, and kinematic viscosity equal to that of CO₂ at 1 atm ("Carbon Dioxide - Dynamic and Kinematic Viscosity" 2015). Cp value used was calculated for a gas mixture of 97 percent CO₂ and 3 percent N₂ (T Spohn, Breuer, and Johnson 2014). The shape of the aerobot was assumed to be cubic with a D value of 1. For estimating the Reynolds number, a relative wind velocity of 5 m/s was used. Although relative wind speed could be estimated to be negligible for an aerobot cruising at wind velocity, it is likely that the momentum of the craft would result in some small relative wind speed as the system moves in locally varying wind directions. Put it another way, as local winds change direction slightly with time, the aerobot would take time to adjust to the changes in wind velocity and thus would have some small relative velocity. Furthermore,

assuming a small relative wind velocity helps to offset the lack of analysis of natural convection. Given this, a mild fluid velocity of 5 m/s compared to the ground relative wind velocity of 80 m/s at 50 km appears reasonable. Another concern was the relative differences in fluid motion across the different surfaces. To model these differences, the wind direction was assumed to be perpendicular to one face of the aerobot. For the front, top, and bottom faces, Nusselt number was calculated with the following formula:

$$Nu_D = CRe_D^m Pr^{1/3}$$

Where Re_D is the Reynolds number, Pr is the Prandtl number, and the values of C and m are geometry-dependent constants. Due to difference in the magnitude of the Reynolds number, the two side faces utilized the more general formula:

$$Nu_D = 0.3 + \frac{0.62Re_D^{1/2}Pr^{1/3}}{[1+(\frac{0.4}{Pr})^{2/3}]^{1/4}} [1 + (\frac{Re_D}{282,000})^{5/8}]^{4/5}$$

Where Re_D and Pr are once again the Reynolds and Prandtl numbers respectively. Finally, the back face was modeled as experiencing negligible windflow and thus has no convective heat transfer.

Conduction between the surface and the interior was estimated with a conduction coefficient of silverized Teflon of 0.035 W/m² (Donabedian 2003). The surface temperature was derived from a thermal equilibrium analysis of heat flow. This included heat transferred to the inside of the aerobot through conduction. The remaining excess heat must thus be removed through the active cooling system. Internal heat generation was derived from the power draw to the on board electronics, assuming all power was dissipated as heat.

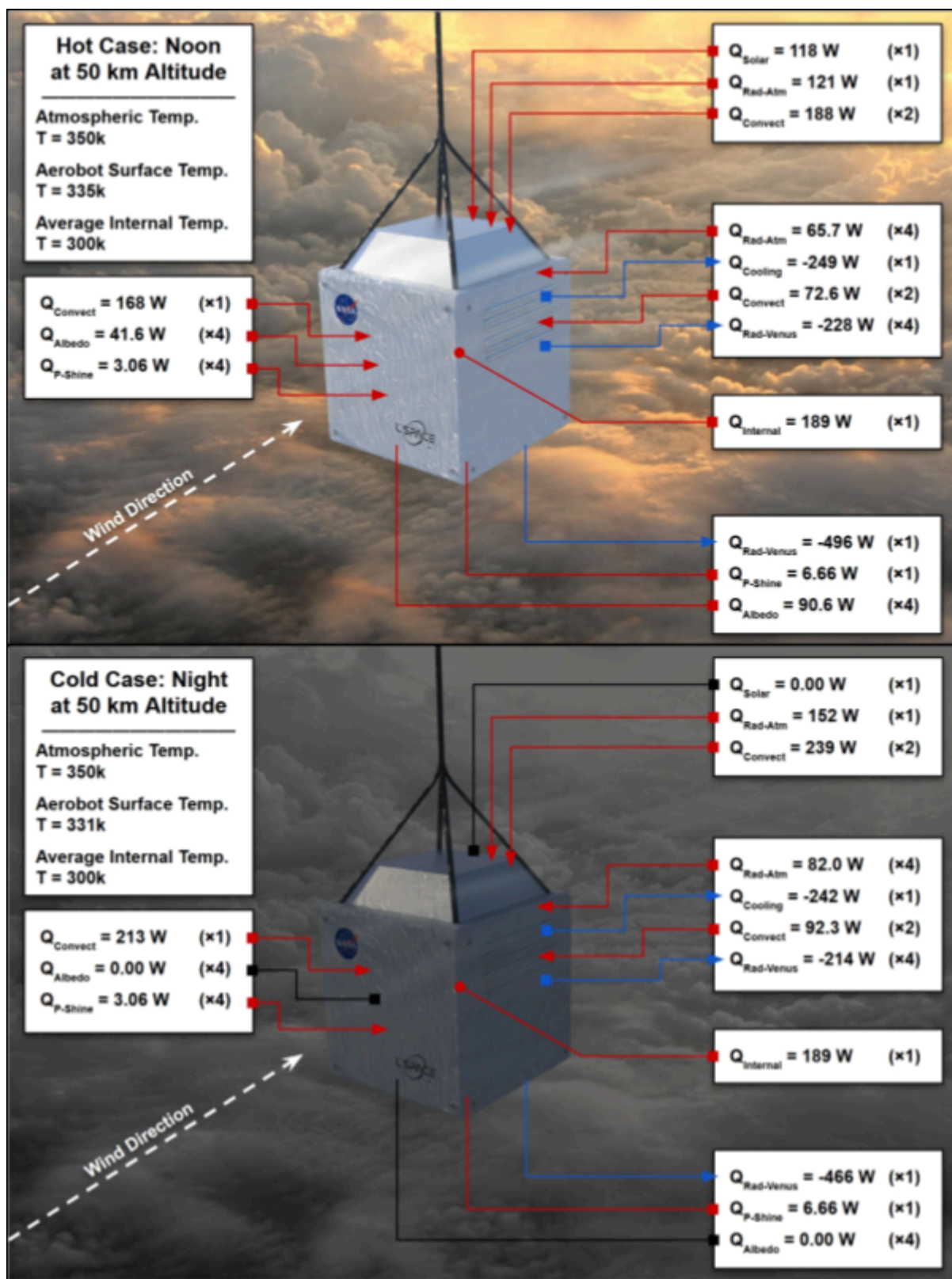


Figure 6: Heat Flow Diagram

2.1.4.1. Thermal Control Subsystem Requirements

Thermal control requirements are listed in Table 10. These requirements were derived to characterize success criteria for the overall subsystem. Requirement TS:1.0 defines the main purpose of the thermal subsystem while its child requirements give clarity on how this success state may be reached. Requirements TS:1.1 and its child TS:1.1.1 define the ideal reflective properties and longevity of the aerobot coating for minimum radiative heat gain. Requirement TS:1.2 characterizes proper application of the aerobot insulation. Requirement TS:1.3 further specifies the appropriate temperature field within the aerobot. Lastly, requirement TS:1.4 ensures that the active cooling system will be capable of offsetting the expected heat loads to maintain a stable temperature.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
TS:1.0	The thermal subsystem shall maintain the aerobot at a uniform internal operating temperature of 300k +/- 7k for a minimum of 36 hours.	The science instruments selected have a median optimal operating temperature of 300k. Thus, the internal temperature of the aerobot should not vary greatly from this number during its expected lifetime.		T:1.1-1.5	Test	Met
TS:1.1	The outside of the aerobot shall be coated with a reflective finish, with a minimum emissivity of 0.8 and maximum absorptivity of 0.15.	To control the amount of heat absorbed and reflected by the aerobot, an outer coating with precise properties is required.	T:1.0	T:1.1	Analysis	Met

TS:1.1.1	The coating of the aerobot shall withstand exposure to Venus atmospheric conditions including pressures, temperatures, and an air mixture of 80% sulfuric acid for 48 hours.	To maintain performance and the thermal equilibrium enabled by the coating throughout the mission lifecycle	T:1.1		Test	Met
TS:1.2	The aerobot shall have an insulating layer uniformly distributed around the body perimeter and a minimum of 9.5 cm thick.	Insulation material thickness affects heat transfer rate to internal components.	T:1.0		Inspection	Met
TS:1.3	Local internal temperatures shall not deviate by more than 10 K from the mean internal temperature.	All components should be kept near the determined optimal operating temperatures. Hot or cold spots within the aerobot system may lead to instrument malfunctions.	T:1.0		Test	Met
TS:1.4	The active cooling subassembly shall be capable of continuously removing 250 Watts of heat from the aerobot under Venus-like conditions.	If passive measures are not sufficient, this component should be able to keep heat flow low enough such that the aerobot stays within its optimal operating temperature limits during the mission lifetime.	T:1.0		Demonstration	Met

Table 10: Thermal Control Subsystem Requirements

2.1.4.2. Thermal Control Sub-Assembly Overview

Due to the thick atmosphere and cloud layers of Venus, solar intensity at the cruising altitude of 50 km is 20 to 50 percent of the intensity outside the atmosphere (Landis, Colozza, and LaMarre 2003). Because Venus is much closer to the Sun than Earth, the solar heating at this altitude remains a large contributor to the heat flow even at such diminished intensities. Other sources of radiative heat transfer include albedo, planet shine, and the hot gasses of the atmosphere themselves, which at 50 km of altitude glow in the infrared with a temperature of 350 k (T Spohn, Breuer, and Johnson 2014). To keep the internal system temperature low, heat radiated from the aerobot must be maximized, while heat absorbed should be minimized.

Towards this end, many coatings exist that have high emissivity and low absorptivity values. The most common such coatings are various forms of white paint, which has long been used in the space industry to regulate temperature. For the Venus case the selection of an outer coating is complicated by the presence of sulfuric acid in the atmosphere. Any coating materials chosen must have a level of acid resistance to protect the underlying vessel and retain its desired properties for the mission duration. After much research and a trade study, it was decided that a 5 mil (0.127 mm) layer of silverized Teflon achieved the desired optical and chemical properties with an emissivity of 0.88 and absorptivity of 0.09 (Donabedian 2003).

Finishes are not just important for the exterior of the aerobot. Alongside the heat distribution subassembly, internal finishes can be used that aid in radiative heat transfer between components within the aerobot, thus creating greater uniformity in internal temperatures. Following common practice in aerospace applications, black paint was chosen for the bulk of the interior components. Much like white paint, black paint possesses a high emissivity, but it also has high absorption rates. These properties make components coated with black paint to act closer to black bodies, absorbing and re-emitting most radiation such that heat is easily distributed throughout the overall system.

Although insulation does not directly remove heat from the system, it does slow heat flow from the outer shell to the internal components. In turn, greater levels of insulation allow surface temperatures to be kept higher than the desired operating temperature of 300 k. Higher surface temperatures then help to decrease heat intake by radiation and convection from the atmosphere. Because solar flux is not affected by higher surface temperatures, insulation can permit the surface temperature to be kept even above that of the surrounding environment. In this case it is even possible to lose

heat to the atmosphere. To maximize internal volume for science instruments while providing the appropriate level of insulation, the amount of insulation was limited to a shell with a thickness of 10 cm. The shell makes up the bulk of the aerobot exterior and is supported by the aerobot frame. The insulating material type is foam insulation with a conductivity coefficient of 0.035 W/m^2 (Donabedian 2003).

At 50 km of altitude, passive components are not sufficient to keep the aerobot at the desired internal temperature. An active cooling solution must thus be used to remove excess heat from the system. For this a refrigeration cycle will be powered through small fluid pumps. With the atmospheric and internal temperatures, the maximum theoretical efficiency of the heat pump can be calculated from the Carnot efficiency given by:

$$COP_{Carnot} = 1 - \frac{T_{low}}{T_{high}}$$

Here, T_{low} is the temperature of the system of 300 k, and T_{high} is the atmospheric temperature of 350 k. This results in a coefficient of power (COP) of 6. Estimating a true efficiency of half that of Carnot results in a COP of 3 and a maximum power draw of 83 W. Mechanically, the heat pump would consist of four major components. Fluid begins at a temperature below 300 k as it flows through a strategically placed pipe array within the inside of the aerobot. After absorbing some internal heat, fluid is passed through a compressor which suddenly raises its temperature to above 350 k. Fluid then moves through pipes that penetrate the outer shell and are exposed to the outer atmosphere. In this step, heat is expelled via radiation and convection to the surroundings. Next, fluid moves back to the interior of the aerobot but passes through bends in piping that act as throttles and return the fluid to sub-300 k temperature. This cycle repeats in a steady state to expel heat from the aerobot internals.

If the aerobot were designed to operate under the full possible altitude range of up to 70 km, an extensive heating system would have been necessary for operation during a cold case scenario. Given that the mission target set by the science team of a 50 km cruising altitude is to be maintained throughout the mission, designing the aerobot system to operate in such a cold scenario would be extremely inefficient. Nevertheless the aerobot must pass through these colder regions during its descent to the cruising altitude. Because the heat loss during this process is minimal, it is possible to maintain a steady internal temperature through the use of phase change materials. Estimating a descent time from the mission timeline of the Vega 2 balloons yields a safe upper bound of 10 minutes ("NASA - NSSDCA - Spacecraft - Details," n.d.). Assuming the worst possible conditions (night time, 230 k atmosphere throughout) for the duration of descent leads to an estimated heat loss of 51 kJ. Using Lithium Metaborate Octahydrate ($\text{LiBO}_2 \cdot 8\text{H}_2\text{O}$), with a melting enthalpy of 289 kJ/kg, results in only 0.176 kg

of material needed to offset the heat lost during descent (Hirschey et al., n.d.). With a melting temperature of 298.9 k, this material would begin to undergo phase change once the aerobot reaches this internal temperature. As the material freezes, it releases heat under an exothermic process, keeping the temperature from dropping below its melting point as long as enough material is present. After the descent phase, the material absorbs heat and returns to its liquid state for the remainder of the mission. No further heating is required because the cold case at 50 km of altitude is one of net heat gain before active cooling systems are accounted for.

Due to asymmetries in the heat flows from sunlight, planet shine, convection, and components, a heterogeneous temperature distribution would be expected within the aerobot. To reduce this concern and ensure that all instruments are working under the desired operating temperature of 300 k, a heat distribution system is needed within the aerobot. Though efficient, a fluid pump heat distribution system would dramatically increase complexity and risk of the system. The Soviets, for instance, chose against such systems in later Venera missions and instead opted for much simpler gas based systems. For a pressurized and interconnected internal cavity, one simple way of dispersing heat is through the use of fans. Similar to applications for computer electronics on Earth, several small fans can be used to promote airflow throughout the system and create a uniform temperature distribution.

To ensure active management of internal operating temperatures by the cooling system, and a uniform temperature distribution by the fans, the aerobot must monitor local temperatures throughout its internal structure. This monitoring will be carried out by an array of Model 3200 High Precision Honeywood thermostats ("Precision & High Reliability Thermostats" n.d.). Commonly used for aerospace applications, such devices will communicate with the on-board computer. If temperature distribution is fairly uniform across the aerobot as measured by the thermostats, the computer would conserve power by reducing fan speed. On the other hand, if hotspots are formed, fans near that area may have their speed increased to alleviate the issue. In this way, the heat distribution system can dynamically adjust to current conditions for optimal performance. Similarly, the computer communicates with thermostats and the heat pump to vary cooling rates as necessary. For this case, the computer would wait until temperatures deviate by +/- 5 k from the desired temperature of 300 k before changing heat pump power.

A breakdown of the TRL, mass, dimensions, and power draw of each thermal subsystem subassembly is given in Table 11 below. Although the majority of the thermal subsystem components have seen frequent use in aerospace applications, the lack of atmospheric and lander missions to Venus in the past decades means that none of the components can be considered heritage or given high TRLs. Although the overall aerobot thermal TRL is a modest four, in lab testing, inspections, and experiments can

be used to significantly raise this TRL score before launch.

Components	TRL	Mass (kg)	Dimensions	Power Draw (W)
Outer Coating	4	0.878	Coating of 5 mil thickness around the aerobot	0
Insulation	4	2.32	10 cm (shell along inner panels perimeter)	0
Heat Pump	5	1.36	2 x (7.06 cm x 7.06 cm x 18.8 cm)	83
Thermostats	4	0.152	12 X (1.537 cm x 1.537 cm x 0.965 cm)	0.1
Fans	5	0.91	2x (10.7 cm x 7.62 cm x 7.62 cm)	4.4
Phase-Change Material	5	0.176	2x (5 cm x 5 cm x 2.5 cm)	0
Thermal Subsystem Net	4	5.796	79,622 cm ³	88

Table 11: Thermal Control Subsystem Components

Alongside thermal flow analyses, the thermal subsystem was designed with the aid of several trade studies. These studies allowed for the selection of the best possible components to achieve the goals of the subsystem. In conducting such studies two main areas of consideration emerge: the criteria under comparison and the score allotment method.

As described in Table 13, the three criteria of importance in deciding between possible outer coatings were performance, reliability, and durability. Though performance is of great importance, the differences between the final heat flows between the different options were found to be only a few watts apart, with the worst performing coating, Titanium Oxide, having an active cooling requirement only 5.44 W greater than that of the best performing coating, BS-6. For this reason the performance

category was given a modest weight of 30 percent. Similarly, though reliability is of great importance, the TRL of each of these coatings can be raised through routine testing in the laboratory. Additionally, none of the coatings can be considered heritage and thus are at similar reliability levels, earning this category a lower weighting of 20 percent. Perhaps the biggest concern for the mission, given the harsh Venusian conditions, is that the coatings can survive and retain their performance. Not only will coatings serve thermodynamic purposes, but they will also help to protect the underlying aerobot from the sulfuric acid present in the atmosphere of Venus. If the coatings are to completely fail, it is likely that the entire mission would be jeopardized. Given this importance, the durability category was given the highest weight of 50 percent.

Each coating candidate was selected to have high emissivity, low absorptivity, and some level of recorded resilience to sulfuric acid. Performance grades were derived by using the individual emissivity and absorptivity values of each coating (see Table 12) in the heat flow analysis to determine how much cooling would be needed. Reliability grades were derived from the TRL of each component. While BS-6 and silverized Teflon have seen their optical and chemical properties tested in the lab, Titanium Oxide has not been given as much attention in contexts similar to this use case. Specifically, only information about its optical properties was found. For durability, Teflon stands as one of the most tried and tested, acid resistant materials (“Teflon PTFE Chemical Compatibility Reference Chart” 2019). BS-6 was shown to survive under 24 hrs of sulfuric acid exposure, but saw a 22.6% degradation of its optical properties (Azad, Long, and Warner 2022). Finally, no information on Titanium Oxide’s resistance to sulfuric acid could be found. Ultimately it was decided that silverized Teflon was the best option, and this was not different when differing weights were tested.

Properties	BS-6 White Paint	Teflon with Silver Backing	Titanium Oxide White Paint
Emissivity	0.974	0.78	0.9
Absorptivity	0.087	0.05	0.2

Table 12: Properties of Outer Coating Options

Outer Coating						
Criteria	Explanation	Grade	Weight	BS-6 White Paint	Teflon with Silver Backing	Titanium Oxide White Paint

Performance	Heat rate alteration in W. Req. T:1.0	10 = high, 5 = medium 1 = low 0 = Fail	30%	10	9	5
Reliability	Measured by component TRL. Mission success is dependent upon consistent performance of components.	10 = high, 5 = medium 1 = low 0 = Fail	20%	4	4	2
Durability	Rate of degradation under Venus atmospheric conditions. Req. T:1.1	10 = high, 5 = medium 1 = low 0 = Fail	50%	8	10	1
		TOTAL S:	100%	78.00%	85.00%	24.00%

Table 13: Outer Coating Trade Study

For the insulation material selection trade study (Table 15), the cost, weight, and performance of each candidate material was considered. Due to the significant impact of insulation on the amount of active cooling needed, performance was designated the most important category and given a weighting of 50 percent. Next, weight in the aerobot is limited and should ideally be reserved for science instruments as much as possible, thus this category was given a weight of 30 percent. Still important, though less restrictive are the budget limitations. Because most costs will come in the way of personnel and not insulation material, this category was given a lesser weighting of 20 percent.

Batt and foam insulations are both used extensively in consumer and aerospace applications. Thus, both of these materials choices have modest materials costs. Conversely, aerogels are known to have much higher materials costs, especially in aerospace applications (“How Much Does Aerogel Cost?” 2024). Additionally, a large spherical shell of aerogel material would require new manufacturing methods as most

current manufacturers only create smaller flat panels. Due to the delicate nature of aerogel, further considerations would also have to be made in the design of its containment. Combined, these considerations make aerogel the most expensive option. The mass and performance grades of each candidate can be easily determined from the properties given in Table 14. Aerogel has the highest estimated mass at 2.84 kg, while foam has the lowest at 1.37 kg. However, while the conductivities of foam and batt insulation are similar, aerogel processed a clear advantage in insulating ability. Ultimately, this was not sufficient to make aerogel the best choice. Instead, foam insulation was the clear choice, and was unaffected by mild changes to the weights.

Properties	Batt	Foam	Aerogel
Density (kg/m ³)	15	4.75	20
Conductivity Coefficient (W/m ²)	0.03	0.035	0.017

Table 14: Properties of Insulation Options

Insulation						
Criteria	Explanation	Grade	Weight	Batt	Foam	Aerogel
Cost	Cost estimate for material and development in \$. Req. 11	10 = high, 5 = medium 1 = low 0 = Fail	20%	10	9	1
Mass	Mass of insulation required in kg. Req. 1	10 = high, 5 = medium 1 = low 0 = Fail	30%	6	10	5

Performance	Heat transfer coefficient k in W/m^2 . Req. TS:1.0	10 = high, 5 = medium 1 = low 0 = Fail	50%	6	5	10
		TOTAL S:	100%	68.00%	73.00%	67.00%

Table 15: Insulation Trade Study

For the internal cooling component of the thermal subsystem, five categories were considered for weighting: cost, mass, volume, reliability, and performance. Performance and reliability were both weighted at 25 percent because candidates that fail to excel in these categories will fail to meet the core goals of the cooling system. The mass criterion was also given a 25 percent weight due to the restrictive nature of the mission mass requirements. Similar, but slightly less restrictive are the volume requirements which led to a weighting of 10 percent for the volume category. Finally, due to the higher relative complexity of potential cooling solutions, the cost category was given a weighting of 15 percent.

After extensive research, the two cooling solutions considered were phase-change materials and pumped fluid loops. For initial estimates, Lithium Nitrate Trihydrate will be used as the phase material of choice. With a melting temperature of 303 K and upper bound melting enthalpy of 296 kJ/kg, $LiNO_3 \cdot 3H_2O$ is an ideal phase-change material candidate for this application (Hirschey et al., n.d.). Mass estimation for providing cooling throughout a mission time of 48 hrs in a hot case scenario yields a minimum of 87.8 kg of $LiNO_3 \cdot 3H_2O$ needed. Although this is way beyond the mass limitations set by the customer, the required mass could be reduced via a change of mission plans. For instance, the balloon could be made to alternate between altitudes periodically enabling a kind of recycling of the heat absorption that occurs during phase transition. Though mass is much harder to estimate for a pumped fluid loop solution, it is easy to say that it would be much lower than that of a phase-change solution. On the other hand, costs for a phase-change approach would be much lower, with design being very simple and material costs being in the range of a few hundred dollars for several kilograms of material (Hirschey et al., n.d.). Volume occupied by phase change materials is estimated to be on the order of 0.01 m^3 , while for the pumped fluid loop it is on the order of 0.001 m^3 . Both fluid loops and phase-change materials have been previously used in aerospace applications. Phase-change materials specifically have seen extensive use in past Venus missions. However, the TRL of such materials would drop if $LiNO_3 \cdot 3H_2O$ specifically was chosen as it has not seen previous use in Venus missions. Finally, though phase-change

material performance can be increased with material quantity, given the limitations of the mission, pumped fluid loops would achieve a much greater performance than phase-change materials.

Other solutions briefly considered included thermoelectric coolers and heat pipes. Thermoelectric coolers were deemed unfit for this use case because they are designed to cool localized areas and have very low efficiencies (Donabedian 2003). Heat pipes too, were deemed unfit because they rely on capillary forces which would fail under the influence of planetary gravitational fields and the chaotic Venusian atmosphere (Donabedian 2003).

Internal Cooling System					
Criteria	Explanation	Grade	Weight	Phase-Change Materials	Pumped Fluid Loops
Cost	Cost estimate for basic system in \$. Req. 11	10 = high, 5 = medium 1 = low 0 = Fail	15%	10	5
Mass	Mass of components required in kg. Req. 1	10 = high, 5 = medium 1 = low 0 = Fail	25%	0	7
Volume	Volume occupied by components in m ³ . Req. 12	10 = high, 5 = medium 1 = low 0 = Fail	10%	6	10

Reliability	Measured by component TRL. Mission success is dependent upon consistent performance of components.	10 = high, 5 = medium 1 = low 0 = Fail	25%	9	6
Performance	Heat added or removed from the system in W. Req. T:1.0	10 = high, 5 = medium 1 = low 0 = Fail	25%	7	10
		TOTALS :	100%	61.00%	75.00%

Table 16: Cooling System Trade Study

In selecting the heat distribution method, the performance, mass, and reliability of the components were deemed to be of comparable importance. However, mass as a secondary criterion received the lowest weight. Between the air and fluid based options, fluids have a considerably higher heat transfer potential and thus offer greater performance. This can be seen in electronics on Earth where higher power computers are often water or refrigerant cooled unlike typical fan cooled computers. In mass, however, fluid loops suffer due to the equipment necessary to contain the fluid as it travels between components. For the same reason, reliability of fluid loops are lower as they contain more complex components. Pumps, for instance, are more complex than basic fans. Overall, as depicted in Table 17, the simplicity of the air based heat distribution method makes it ideal for this application.

Heat Distribution System					
Criteria	Explanation	Grade	Weight	Air Based	Fluid Loops

Performance	Heat distribution capability measured in W. TS:1.3	10 = high, 5 = medium 1 = low 0 = Fail	40%	5	10
Mass	Mass required to support the system in kg. Req. 1	10 = high, 5 = medium 1 = low 0 = Fail	25%	7	3
Reliability	Measured in component TRL and design complexity. Mission success is dependent upon consistent performance of components.	10 = high, 5 = medium 1 = low 0 = Fail	35%	7	2
		TOTALS :	100%	62.00%	54.50%

Table 17: Heat Distribution System Trade Study

Proper selection of a heating element is crucial to ensure the survival of the aerobot in the earliest stages of the mission. Here, three options were considered: electric heaters, chemical heaters, and phase-change materials (Table 18). Though important, performance comparisons were not made because the low heat output necessary could be easily achieved by all methods. Thus, reliability became the most important factor, followed by mass given the limited mass requirement of the mission. Because of the short use-case control over precise application was not considered crucial. Neither was cost as it is only a secondary consideration.

Development cost would be highest in a chemical solution due to its custom and delicate nature. Electric heaters in comparison could be simple resistors, while off the shelf (OTS) products exist for phase-change materials. In terms of mass, a chemical heater has the advantage due to the high concentration of energy stored within molecular bonds, on the other hand the intermolecular bonds of phase-change

materials do not offer the same energy output. Electric heaters would also be fairly light as simple resistors can make use of the existing electrical infrastructure. Examining control, electric heaters can communicate directly with thermostats to release precise amounts of heating. Chemical heaters, on the other hand, cannot be controlled once the reaction is initiated. Phase-changes do not have real time control, but their activation temperature can be precisely determined through the compound structure. As for reliability, phase-change materials offer a completely passive system and offer the least complexity. They have also seen previous uses in Venus as a whole, though the exact compounds would differ. Electric heaters are also fairly simple but would require active management, and chemical heaters are the most complex as a system must be created to initiate and contain reactions. Unlike for cooling, phase-change materials prove to be the ideal method for temporary heating of the aerobot during its descent stage.

Transitory Heater						
Criteria	Explanation	Grade	Weight	Electric Heater	Chemical Heater	Phase-Change Material
Cost	Cost estimate for basic system in \$. Req. 11	10 = high, 5 = medium 1 = low 0 = Fail	10%	5	2	8
Mass	Mass required for minimum heat generation in kg. Req. 1	10 = high, 5 = medium 1 = low 0 = Fail	30%	7	8	5
Control	Level of control in initiating and fluctuating heat output level. TS:1.0	10 = high, 5 = medium 1 = low 0 = Fail	10%	10	5	3
Reliability	Measured by component TRL and design complexity.	10 = high, 5 = medium 1 = low	50%	6	3	10

	Mission success is dependent upon consistent performance of components.	0 = Fail				
		TOTALS:	100%	66.00%	46.00%	76.00%

Table 18: Transitory Heater Trade Study

Although the aerobot mission is a low cost mission and can thus afford greater risk levels, significant steps should still be undertaken to minimize such risks. Besides thorough tests and inspections, redundant design and recovery procedures will be used to address possible risks for each thermal subsystem component.

The most basic components, coating and insulation, cannot be made redundant in the typical sense as they are structural features of the entire aerobot. It would be possible to render these components redundant by constructing the aerobot such that the other components of the thermal subsystem can maintain thermal equilibrium in the event of total coating or insulation failure. However, the passive and simple nature of the insulation and coating materials make them extremely reliable. These materials can be thoroughly tested under Venusian-like conditions to ensure their performance during the mission. Furthermore, neither component relies on the rest of the aerobot to function properly. Finally, any failures in insulation or the outer coatings would likely be minor. In this case, the aerobot would continue to function as needed because it is designed to tolerate a greater temperature range than is truly expected. Thus, a certain level of caution is already built into the thermal system design. In the unlikely event of a total failure of either the insulation or outer coating components, the aerobot system can temporarily reduce power to science instruments to decrease internal load. At the same time, the in-flight computer can make use of the thermostat array and the pump to increase altitude until thermal equilibrium is reached.

Much like the outer coating and insulation, phase change material works passively and thus has a very low risk of failure. Nevertheless, steps can be taken to increase the success rates of the phase-change components. In estimating the amount of phase change material required, a very worst case scenario was considered such that the aerobot is already designed to contain a surplus of material. Material can thus be separated and stored in multiple containers around the system such that if one container is to fail, the rest are unaffected. This has the added benefit of allowing a more distributed heat release during phase-change. Further risk can be lowered through extensive testing in the laboratory. In the case of a total failure of the phase change materials, the thermostats would detect a temperature drop to below that of the phase-change temperature. In this scenario,

the on board computer could operate the pump to cause a rapid descent in order to minimize exposure to the colder, higher altitudes. Additionally, power to all possible components, such as fans, can be increased to increase the internal heat generated.

The heat distribution system already makes use of a fan array consisting of several fans around the interior of the aerobot. The fans can thus be kept in separate circuits to ensure redundancy in the case of a single fan failure. Because fans rely on moving parts they are more likely to fail than other components of the thermal subsystem. Thus, thorough testing should be conducted to ensure optimal reliability. Ultimately, in the case of several fan failures, it is possible to keep the aerobot operational by dynamically controlling the power flow to each science instrument to avoid hot spot formation. This recovery protocol can be tested in the laboratory as a precaution before launch.

To create redundancy for the heat pump sub assembly, multiple independent heat pumps can be operated in parallel. If, for instance, three parallel heat pump assemblies are run simultaneously, in the event that one heat pump fails, the other two can increase power output to make up for the loss of one. In the scenario that multiple heat pumps fail, the balloon pump can be activated and made to communicate with the thermostat array to bring the aerobot up to altitudes at which temperatures naturally stabilize for the given situation.

Thermostats are crucial components of the thermal subsystem as they communicate with the heat pump and on board computer to maintain the aerobot at the desired internal operating temperature. Three main types of failures tend to occur for thermostats. These are called open, closed, and dithering (Donabedian 2003). Essentially, thermostats may fail in a way that continually reads higher temperatures, colder temperatures, or repeatedly turns on and off (dithering). Failures typically occur due to manufacturing errors and can be alleviated by having many thermostats on independent circuits. Dithering failure can be avoided by setting the dead band temp to greater than 4 degrees. Failures due to freezing can occur but are not expected due to the high temperatures for this application. To create redundancy it was decided to create a thermostat array composed of 6 thermostat locations throughout the inside of the aerobot. Each location sports two independent thermostats. In the case that the two thermostats at one location disagree, it is known that an error must have occurred and the on board computer will elect to ignore temperature readings from that location. In the event of a total thermostat array failure, the aerobot will switch to a theoretical based operating strategy where the adequate power to the thermal and other components is calculated based on known variables and heat flow models.

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

Although the aerobot mission is a low cost mission and can thus afford greater risk levels, significant steps should still be undertaken to minimize such risks. Besides thorough tests and inspections, redundant design and recovery procedures will be used to address possible risks for each thermal subsystem component.

The most basic components, coating and insulation, cannot be made redundant in the typical sense as they are structural features of the entire aerobot. It would be possible to render these components redundant by constructing the aerobot such that the other components of the thermal subsystem can maintain thermal equilibrium in the event of total coating or insulation failure. However, the passive and simple nature of the insulation and coating materials make them extremely reliable. These materials can be thoroughly tested under Venusian-like conditions to ensure their performance during the mission. Furthermore, neither component relies on the rest of the aerobot to function properly. Finally, any failures in insulation or the outer coatings would likely be minor. In this case, the aerobot would continue to function as needed because it is designed to tolerate a greater temperature range than is truly expected. Thus, a certain level of caution is already built into the thermal system design. In the unlikely event of a total failure of either the insulation or outer coating components, the aerobot system can temporarily reduce power to science instruments to decrease internal load. At the same time, the in-flight computer can make use of the thermostat array and the pump to increase altitude until thermal equilibrium is reached.

Much like the outer coating and insulation, phase change material works passively and thus has a very low risk of failure. Nevertheless, steps can be taken to increase the success rates of the phase-change components. In estimating the amount of phase change material required, a very worst case scenario was considered such that the aerobot is already designed to contain a surplus of material. Material can thus be separated and stored in multiple containers around the system such that if one container is to fail, the rest are unaffected. This has the added benefit of allowing a more distributed heat release during phase-change. Further risk can be lowered through extensive testing in the laboratory. In the case of a total failure of the phase change materials, the thermostats would detect a temperature drop to below that of the phase-change temperature. In this scenario, the on board computer could operate the pump to cause a rapid descent in order to minimize exposure to the colder, higher altitudes. Additionally, power to all possible components, such as fans, can be increased to increase the internal heat generated.

The heat distribution system already makes use of a fan array consisting of several fans around the interior of the aerobot. The fans can thus be kept in separate circuits to ensure redundancy in the case of a single fan failure. Because fans rely on moving parts they are more likely to fail than other components of the thermal subsystem. Thus, thorough testing should be conducted to ensure optimal reliability. Ultimately, in the case of several fan failures, it is possible to keep the aerobot operational by dynamically controlling the power flow to each science

instrument to avoid hot spot formation. This recovery protocol can be tested in the laboratory as a precaution before launch.

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Thermostats are crucial components of the thermal subsystem as they communicate with the heat pump and on board computer to maintain the aerobot at the desired internal operating temperature. Three main types of failures tend to occur for thermostats. These are called open, closed, and dithering (Donabedian 2003). Essentially, thermostats may fail in a way that continually reads higher temperatures, colder temperatures, or repeatedly turns on and off (dithering). Failures typically occur due to manufacturing errors and can be alleviated by having many thermostats on independent circuits. Dithering failure can be avoided by setting the dead band temp to greater than 4 degrees. Failures due to freezing can occur but are not expected due to the high temperatures for this application. To create redundancy it was decided to create a thermostat array composed of 6 thermostat locations throughout the inside of the aerobot. Each location sports two independent thermostats. In the case that the two thermostats at one location disagree, it is known that an error must have occurred and the on board computer will elect to ignore temperature readings from that location. In the event of a total thermostat array failure, the aerobot will switch to a theoretical based operating strategy where the adequate power to the thermal and other components is calculated based on known variables and heat flow models.

2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

Cascon Pumps was chosen as the primary supplier of the cooling pump for the thermal subsystem. With pumps as their specialty, established aerospace experience, and an impressive existing product line up including their 100 Series Fuel and Cooling Pumps, Collins Aerospace is the ideal candidate for the aerobot cooling pumps (“Aerospace Archives - Cascon Pumps” 2015). Crane Aerospace and Electronics also has great experience in thermal control systems and cooling pumps and will thus be assigned as the backup supplier (“Crane Aerospace & Electronics Aerospace Thermal Management” 2025). Based on the industry-wide trends of pump lead times and the customized nature of the active cooling system, lead times for the cooling pumps will be on the order of 40 weeks (Ferman 2018).

Thermal insulation for the aerobot will be provided by Polymer Technologies due to their expertise with the desired insulation type. Polymer Technologies provides foam

insulation with high heat resistance, such as their POLYDAMP® Polyimide Foam product, which is ideal for applications on Venus (“Thermal Insulation Materials | Foam by Polymer Technologies” 2015). Boyd Corporation will serve as the backup supplier as they also provide extensive foam insulation services (“Solimide® Insulation Foams” 2024). Boyd is also a big player in the aerospace industry, having previously worked as a contractor for NASA in support of space missions. Given that the insulation can be ordered on an OTS basis and be modified in-house to fit to the aerobot design, a lead time of only 4 weeks is expected.

DUNMORE will provide the reflective thermal coating as they have thorough experience in the radiative thermal management industry and offer several options for 5 mil and 10 mil silverized FEP coatings which perfectly match the needs of the thermal subsystem (“Silver Teflon™ Film & Silver FEP Film | Dunmore Aerospace,” n.d.). Given their experience in the space industry and with metalized FEP coatings, Sheldahl will serve as the backup supplier (“FEP Tape for Radiator Coating - Sheldahl” 2023). Due to the OTS nature of the product, a lead time of only 4 weeks is expected.

Phase change material will be supplied by ERG Aerospace. ERG was chosen as the primary supplier because of their unique offering of foam based thermal capacitors with high heat transfer rates (“Phase Change Material (PCM)” 2025). The backup supplier will be Laird as they also have experience with phase-change materials (“Phase Change” 2019). The phase-change material is expected to be largely OTS with minor modifications, therefore a 6 week lead time is estimated.

Parker Aerospace will provide low mass vaneaxial OTS fans for the internal heat distribution (“Military and Aerospace Fans - Parker Defense Systems Division” 2025). Their extensive aerospace and military applications, along with an existing array of product options make them ideal partners. For the same reasons, AMETEK Rotron will serve as backup suppliers (“Aerospace Fans” 2025). Lead time has been estimated to be 30 weeks from the AMETEK Rotron ML8L fan (“ML8L,115AC,N,1891DF” 2025).

Thermostats will be supplied from Honeywell’s aerospace-grade 3200 Series (“Precision & High Reliability Thermostats,” n.d.). As mentioned in NASA’s Thermal Engineering Handbook, Honeywell has a long history with manufacturing thermostats for space applications. The backup thermostats will be the M2 models from Sensata Technologies, another company with experience in this area (“M2 Series Space Flight Thermostats G311P641/02 | Sensata Technologies” 2024). According to Sentata, expected lead times for OTS products like these thermostats can be as short as 6 weeks (“Precision Thermostats | Sensata Technologies” 2020).

2.1.4.5. Thermal Control Subsystem Verification Plans

It is imperative that the proper construction and functionality of the aerobot thermal subsystem is confirmed before launch, at which point only software solutions will be possible for addressing imperfections. Thus, a verification method has been assigned to each thermal subsystem requirement (Table 19). Verification will be an ongoing process from the beginning of construction and until launch. Beyond the examples provided, every component will be independently verified before and after its addition to the aerobot. The aerobot will be unable to launch unless every component passes verification.

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
TS:1.0	The thermal subsystem shall maintain the aerobot at 300 k +/- 7k for a minimum of 36 hours.	Test	For proper functioning of the payload, the aerobot must maintain the chosen temperature range. This can be easily verified through tests of the temperature under expected conditions.	The aerobot will be exposed to Venus-like conditions in a hot room while its temperature is measured by the installed thermometers and separate lab thermometers for reference. Temperature fluctuations will be tracked over a period greater than the expected mission lifespan.
TS:1.1	The surface finish shall have minimum emissivity of 0.8 and maximum absorptivity of 0.15.	Analysis	The properties of the outer coating will affect the heat transfer rates to the aerobot. Lab analyses of the coating will reveal whether it falls within appropriate levels.	Samples of the outer coating will be exposed to visual and infrared wavelengths in the lab. Spectrography of the reflected wavelengths will reveal the material properties.

TS:1.1.1	The coating of the aerobot shall withstand an air mixture of 80% sulfuric acid for 36 hours.	Test	Due to the harsh conditions in the atmosphere of Venus, the properties of the outer coating will degrade with time. Understanding the nature of this degradation will be critical for the lifespan of the mission. The longevity of the outer coating can only be confirmed with tests that expose it to Venus like conditions.	Samples of the aerobot shell coated in the reflective finish will be placed in a sulfuric acid chamber. Material properties of each sample will be tested periodically for a period greater than 36 hours. The rate of degradation will be measured.
TS:1.2	The insulating layer shall have a minimum thickness of 9.5 cm.	Inspection	The effectiveness of insulation is mostly a function of its proper application. Simply assessing application quality will thus be sufficient.	Each side of the aerobot frame will be independently inspected. Lasers will shine horizontally across the surface of the insulating material such that deviations in thickness will cause shadows.
TS:1.3	Internal temperatures shall be uniform to within 10 k.	Test	Temperature deviations could cause malfunctioning of science instruments and other aerobot systems. The completed aerobot must thus be tested to ensure a uniform temperature distribution.	The aerobot will be imaged in the infrared while its instruments are active. Sustained deviations in temperature and their magnitudes will be monitored.

TS:1.4	The active cooling system shall have a sustained heat removal rate of 250 Watts.	Demonstration	Proper functioning of the active cooling system can only be proven by demonstrating proper operation in tandem with the entire thermal system.	The cooling pump will be subjected to the maximum expected heat load and the equilibrium temperature will be measured to determine the heat removal capacity.
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Table 19: Thermal Control Subsystem Verification Plans

2.1.5. Payload Subsystem Overview

The Payload subsystem consists of two instrument subsystems: the Visible and Infrared Thermal Imaging Spectrometer (VIRTIS) and the Infrared Tunable Laser Spectrometer (IRTLS). The science instrumentation within the aerobot system was selected based on their ability to accurately obtain high-resolution data of the features of interest throughout Maat Mons to fulfill the mission's science objectives outlined in the STM for answering the Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023-2032 Question 6.4: *"How do planetary surfaces and interiors influence and interact with their host atmospheres?"* Due to a lack of readily available knowledge, estimates based on overall vehicle and instrument size were made based on instrument designs of heritage missions for the dimensions of VIRTIS and IRTLS (Table 20). Mass and power draw for VIRTIS was drawn from SatCatalog. As the instrument resides completely within the aerobot system, planetary environmental hazards that were outlined within the MCR such as varying high speeds winds of the Venusian lower latitudes and altitudes are mitigated passively for the payload subsystem by the aerobot system as a whole. Due to wind speeds varying. Internal reflector covers of individual instrument components, as well as the entire payload subsystem within the aerobot allow for protection from potential corrosion of material from limited interactions with the highly acidic sulfuric clouds that cover Venus.

VIRTIS is a heritage instrument that has provided spectroscopy data for both the drastically different environments of a comet for ESA's Rosetta mission and Venus for their Venus Express mission ("ESA Science & Technology - Instruments" 2019). This instrument is attractive for understanding planetary bodies due to the housing of the High Resolution Split Spectroscopy (VIRTIS-H) instrument and Mapping instrument (VIRTIS-M) that collect data simultaneously (Piccioni et al., n.d.). While they only vary slightly (Piccioni et al., n.d.), the design of the VIRTIS subsystem for the Venus Express mission is analyzed for this mission instead of Rosetta's VIRTIS subsystem due to the

environmental factors of Venus being accounted for in this configuration. Thermal dissipation through cooling fan systems within VIRTIS and a slightly reduced repetition time between observations (Piccioni et al., n.d.) reduces the risk of damage to the instrument due to overheating in Venus' extreme conditions of up to 470 °C and 95.6 bar at the surface (NASA 2025). This allows the instrument to function at its ideal operating temperature of 80-150 K (Piccioni et al., n.d.).

Overall, the instrument is an imaging spectrometer with three focal planes across the two channels. The mapping channel has two 2-D focal planes in the visible (0.25-1 μ m) and IR (1-5 μ m) ranges and the spectroscopic channel has a single, high-resolution plane at 2-5 μ m (Piccioni et al., n.d.). VIRTIS-M outputs maps in the visible and IR ranges as stacked monochromatic 2-D images in the x- and y-planes that have a spatial resolution of 256 x 256 pixels (Piccioni et al., n.d.) at a moderately but relatively low spectral resolution compared to VIRTIS-H. Utilizing a Shafer telescope and Offner Relay spectrometer, the VIRTIS-M collects data through scans by a rotating primary mirror within multiple observation modes (Piccioni et al., n.d.). This mapping tool uses two detectors, the Thomson TH7896 CCD for visible light and HgCdTe for infrared (Piccioni et al., n.d.). For the team's science objectives, the VIRTIS-M will be used to capture spectral imaging to examine the impact of atmospheric chemistry on the surface through weathering. Volcanic deposits will be assessed for chemically driven weathering products such as iron oxides and sulfates. VIRTIS-H is designed to collect high resolution spectral data in a small central area of each image taken by VIRTIS-M using an off-axis parabolic mirror and an Echelle spectrometer coupled with the same HgCdTe detector used by VIRTIS-M (Piccioni et al., n.d.). Equipped with a prism-grating cross-disperser, spectral orders are separated and validated with internal calibration lamps and filters (Piccioni et al., n.d.). Electronics boards are used to drive the detectors and arrays of the two channels and to ensure they are running as they should be (Piccioni et al., n.d.). The optics heads that are responsible for taking the measurements are situated within an Optics Module that is designed to reduce thermal background radiation to the measurements by having a warm enclosure plate, called the Pallet, that can withstand the heat of the spacecraft and hold two cryocoolers referred to as the Cold Box (Piccioni et al., n.d.). This Cold Box is thermally disconnected from the aerobot to allow for passive cooling of the two optical heads to below 150 K (Piccioni et al., n.d.). The VIRTIS-H and -M channels have individual retro-reflector covers to protect the instruments from thermal emissions when in use and to preserve the Cold Box environment when they are not in use (Piccioni et al., n.d.).

IRTLS allows for sensitive, direct, and non-invasive in-situ measurements of IR absorption spectra of planetary bodies (Tunable Laser Spectrometers | Capabilities" 2022). The miniaturization of this instrument makes it ideal for space exploration with Discovery Class missions, with design based on the TLS in NASA's Mars Curiosity Rover (Webster et al. 2023). The instrument produces a laser with a linewidth of 1/20th

of molecular linewidth ($<0.02\text{ cm}^{-1}$), allowing for high sensitivity of absorption depths for organic molecules (Webster et al. 2023). The laser spectroscopy is not dependent on mass ratio dynamic ranges, making tuning and calibration of the instrument relatively simple (Webster et al. 2023). The instrument's laser source is tuned to an individual roto-vibrational line in an IR absorption band of the target gasses (for this mission, that is O, C, N, and H) and is frequency modulated at 2 kHz ("Airborne Tunable Laser Absorption Spectrometer | NASA Airborne Science Program" 2022). Once the instrument emits the laser beam, the assigned target IR gasses are detected by initial measurements of the fractional absorption of the tunable diode laser beam. This detection is completed once the absorbed laser beam passes through a multipass Herriott Cell within the instrument that contains an atmospheric sample at ambient Venusian pressure ("Airborne Tunable Laser Absorption Spectrometer | NASA Airborne Science Program" 2022). The detection of the laser's amplitude after absorption at 2kHz and 4kHz outputs the first and second harmonics of the generally weak absorption feature at a high sensitivity ("Airborne Tunable Laser Absorption Spectrometer | NASA Airborne Science Program" 2022). The main beam is split and half of it is taken through a short cell that contains a known amount of the target gasses for a reference absorption spectra measurement to allow for calibration and synchronization of the laser frequency for each absolutely gas concentration measurement ("Airborne Tunable Laser Absorption Spectrometer | NASA Airborne Science Program" 2022). Unlike the VIRTIS subsystem, the IRTLS subsystem is not designed with an internal cooling system. The IRTLS must rely on the thermal subsystem within the aerobot to ensure the instrument is staying cool from the extreme Venusian environmental conditions during data collection.

Components	TRL	Mass (kg)	Dimensions LxWxH (cm)	Power Draw (W)
VIRTIS	7	27	50 x 50 x 40	50
IRTLS	8	5	10 x 40 x 10	25
Total	7	32	60 x 50 x 40	75

Table 20: Payload Subsystem Specifications

2.1.5.1 Science Instrumentation Requirements

The Mission Task is to develop a small, low-cost robotic Aerobot System to help the scientific community improve the understanding of Venus. The system must operate within extreme environmental conditions while collecting data related to both atmospheric composition and geological processes to fulfill the mission objectives. The major environments the subsystem must withstand are the shock and vibration of the launch from Cape Canaveral, Florida, increasing temperatures and pressure up to 470°C and 95.6 bar with decreasing altitude in Venus' atmosphere (NASA 2025) ("The Unexpected Temperature Profile of Venus's Atmosphere," n.d.). The Payload Instrumentation Requirements detailed in Table 21 have been derived from the science goals and objectives detailed in the STM, which are restrained by the customer's Mission Task and Initial Mission ConOps.

The science goal is to provide information for Question 6.4 of the 2023-2032 Decadal Strategy: *"How do planetary surfaces and interiors influence and interact with their host atmosphere?"* The mission's science objectives to determine the concentration of sulfur dioxide and carbon dioxide in volcanoes and mantle plumes (SO1), the impact of atmospheric chemistry on surface weathering (SO2), and how Venus's atmosphere has changed over time to investigate ancient tectonic activity (SO3) provide scientific data to fulfill the science goal.

The payload subsystem requirements ensure that the Aerobot System aligns with NASA's Discovery Program objectives while adhering to physical, operational, and scientific constraints. These requirements will guide the design and subsystem development to ensure minimum payload success

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Req met?
SCI-1.1	Science instrumentation shall collect absorption spectra from 70-40 km above Venus' surface	To ensure the aerobot has descend below the main cloud haze to begin collecting meaningful science data	MG-1.0	SCI-1.1.1 SCI-1.1.2 SCI-1.1.3	Demonstration	Met
SCI-1.2	The aerobot shall be restrained to a horizontal distance of 1600 km from the Maat Mons caldera summit	Staying within the radius of Maat Mons allows for minimum Payload Success for the mission's science goals	MG-1.0	-	Demonstration	Met
SCI-1.1.1	VIRTIS-M shall capture spectral images at 290-4300 nm within a 5 km horizontal radius around Maat Mons' caldera	Visible and IR spectral images of the highest elevation in the region fulfill Science Objective 2	SCI-1.1	-	Analysis	Met
SCI-1.1.2	VIRTIS-H shall measure absorption spectra at 1870-4300 nm at 70-50 km altitude within the 30 km radius of Maat Mons' caldera summit	IR absorption spectra above the volcanic vent fulfill Science Objective 1	SCI-1.1	-	Analysis	Met
SCI-1.1.3	IRTLS shall collect stable isotope ratios for C, N, O, and H at Maat Mons' flanks at a resolution of 1 ppm	Isotope fractionation of varying aged geology at the specified resolution fulfills Science Objective 3	SCI-1.1	-	Analysis	Met
SCI-1.3	The payload subsystem shall be designed to reduce EM interference	Understanding the effects of emitted and received EM energy to/from the Payload subsystem is necessary for meaningful data collection	MG-1.0	SCI-1.3.1 SCI-1.3.2	Test	Met
SCI-1.3.1	The payload subsystem shall produce minimal EM emissions	EM emission from the payload subsystem would negatively affect the performance of other subsystems within the aerobot	SCI-1.3	-	Test	Met
SCI-1.3.2	The payload subsystem shall withstand 1 kHz of external EM energy	The payload subsystem must withstand natural emissions of the Venusian environment and potential emission from nearby aerobot subsystems to ensure the instrument can perform during data collection	SCI-1.3	-	Test	Met

Table 21: Payload Subsystem Requirements derived from the higher level instrument requirement outlined in the STM

2.1.5.2 Payload Subsystem Recovery and Redundancy Plans

The payload subsystem maintains an acceptable level of redundancy whilst working within the constraints of the lower mission budget and weight. While there are only three instruments and two of them are in reality one subsystem, each one is capable of partially contributing to the objective if another fails. VIRTIS-M maintains the highest level of redundancy for its intended task since both VIRTIS-H and IRTLS are able to partially cover for it. The other two instruments are covered by each other with VIRTIS-M making no contributions to reliability.

In the case of a system failure, it is still possible to attempt recovery through the system architecture provided by the command and data handling subsystem. While the chance of this is anticipated to be extremely low due to verification demonstrations planned, it is still very important that recovery may be attempted in the worst case situation. A failure of the VIRTIS system may result in both instruments going offline, in which case only software troubleshooting may take place. This goes as well for the IRTLS, in which a hardware issue due to a physical stimulus will render the subsystem irrecoverable. It is for that exact reason that preliminary validation of the payload subsystem must be completed in great detail to ensure the success of the mission.

Science Measurement Requirements		Baseline Performance Requirements		Instrument		
Physical Parameters	Observables			VIRTIS-H	VIRTIS-M	IRTLS
Identify gas concentrations of CO ₂ and SO ₂	Collect absorption spectra of CO ₂ and SO ₂ in the 1870 to 4300 nm range over 5 km ²	Wavelength range:	1870 - 4300 nm			
		Integration time:	1 sec			
		Spectral Resolution:	1300 - 3000 $\lambda / \Delta\lambda$			
		Sensitivity:	$1.2 * 10^{-4} \text{ Wm}^{-2}\text{sr}^{-1}\mu\text{m}^{-1}$			
Identify volcanic deposits for atmospheric weathering products such as Fe oxides and sulfates	Capture spectral imaging in the 290 - 4300 nm range over 5 km	Wavelength range:	290 - 4300 nm			
		Integration time:	1 sec			
		Spectral Resolution:	100-500 $\lambda / \Delta\lambda$			
		Spatial Resolution:	0.25 mrad (250 m at 1000 km)			
Identify the isotopes fractionation of H, C, N, and O on faulted geology	Measure stable isotope ratios of D/H of carbon-21, oxygen-18, and nitrogen-15 over 1-2 km	Wavelength range:	1000-25000 nm			
		Integration time:	1-100 ms			
		Spectral Resolution:	0.0001 cm^{-1}			
		Sensitivity:	1 ppm			

Table 22: Instrument Redundancy Table for Payload instrumentation.

Full circles = instrument can fully fulfill objective, half filled circle = instrument partially contributes to objective, quarter filled circle = does not meet baseline performance requirements

2.1.5.3 Payload Subsystem Manufacturing and Procurement Plans

The Payload subsystem instrumentation within the aerobot system was selected based on their ability to accurately obtain high-resolution data of the features of interest throughout Maat Mons to fulfill the mission's science objectives outlined in the STM for answering the Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023-2032 Question 6.4: *"How do planetary surfaces and interiors influence and interact with their host atmospheres?"*

The VIRTIS subsystem is a complex and powerful instrument that contains the VIRTIS-H and VIRTIS-M for a range of spectral analyses. As the instrument has been utilized for the ESA's Rosetta mission and Venus Express mission, manufacturing of the VIRTIS subsystem is based on these missions. For the Rosetta spacecraft, the manufacturing of VIRTIS was completed by Leonardo Airborne and Space Systems ("VISIBLE and INFRARED THERMAL IMAGING SPECTROMETER" n.d.), and will be the primary contractor for manufacturing the VIRTIS subsystem in this mission. In the event that Leonardo is not able to be the contractor of the VIRTIS subsystem for this mission, the team will turn to the Southwest Research Institute (SwRI) as a secondary manufacturer for the subsystem development. SwRI is a leader in ground and space instrumentation and a longstanding NASA partner ("Space & Ground-Based Instrumentation | SwRI" 2022), meaning they are a reliable partner for manufacturing instrumentation for the mission. The environments of the Rosetta and Venus Express missions were vastly different to the team's mission, meaning complete heritage of the previously manufactured instruments are likely not appropriate for this mission. Because of this, utilizing contractors to manufacture a VIRTIS subsystem for this aerobot mission allows for less risk to the ability of the subsystem to withstand verification testing and the Venusian atmospheric conditions during ConOps. The standard lead time provided by contractors of the VIRTIS subsystem is 12 to 20 weeks ("SP Scientific Virtis VirTual 50L Lyophilizer with Encore Software" 2023). Given the complex nature of this instrumentation and the possibility of a new design to be needed due to customer requirements for the aerobot and for successful verification testing, extending lead time to 12 to 50 weeks is appropriate to reduce the risk of delays to the mission schedule if design updates are required after verification testing.

The IRTLS subsystem will be manufactured by the contractor Toptica Photonics. Toptica Photonics has shown to produce reliable and innovative laser technology ("Tunable Lasers for Mid-Infrared Spectroscopy | TOPTICA Photonics AG" 2023). Because of this, they have the capability of manufacturing the IRTLS subsystem for the team's mission. In the event that Toptica Photonics is not able to manufacture the subsystem, the Japanese company Yokogawa is a viable secondary manufacturer. Yokogawa boasts similar expertise to Toptica Photonics in laser technology, meaning they have the capability of manufacturing the IRTLS subsystem if needed ("Tunable Diode Laser Spectrometers | Yokogawa America" 2019). Industry standard lead times of 12 to 14 weeks for IR laser instrumentation have been estimated by the contractor THORSLABS for their MIR laser systems ("Turnkey MIR Laser Systems" 2017). An estimated lead time of 12 to 20 weeks reduces the risk to the mission schedule if unexpected delays occur during the manufacturing timeline.

2.1.5.4 Payload Subsystem Verification Plans

Both VIRTIS and IRTLS are legacy systems that have been used on past missions to great effect. VIRTIS was used on the Rosetta and Venus Express missions with a modified version being used on the Dawn mission (“VISIBLE and INFRARED THERMAL IMAGING SPECTROMETER” n.d.). IRTLS will be based on the TLS system on the Curiosity Rover currently on Mars (“Tunable Diode Laser Spectrometers | Yokogawa America” 2019). While these are systems that have provided successful missions, they will be operating in an environment that is extremely different from the planetary bodies they have been in before. It will be necessary to conduct verification testing on the systems to ensure their ability to survive the extreme pressure and temperature of Venus, operate within the lower stratosphere, and gather meaningful data. To determine the capability of the system’s resistance to EM waves, testing for susceptibility as set in the *INTERFACE STANDARD REQUIREMENTS for the CONTROL of ELECTROMAGNETIC INTERFERENCE CHARACTERISTICS of SUBSYSTEMS*. The capability of the systems to make through the launch phase will be tested according to the standards of *Rocket launch-induced vibration and ignition overpressure response* by Caimi et. al., a technical report by NASA. Further testing, analysis, and demonstration of achieving resolutions necessary to recover the proper data will be performed and are noted in the table below.

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
SCI-1.1.3	The IRTLS subsystem resolution shall be verified by test. The test shall determine perform a data error and accuracy test after having the instrument measure the outgassing in a simulated site environment with a thermal vacuum chamber. The test shall be considered successful if the maximum analyzed isotope resolution is 1 ppm.	Test	To collect meaningful data for stable isotope ratios in planetary materials, a resolution of 1 ppm is required to differentiate the source of the isotopes.	A Test within a thermal vacuum chamber will allow for determining the relationship of IRTLS accuracy and error with increasing pressure and temperature.
SCI-1.1.1	The VIRTIS-M subsystem shall be developed to produce spectral images with a spectral resolution that allows for meaningful scientific interpretation and analysis.	Analysis	The subsystem collects data at a lower spectral resolution than the VIRTIS-H subsystem. Verification will ensure the instrument's ability to provide data of the minimum required resolution.	An analysis of the instrument's heritage performances at varying elevations and planetary environments will determine the instrument's ability to provide meaningful spectral maps from ~50 km above the imaging site.
SCI-1.5	The Payload subsystem shall transfer the scientific data to the Computer Hardware subsystem as data is collected.	Demonstration	This relationship is imperative to the success of the mission. Demonstrating that there is minimal risk for failure of transfer will greatly increase the chance of success for the mission.	A demonstration on the ability to transfer large datasets from the Payload subsystem to the Computer Hardware system will be complete once implementation of both subsystems has been completed.
SCI-1.1.2	The VIRTIS-H subsystem shall be developed to collect high resolution absorption spectra in the 1870-4300 nm range.	Analysis	The subsystem has shown to collect high resolution spectral data. Verification will ensure the instrument's ability to produce data for at least the minimum payload success.	An analysis of the instrument's heritage performances at varying elevations and planetary environments will determine the instrument's ability to provide meaningful spectral maps from ~50 km above the imaging site.

SCI-1.4	The Payload subsystem's ability to withstand the shock and vibration of launch shall be verified by test. The test shall use a vibration table and fixture for 3-axis testing within an acoustic room emitting up to 200 Hz. The test shall be considered successful if the subsystem survives the test with no failures.	Test	The subsystem has to withstand the launch environment for any sort of success to occur within the mission.	A Test including a vibration table within an acoustic chamber will allow for confidence that the subsystem will survive the launch environment of high shock and vibration.
SCI-1.3.1	The Payload subsystem will be verified by test to see if EM radiation occurs with use. The test shall measure if there is a presence of EM emissions with a sensitive receiver, anechoic chamber, and antenna with known gain. The test shall be considered successful if there are no detectable EM emissions from the subsystem when functioning.	Test	It is important to understand the base emissions for calibrating measured data and to make sure subsystems are not affecting other subsystems ability to conduct tasks due to EM radiation.	A Test within an anechoic chamber, antenna with known gain, and sensitive receiver to measure EM emissions will measure the presence of possible emissions from the Payload subsystem.
SCI-1.3.2	The Payload subsystem conduction susceptibility shall be verified by test. The test shall determine if external EM emissions affect the ability for the subsystem to perform required tasks in an anechoic chamber within a conduction susceptibility matched "box" of known conduction. The test shall be considered successful if the subsystem is not affected by 1 kHz rate pulse signals	Test	The subsystem's ability to maintain function when exposed to EM radiation when in close proximity to other subsystems and extreme thermal environments is important to test for success of the mission.	A Test within an anechoic chamber and conduction susceptibility matched box will determine the Payload subsystem's ability to withstand EM emissions that mimic real planetary and system environments.

Table 23: Payload Subsystem Verification Matrix

2.2. Interface Control

The interfaces of the internal components of the aerobot are represented in Figure 7. Colors and opacities of components in the figure were chosen for visualization purposes only. All components are secured directly to the aerobot frame, with most using bolted connections. Phase change material and thermometers are secured with the use of adhesives. The science instruments sit in the center of the gondola, creating a shell cavity of air where all other components are placed. This configuration ensures ample air flow between components for even heat distribution.

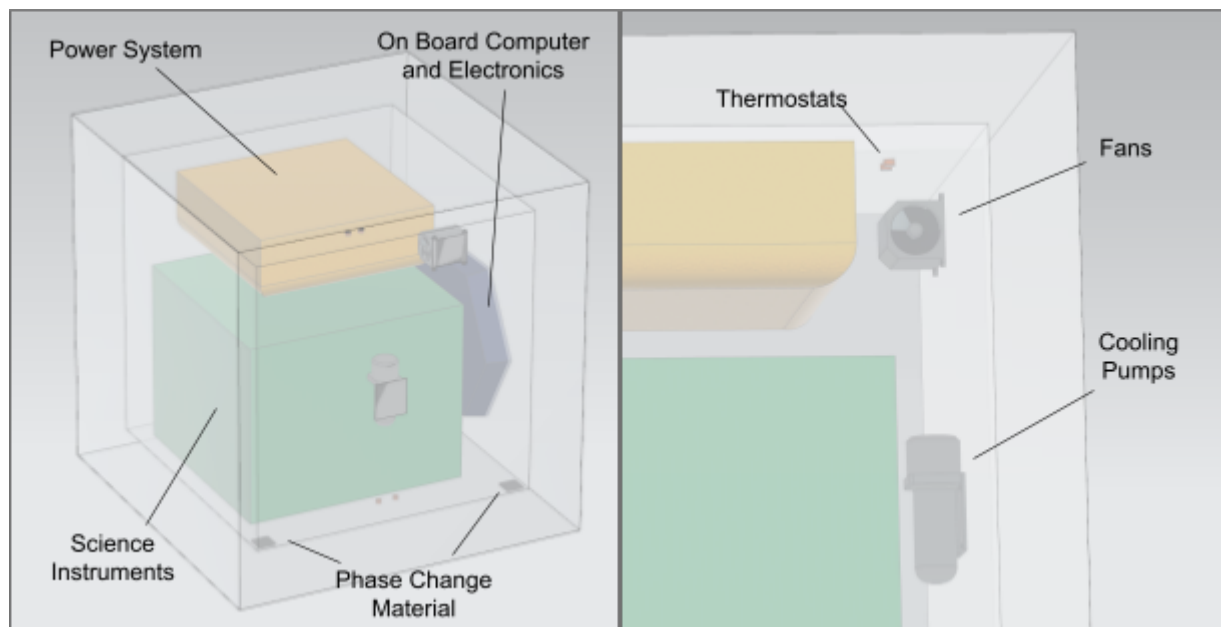
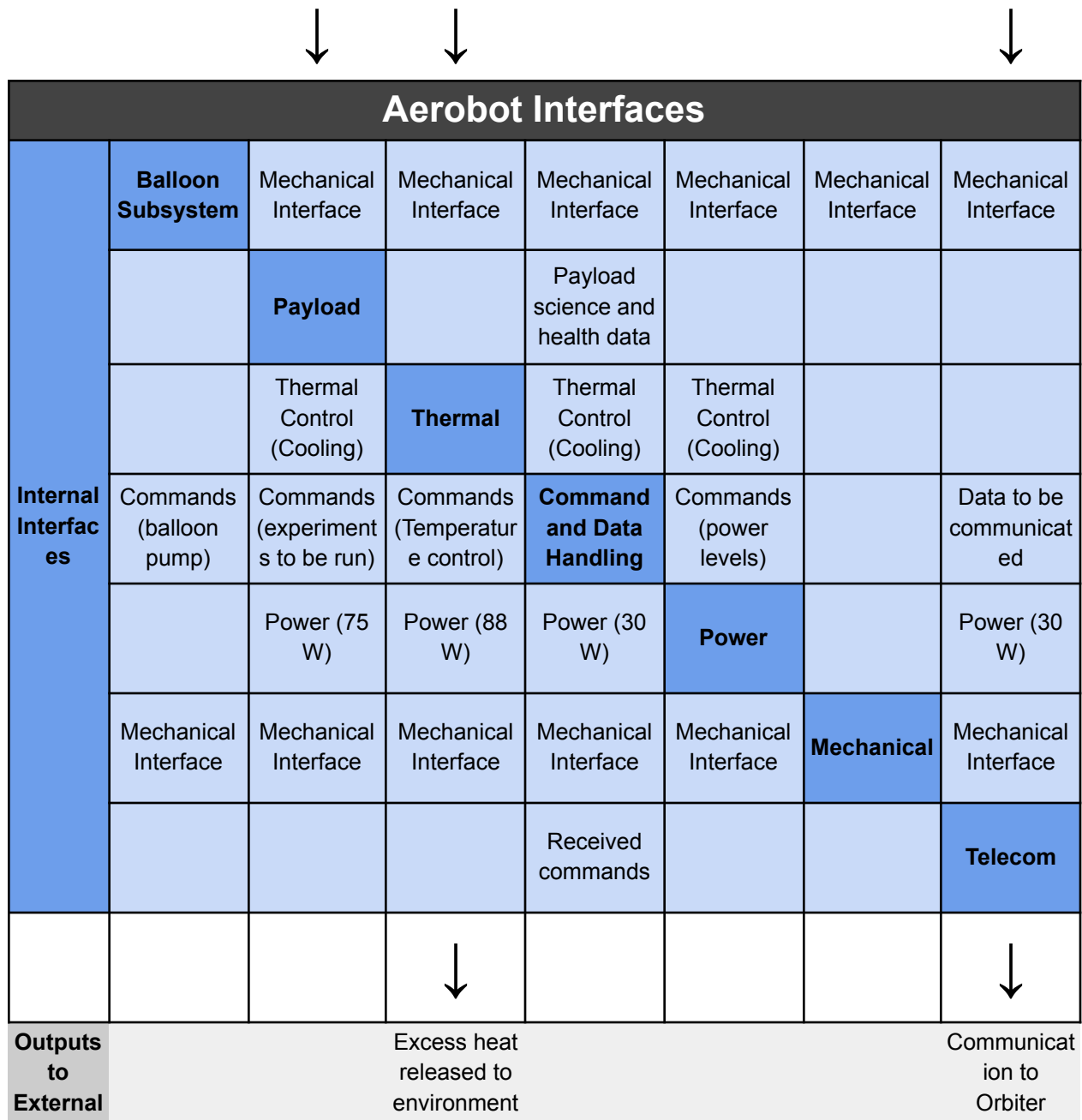


Figure 7: CAD Interface Assembly

The interfaces between subsystems are given at a high level by the N² diagram (Table 24), and at a subassembly level by the block diagram (Table 25). The power system applies interfaces with every other subsystem to supply and manage power levels. The thermal subsystem provides thermal control to all other subsystems with the exception of the communications antenna and the balloon subsystem. One of the more isolated subsystems, telecommunications provides received data to command and data handling. Also fairly isolated is the balloon subsystem which interfaces mainly mechanically with the rest of the aerobot. Much like telecommunications, the payload subsystem sends gathered data to command and data handling. Finally, the most interconnected subsystem, command and data handling, commands the balloon pump, science instruments, thermal decision making, component power levels, mechanical subcomponents, and data to be transferred to the orbiter.

Inputs from External		Sample and Visual Data Gathered	Surrounding Heat Flow				Communication From Orbiter
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Table 24: N² Diagram

In the subcomponent view provided by the block diagram, it can be seen that command and data handling is broken up into two main subcomponents, a data recorder which stores data locally, and a processor which makes autonomous decisions and carries out operations. The breakdown of the thermal subsystem reveals the intercommunication between the thermostat array, the cooling system, the heat distribution system, and command and data handling. This web enables the thermal subsystem to operate dynamically to maintain the aerobot at its desired temperature. This subcomponent view also reveals the individual science instruments, which all

gather data primarily through light. Also visible are the important interactions between aerobot subsystems, the environment, and the orbiter.

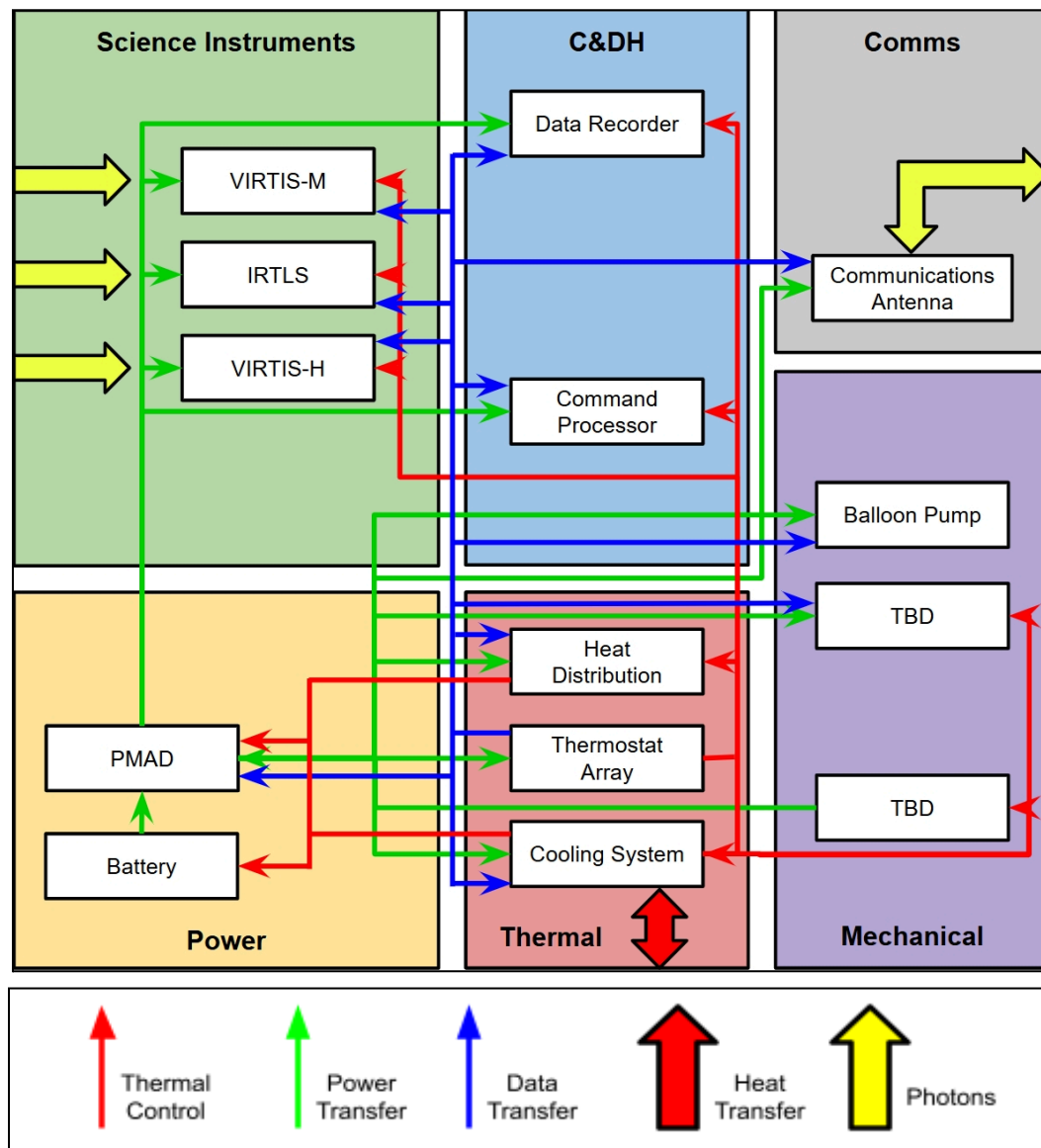


Table 25: Block Diagram

3. Science Mission Plan

3.1. Science Objectives

The primary science objectives of the mission are to (1) quantify the abundance and distribution of sulfur dioxide (SO₂) and carbon dioxide (CO₂) in volcanic emissions and mantle plumes, (2) identify surface weathering products indicative of atmospheric-surface chemical interactions, and (3) determine the isotopic composition of key atmospheric species to assess the evolutionary history of Venus's atmosphere and tectonic processes. These objectives directly address Question 6.4

of the 2023-2032 Planetary Science and Astrobiology Decadal Survey: *“How do planetary surfaces influence and interact with their host environment?”*

Measurement of SO₂ and CO₂ will be conducted using the Visible and Infrared Thermal Imaging Spectrometer High Resolution Subsystem (VIRTIS-H), optimized for spectral acquisition between 1870-4300 nm with a spectral resolution of 1300-3000 $\lambda/\Delta\lambda$, an integration time of 1 seconds, and a sensitivity of $1.2 \times 10^{-4} \text{ W}\cdot\text{m}^{-2}\cdot\text{sr}^{-1}\cdot\mu\text{m}^{-1}$. These observations shall support the detection of active volcanic degassing and the spatial mapping of plume activity over areas of at least 5 km².

Detection of surface weathering products such as iron oxides and sulfates shall be performed using the Visible and Infrared Thermal Imaging Spectrometer Mapping Subsystem (VIRTIS-M). This instrument operates across 290-4300 nm range with a spectral resolution of 100-500 $\lambda/\Delta\lambda$, an integration time of 1 second, and a spatial resolution of 0.25 mrad (corresponding to 250 meters at 1000 km altitude). These measurements shall enable mineralogical characterization of chemically altered volcanic terrains, supporting assessment of thermochemical weathering processes.

Isotopic fractionation of hydrogen, carbon, nitrogen, and oxygen shall be characterized using an Infrared Tunable Laser Spectrometer operating between 1000-30,000 nm with a spectral resolution of 0.0001 cm⁻¹ and sensitivity of 1 ppm. Targeted isotopes include deuterium/hydrogen (D/H) of carbon-21, oxygen-18, and nitrogen-15 over 1-2 km. These measurements shall enable high-precision analysis of stable isotope ratios to reconstruct past atmospheric escape processes, mantle outgassing, and the role of tectonic resurfacing in volatile cycling.

Together, these objectives and associated measurement instruments are designed to yield a multi-scale, spectroscopically driven understanding of Venus's active geochemical environment and its implications for terrestrial planet evolution.

3.2. Experimental Logic, Approach, and Method of Investigation

The Venus mission is designed to characterize the interactions between the surface and atmosphere of Venus by measuring volcanic gas emissions, detecting chemical weathering products, and quantifying isotopic ratios of atmospheric constituents. Section 3.2 requires a detailed explanation of how the science payload will collect these deliverables through instrument integration, data collection protocols, subsystem coordination, and site-specific targeting. This section further demands clarification of how instrument operations are sequenced, how each system supports data acquisition under Venusian conditions, what experimental logic governs timing and targeting, and how the approach constitutes a cohesive investigation. The full payload integration is designed to function throughout the 36-hour descent profile, acquiring data from a sequence of altitudes and terrains identified for high science return.

Following deployment from the carrier platform, the aerobot shall stabilize at an initial float altitude of 70 km and begin controlled descent at a mean vertical rate of 0.23 m/s, as outlined in Section 1.5 (Concept of Operations). Payload deployment logic is based on altitude-triggered thresholds, with each instrument initialized at its optimal operating

band based on spectral transparency, atmospheric stability, and target visibility. Subsystems described in Sections 2.1.1, 2.1.2, and 2.1.3—including mechanical, power, and CDH—maintain instrument stability, data handling capability, and mechanical alignment throughout operation. The mechanical subsystem (Section 2.1.1.2) provides structural integrity, vibration protection, and altitude control via the helium balloon and active buoyancy pump. The power subsystem (Section 2.1.2.2) uses lithium-sulfur batteries and supercapacitors to support peak operational loads during instrument deployment and communication windows. The CDH subsystem (Section 2.1.3) coordinates instrument operations, buffers acquired data, and queues prioritized transmission.

At approximately 68 km, VIRTIS-H initiates high-resolution spectral scanning. The instrument operates in the 1870–4300 nm range with a spectral resolution between 1300 and 3000 $\lambda/\Delta\lambda$, targeting SO₂ and CO₂ absorption features. The initial observational plan includes broad atmospheric transects over quiescent and active volcanic zones. Integration time is set to 1 second, with spectral scans executed continuously. Targeting is focused on emission zones proximal to Maat Mons and Ozza Mons, as mapped in the JMARS Zoomed-Out and Zoomed-In maps. The experimental logic dictates repeating spectral passes across defined longitudinal corridors while descending, enabling vertical profiling of gas concentration gradients. Scans are timed relative to predicted plume dispersal zones using pre-loaded atmospheric circulation models. Data are batched and stored in redundant onboard memory with diagnostic flags tracking line intensity variance and baseline spectral drift. 63 km, VIRTIS-M is activated to begin multispectral imaging of surface units within the 290–4300 nm band. This subsystem maps surface mineralogy with a resolution of 0.25 mrad (250 m at 1000 km), enabling mineral identification related to surface weathering, including sulfates and iron oxides. Initial observations prioritize the western caldera of Maat Mons, followed by traverse paths along the eastern rift valleys of Ozza Mons. Integration time is set to 1–2 seconds per frame, with a rotational scan cadence that repeats every 30 minutes. Image tiles are georeferenced through internal gyroscopic and altimetric data, then compared against onboard mineral spectral libraries for in-situ classification.

The experimental logic includes comparative analysis between highland and lowland units, allowing detection of alteration gradients. CDH logic prioritizes image telemetry based on spectral richness and coherence with preceding frames.

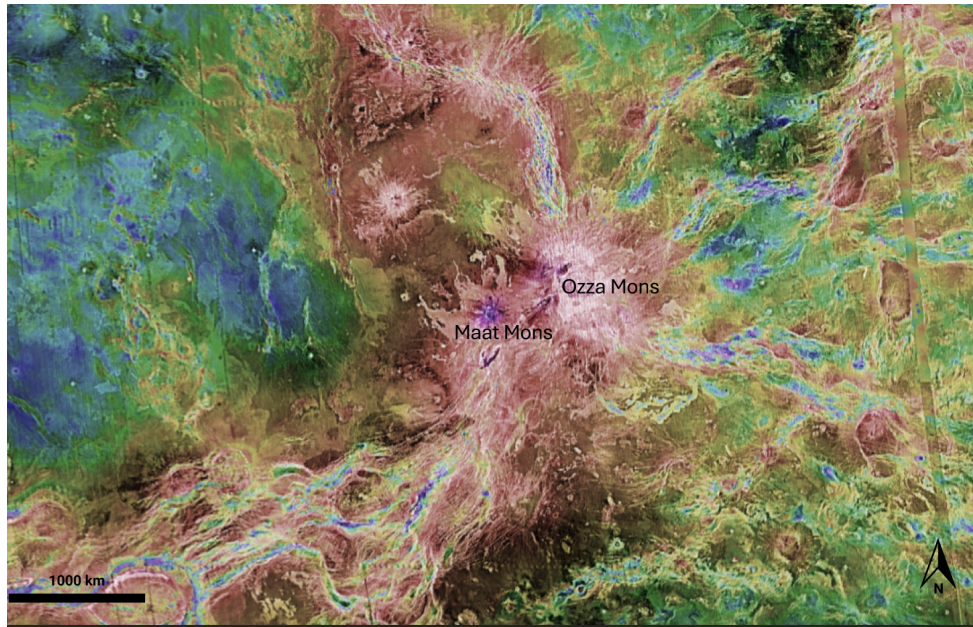


Figure 8: JMARS Zoomed-out map of Maat Mons and Ozza Mons

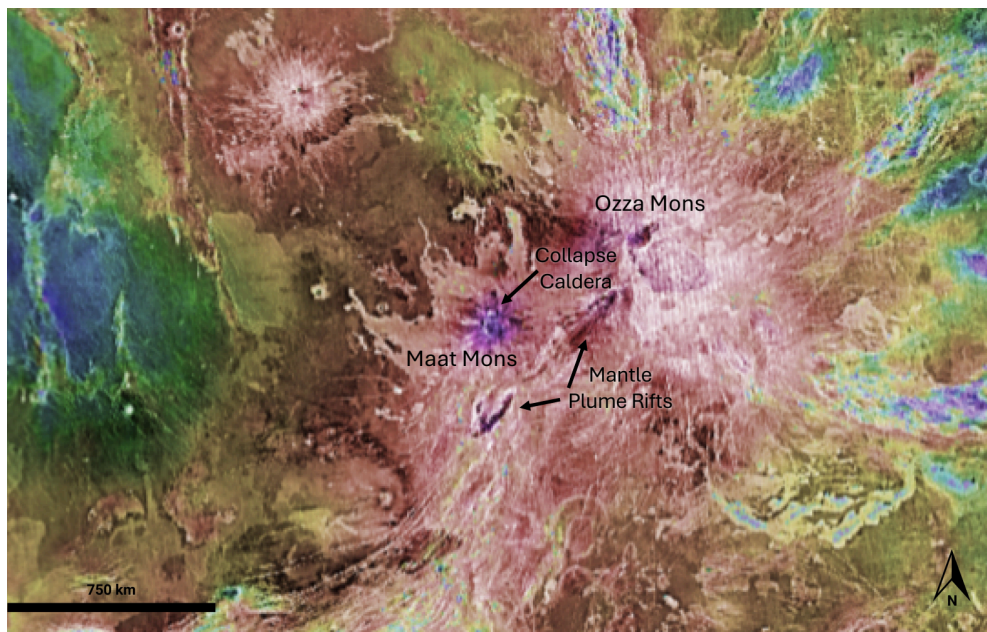


Figure 9: JMARS Zoomed-In map of Maat Mons and Ozza Mons

At 50 km, the IR Tunable Laser Spectrometer is initialized for isotopic analysis. The system operates between 1000–30,000 nm with a spectral resolution of 0.0001 cm^{-1} , measuring D/H. The calibration protocol is conducted using onboard reference cells containing known isotopic mixtures. Measurements are taken in vertical intervals of 1 km as the aerobot descends, generating isotopic profiles to track variations in fractionation ratios. The laser spectrometer uses modulated tuning and lock-in detection to extract signals from background noise. Environmental control variables

include laser temperature stability, optical path length, and internal pressure control. Observations are prioritized over regions with strong geologic context, particularly above active plumes and tectonic scarps. Experimental control logic manages real-time signal quality checks and queues data segments for downlink in priority order.

Throughout the mission, all three instruments operate under a coordinated observational sequence governed by the CDH system. Instrument-specific calibration events occur at regular intervals based on accumulated time, environmental shifts, and internal diagnostic triggers. Thermal stability is managed by layered insulation and heat-dissipating fins integrated into the payload structure, while regulated power distribution ensures each subsystem receives required voltage levels without fluctuation. Real-time health telemetry is collected every 10 minutes and used to validate instrument output stability. If any subsystem experiences deviation beyond thresholds set in Section 3.3 (Payload Success Criteria), failover logic initiates recalibration or conditional shutdown.

Scientific site selection is based on surface units visible along the aerobot's descent trajectory and defined in Section 1.3. These include volcanic edifices, caldera interiors, tectonic boundaries, and inferred plume source regions. The JMARS Zoomed-Out and Zoomed-In maps illustrate the observational corridor covering Maat Mons, Ozza Mons, and associated rift structures. These regions are selected for their potential to yield high-quality gas emissions, altered mineral surfaces, and isotopically distinct atmospheric layers. The approach allows each instrument to capture its respective deliverables under controlled conditions while providing temporal and spatial overlap for integrated analysis.

The mission's experimental logic centers on descending through key observational zones at controlled rates, with instruments cycling through calibrated acquisition modes tailored to Venus's stratified environment. Subsystem coordination enables adaptive response to environmental variability while maintaining scientific integrity. Data is processed and prioritized onboard, ensuring that science return is maximized within power, bandwidth, and thermal constraints.

3.3. Payload Success Criteria

The scientific payload onboard the Aerobot consists of three integrated instruments, each contributing to key mission objectives related to Venus's atmospheric composition, surface weathering processes, and isotopic signatures. Instrument success is evaluated using defined thresholds for minimum operation, optimum performance, and stretch goals exceeding design expectations in the Success Criteria (SCT). Failure modes are characterized by conditions preventing data return or resulting in data below the minimum thresholds for scientific use.

Category	Minimum Success	Optimum Success	Stretch Goal
VIRTIS-H	Detect CO ₂ and SO ₂ spectra at 1300-3000 $\lambda/\Delta\lambda$, 1.2×10^{-4} Wm ⁻² sr ⁻¹ m ⁻¹ , over $\geq 5\text{km}^2$	Detect both gasses at 3000 $\lambda/\Delta\lambda$ across multiple volcanic sites (e.g., Maat Mons, Ozza Mons)	Resolve gas gradients $<2\text{km}^2$ by detecting plume structure in 3D

VIRTIS-M	Collect spectral imaging at 290-4300 nm, $\geq 100 \text{ \AA}/\Delta\lambda$, 250 m spatial resolution over 5 km	Identify Fe-oxides and sulfates with $\geq 300 \text{ \AA}/\Delta\lambda$; full surface mapping of volcanic deposits	Imaging $>5000 \text{ nm}$, $500 \text{ \AA}/\Delta\lambda$ by detecting chemical gradients on slope profiles
Infrared Tunable Laser Spectrometer	Measure D/H and other isotope, 0.0001 cm^{-1} resolution, 1 ppm sensitivity	Resolve D/H of carbon-21, oxygen-18, and nitrogen-15 with $<1\%$ error over 1-2 km	Repeat isotopic sampling

Table 26: Success Criteria Table with the mission's instruments, respectively.

The Visible and Infrared Thermal Imaging Spectrometer High-Resolution Subsystem (VIRTIS-H) is designed to quantify concentrations of carbon dioxide (CO_2) and sulfur dioxide (SO_2) in volcanic plumes. Minimum success requires acquisition of absorption spectra in the 1870-4300 nm range with a spectral resolution of at least $1300 \text{ \AA}/\Delta\lambda$ and a sensitivity above $1.2 \times 10^{-4} \text{ W}\cdot\text{m}^{-2}\cdot\text{sr}^{-1}\cdot\mu\text{m}^{-1}$ over a spatial footprint of 5 km^2 . Optimum success includes resolution up to $3000 \text{ \AA}/\Delta\lambda$ and consistent detection of volcanic gases from Maat Mons and Ozza Mons. The stretch goal involves resolving spatial gradients in gas composition at $<2 \text{ km}^2$ resolution, enabling modeling of plume dispersion. Failure modes include detector instability, spectrometer misalignment, or loss of thermal control preventing instrument operation.

Successful data collection by VIRTIS-H supports detection of present-day volcanism and improves current models of atmospheric replenishment through mantle outgassing. These measurements provide critical context for understanding the role of volcanic processes in Venusian atmospheric dynamics and habitability potential (Piccioni et al., n.d.).

The Visible and Infrared Thermal Imaging Spectrometer Mapping Subsystem (VIRTIS-M) provides spatially resolved spectral imagery of Venus's surface to detect chemical weathering products linked to atmospheric interactions. Minimum success is defined by obtaining spectral imaging in the 290-4300 nm range with a spectral resolution $\geq 100 \text{ \AA}/\Delta\lambda$ and a spatial resolution $\leq 0.25 \text{ mrad}$ (250 m at 100 km altitude), covering a minimum of 5 km of surface. Optimum success includes identifying Fe oxides and sulfates using $\geq 300 \text{ \AA}/\Delta\lambda$ across major volcanic terrains. The stretch goal involves mineralogical mapping beyond 5000 nm with resolution $\geq 500 \text{ \AA}/\Delta\lambda$ and high spatial fidelity across varied geologic terrains.

Failure conditions include loss of optical integrity, pointing instability, or insufficient image contrast. VIRTIS-M data are essential for accessing the extent and distribution of surface alteration, characterizing the interaction between atmospheric chemistry and surface mineralogy, and constraining the chemical evolution of Venus's lithosphere (Piccioni et al., n.d.).

The Infrared Tunable Laser Spectrometer is optimized to detect stable isotopic ratios of hydrogen, carbon, nitrogen, and oxygen in the Venusian atmosphere. Minimum success requires detection of D/H and at least one additional isotopic ratio with spectral resolution of 0.0001 cm^{-1} and sensitivity of 1 ppm across a 1-2 km atmospheric area. Optimum success includes full detection of D/H of carbon-21, oxygen-18, and nitrogen-15 with $<1\%$ measurement uncertainty. The stretch goal involves repeated temporal sampling and spatial correlation of isotopic gradients with

surface features detected by VIRTIS instruments

Failure may result from loss of laser tuning functionality, detector failure, or degradation of spectral resolution. High-precision isotopic analysis is required to assess atmospheric escape, evaluate volcanic outgassing history, and model tectonic and geochemical evolution of Venus's crust and mantle (NASA MDL, 2022).

The payload deliverables from the Aerobot represent a stepping stone in understanding Venus as a geologically and chemically active planetary body. High-resolution spectroscopic data from VIRTIS-H shall enable direct detection of volcanic activity, confirming or refuting hypotheses of ongoing geodynamic processes. VIRTIS-M's surface mapping capabilities shall yield the first aerobot-based mineralogical survey of weathering patterns, offering insight into the chemical interactions between the dense Venusian atmosphere and surface lithology. The Infrared Tunable Laser Spectrometer shall return the accurate isotopic measurements of volatile elements in Venus's atmosphere to date, supporting models of atmospheric loss, resurfacing history, and tectonic evolution. Together, these instruments contribute a unique dataset that bridges atmospheric science and planetary geology.

3.4. Testing and Calibration Measurements

Following deployment within Venus's atmosphere, all instruments shall undergo preliminary testing and calibration to ensure functionality and data collection under operational conditions. These procedures are independent of pre-launch verification and validation tasks and are designed to confirm instrument performance in situ. Calibration operations shall occur within specific altitude ranges as outlined in the Concept of Operations (Section 1.5) and shall precede nominal science operations. Calibration shall be executed autonomously and verified through diagnostic data transmitted during communication windows.

At altitudes above 60 km, VIRTIS-H shall initiate spectral calibration using internal calibration lamps to adjust for potential wavelength drift and detector gain instability. Scans shall be conducted across predetermined atmospheric windows to confirm proper alignment of absorption bands. Calibration targets include regions with low aerosol opacity and modeled SO₂/CO₂ signatures. Control variables include integration time, detector temperature, and spectral line positioning. Calibration shall be considered valid if measured absorption peaks fall within ± 1.5 nm of expected positions and radiometric sensitivity remains within defined operational thresholds.

VIRTIS-M calibration shall be conducted near 60 km altitude by imaging topographically flat terrain with uniform reflectance characteristics. Spectral calibration shall be performed using onboard blackbody sources and wavelength reference lamps to ensure accurate spectral mapping. Spatial fidelity shall be verified by evaluating frame-to-frame registration and geometric consistency across image tiles. Control variables include scan direction, solar incidence angle, detector thermal response. Calibration success is defined by spatial coherence within 1 pixel deviation and spectral features consistent with onboard mineral standards.

Between 50 and 40 km, the Infrared Tunable Laser Spectrometer shall

perform calibration using sealed onboard reference cells containing known isotopic ratios. Calibration shall validate spectral features, laser tuning accuracy, and detector sensitivity. Line center deviations shall be assessed across D/H ratios. Acceptable tolerance is $\leq 0.0001 \text{ cm}^{-1}$. Control parameters include modulation frequency, optical path length, and environmental pressure. Spectral integrity shall be confirmed if retrieved isotopic ratios match onboard reference standards within $\pm 1\%$ deviation.

Calibration operations are designed to occur early in the 36-hour descent phase, prior to entry into denser atmospheric layers. Altitude targeting for each calibration sequence follows the operational profile defined in Section 1.5. Instruments failing to meet internal thresholds during calibration shall initiate repeat cycles until nominal performance is achieved or the next scheduled window is reached.

3.5. Precision and Accuracy of Instrumentation

3.6. Expected Data & Analysis

The VIRTIS-M instrument was used on the Venus Express mission to characterize the planet's mesosphere through night-side thermal emission spectra, providing insight into temperature structure and atmospheric circulation (Grassi et al. 2010). Although the focus of that study was atmospheric rather than surface analysis, the instrument's spectral sensitivity from 1 to 5 μm overlaps with wavelengths used to detect thermally altered surface materials. Team 10's Venusian aerobot mission will use VIRTIS-M in a similar spectral window to identify surface weathering products like iron oxides and sulfates. However, unlike Venus Express's orbital measurements with limited spatial resolution, this mission collects high-resolution hyperspectral data while floating within the lower atmosphere. The proximity to the surface shall reduce scattering and increase the signal-to-noise ratio, improving the detection of subtle absorption features linked to chemical alteration. Once corrected for atmospheric interference, the spectral data shall distinguish and characterize degrees of weathering. Areas near volcanic structures like Maat Mons are expected to exhibit strong signals of iron-bearing minerals and sulfates, suggesting prolonged interactions with Venus's sulfur-rich atmospheric gases. This would support a model in which volcanic degassing has driven long-term surface-atmosphere interactions, helping reconstruct the geochemical history of Venus.

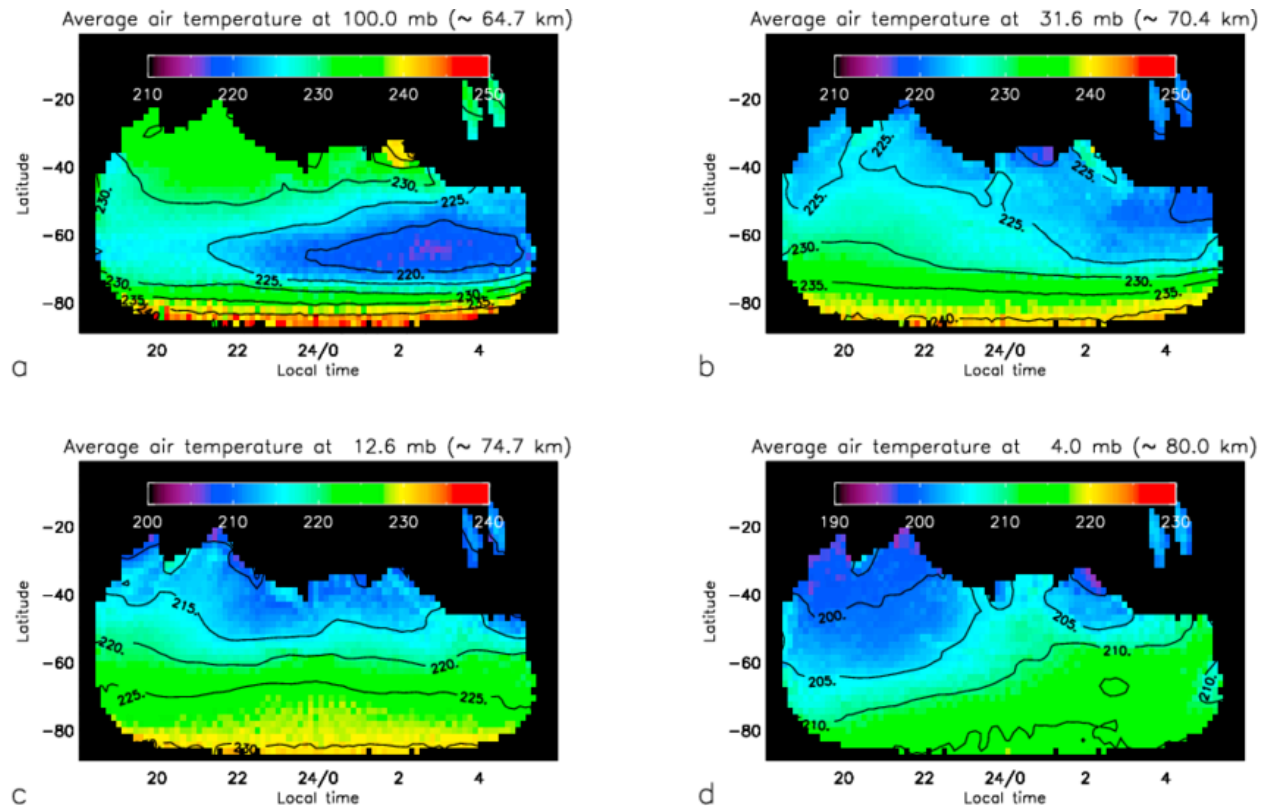


Figure 10: Mean temperature field of the Venusian mesosphere as a function of latitude and local time at four pressure levels (Grassi et al. 2010).

The VIRTIS-H was used to study comet 67P/Churyumov-Gerasimenko to understand the thermal emission and spectral slope of the dust coma (Bockelée-Morvan et al. 2019). In this study, the spectra revealed phase reddening, temperature variations dependent on elevation, and a measurable bolometric albedo (Bockelée-Morvan et al. 2019). Similar to the mission study of comet 67P, team 10's Venusian mission shall analyze the gas emission and compositions of carbon dioxide and sulfur dioxide in similar wavelength ranges of 2-5 μm , detecting trace gases. However, in the comet study, the environmental conditions are completely different from those of Venus. Due to Venus's hot atmosphere, significantly higher temperatures and pressures are expected to require complex data processing tools to remove noise from collected data. The spectral analysis is anticipated to be more of a blended continuum with higher absorption dips for the studied carbon dioxide and sulfur dioxide, once noise is removed. As the aerobot is in-situ and within a hot and dense planet, the data shall likely be richer, denser, and provide insight into Venus's current volcanic processes. Understanding the emissions of carbon and sulfur dioxide, elevated concentrations around Maat Mons and Ozza Mons would provide strong evidence of active volcanoes and support the hypothesis that Venus is geologically active.

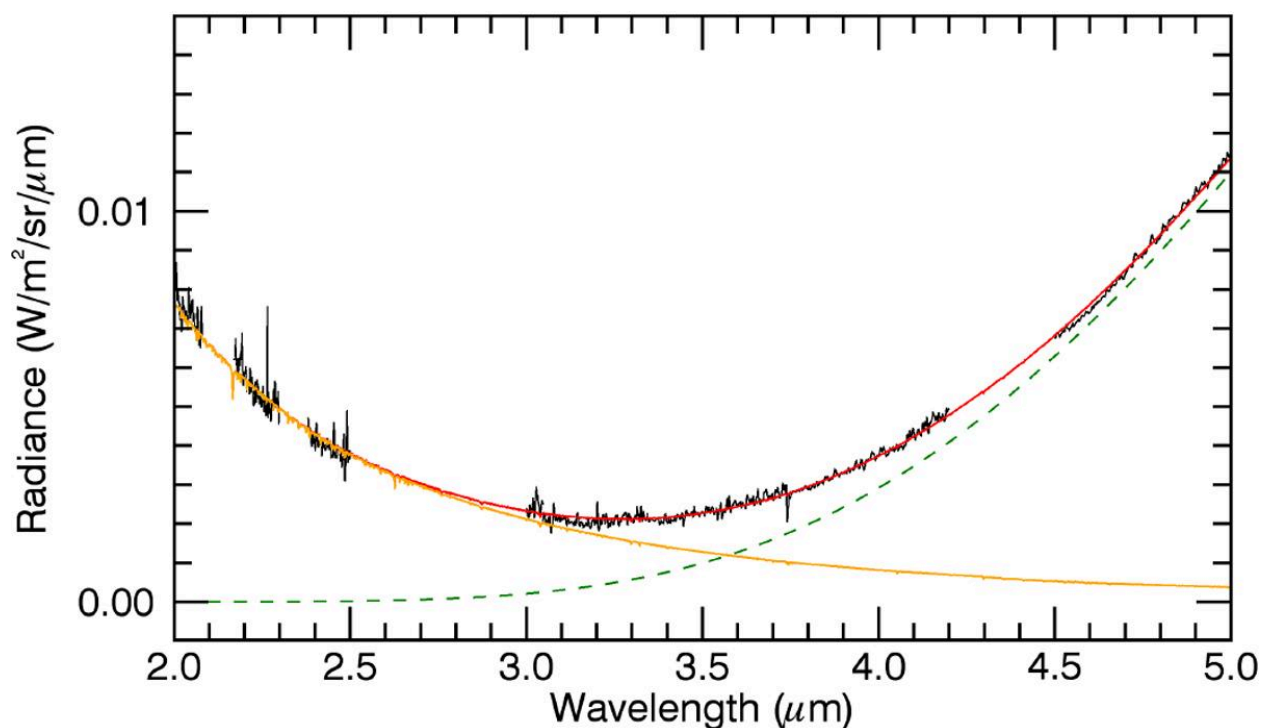


Figure 11: VIRTIS-H spectra of comet 67P/Churyumov-Gerasimenko showing thermal emission features, excluding regions with water and CO₂ emission lines (Bockelée-Morvan et al. 2019).

IRTLS has been previously deployed in planetary missions such as NASA's Curiosity rover on Mars, where it was used to measure isotope ratios of hydrogen, carbon, and oxygen in both atmospheric gases and surface-released volatiles, providing critical insight into atmospheric loss and Mars's hydrological history (Webster et al. 2023). In Team 10's Venus mission, the IRTLS will operate in a similar wavelength range of 1–30 μm and will be used to analyze stable isotope ratios of D/H, carbon-21, nitrogen-15, and oxygen-18 over faulted geological terrains in an attempt to construct Venus's planetary evolution. Unlike the Martian IRTLS, which operated in a relatively cool and low-pressure environment, the Venusian aerobot mission IRTLS must contend with Venusian high temperatures and pressures. The acquired data is expected to showcase different absorption lines for each isotope measured, most likely as narrow bars at specific wavelengths, with each value demonstrating different information regarding Venus. Such values shall be used to trace fractionation patterns by volcanic outgassing, ancient tectonic, or atmospheric processes. Isotopic anomalies over fractured geology or volcanic rift zones would suggest past tectonic activity that contributed to Venus's current atmospheric makeup, supporting hypotheses of a once more dynamic and Earth-like Venusian past.

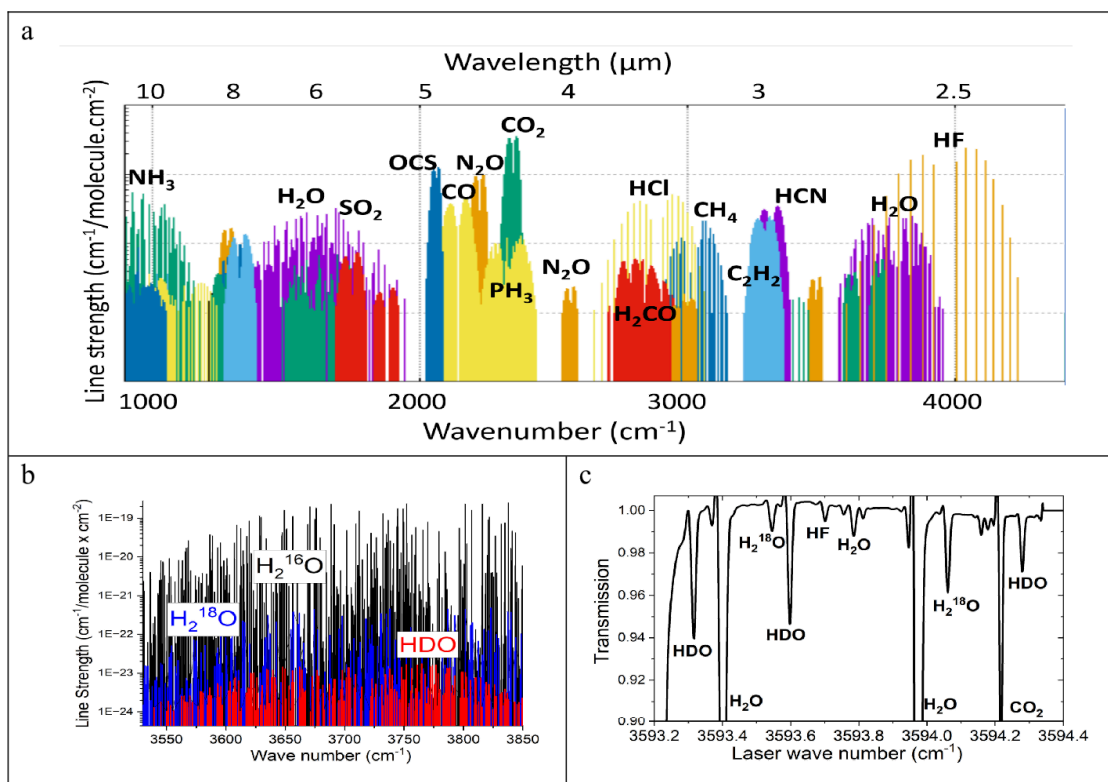


Figure 12: Observed TLS spectra showing infrared absorption features of water and its isotopologues (HDO, H₂¹⁶O, H₂¹⁸O), used for isotope ratio analysis (Webster et al. 2023).

4. Mission Risk Management

4.1. Safety and Hazard Overview

For safety and hazard concerns, our team has adopted a mission-wide safety and hazard management strategy rooted in NASA's system engineering and risk management principles. The team recognizes the challenges of operating in a high-risk planetary environment, and as a result, there is a high priority on early identification, structured analysis, and ongoing mitigation of risks across all mission phases. This includes from design and development to integration, launch, and operations.

To help decrease risk involved, risk identification measures have been structured to begin as early as the concept phase through structured brainstorming, trade studies, and environmental modeling. These efforts are further explored through detailed Failure Modes and Effects Analysis (FMEA), fault tree analysis, and component-level assessments as designs mature and evolve. From here, risks that are identified will be evaluated along two axes: likelihood and consequence. In these two axes, they will be further categorized based on their impact on mission success, schedule, cost, or safety. This is the most effective process as it will not only help

enable the team to focus more resources and time on the most critical threats but also helps the team decide which calculated risks to accept (in areas where margin exists).

As this is a cost-constrained and technology-driven mission, the team acknowledges that some higher-risk elements are necessary and expected. This is especially true for any experimental parts or platforms, new mobility systems, or updated sensor/instrument packages. It is worth mentioning, however, that the approach selected ensures that core functions required for mission success operate within acceptable and defensible risk boundaries. Risk not avoided but has been worked around to satisfy both mission and risk requirements. To support this, heritage components are used wherever feasible and redundancy is applied selectively. In addition to this, all high-priority systems are tested rigorously to meet minimum reliability thresholds.

There has also been an emphasis on hazard analysis as well as it is equally central to system design. Environmental risks associated with the mission destination and purpose, such as extreme thermal conditions, corrosive atmospheres, or high radiation exposure, have been studied and have been incorporated in early modeling efforts. It has also been paramount in material selection, shielding, and system-level redundancies research. As per the standard, launch and deployment hazards are also addressed through established safety reviews and compliance with applicable NASA safety standards.

The team also acknowledges its obligations under planetary protection guidelines, as outlined by COSPAR and NASA. While this mission does not target a surface landing or biological sample return, it still strictly adheres to Category II planetary protection protocols. This includes minimizing forward contamination through rigorous and repetitive cleanliness protocols during integration and ensuring that hardware cannot inadvertently survive atmospheric entry or reach planetary surfaces.

4.1.1. Risk Analysis

A variety of methods are being used to identify, manage, and track risks within the project. To identify risks, brainstorming in a group setting shall be used along with peer reviews of an individual's work and mentor reliance. To broaden the scope of all risks and ensure none are missed, these tactics and others, such as trigger questions will guide the risk strategy of the mission. To manage risks, conventional tools such as risk matrices, documents, and FMEA shall be used. Tracking of risks will see monthly review of these documents to address changes in the mission lifestyle. A risk log will

allow project members to properly log these changes and identify where current risks are. Below, a risk matrix with two risks per subteam and a table explaining those risks can be found. Afterwards, a narrative of how the team plans to mitigate those risks will be provided.

Engineering:

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans
Prioritization order of risks to address	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	What WBS element the risk affects the most	Which team member is assigned responsibility over seeing this risks closure	What effects will this risk cause? An increase in costs, a loss of schedule margin or launch dates, or loss of technical data to be collected?	What is the timeframe until this risk will be addressed? Short = 1-3 months Medium = 3-12 months Long = 12+ months	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]."	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle	Specific actions that will be implemented to mitigate the risk by either lowering the L and/or the C. Add or remove as many mitigation actions as needed and assign a new ID and number (12.3.1) to them. Carry this number across the next few rows where mitigation plans are monitored and controlled
3	R3	1.5.2 Mechanical Subsystem	Akilesh Yelchuru	Scope	Long Term	Material Endurance	Material endurance in Venus is a very important topic as Venus has a corrosive atmosphere additionally traversing various atmospheres and space means the aerobot will experience pressure variance.	1	5	Moderate	Research (R)	Neutral	
1	R1	1.5.2 Mechanical Subsystem	Akilesh Yelchuru	Scope	Long Term	Reliability for Aerobot Pump	The pump should be able to control buoyancy of the aerobot via inflation and deflation at will. Lack of control of the pump from issues such as wire breaking or pneumatic failure would result in loss of control of the system.	2	3	Moderate	Research (R)	Decrease	The verification and validation methods can be utilized for this. For this instance validation is utilized which is the testing of the vehicle under the conditions simulating the final destination and required trajectory to reach there. For this Inspection and Analysis can be utilized. Analysis would use software to anticipate the constraints of a part of interest using the derived data or calculations from other structures. Additionally, Inspection involves visual verification and examination to identify the various design elements and identification from manufacturers. This can be used to ensure the pump has no failures objectively as even from testing not all wear and tear can be examined accurately.
2	R2	1.5.2 Mechanical Subsystem	Akilesh Yelchuru	Scope	Long Term	Inspection and testing	The structures should be placed correctly in a way to accommodate any hinges or joints to be activated easily and as simply as possible. Additionally the tolerances to package the various components should be considered as part of this accommodation. As part of this material from packaging to material of the various other substructures should be considered to avoid any issues such as wear and tear.	2	3	Moderate	Research (R)	Decrease	The verification and validation methods can be utilized for this. For this instance validation is utilized which is the testing of the vehicle under the conditions simulating the final destination and required trajectory to reach there. For this Inspection and Analysis can be utilized. Analysis would use software to anticipate the constraints of a part of interest using the derived data or calculations from other structures (typically a lower system) or verification from end products. Additionally, Inspection involves visual verification and examination to identify the various design elements and identification from manufacturers. Inspections include documentation, various records, and drawings.

Rank	Risk ID #	Risk Owner	Risk Title	Risk Statement	L	C	Rating	Approach	Trend
Prioritization order of risks to address	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	Which team member is assigned responsibility over seeing this risks closure	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]."	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle
1	W1	Eduardo C.	Probability of Overheating	Because of Venus' hot environment and constant solar exposure, if the thermal systems are to fail the aerobot would begin overheating. This could adversely affect all electronics including payload, which would diminish or destroy the ability to complete the mission science goals.	3	5	High	Watch (W)	New

Table 27: Mechanical Risks

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend
Prioritization order of risks to address	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	What WBS element the risk affects the most	Which team member is assigned responsibility over seeing this risks closure	What effects will this risk cause? An increase in costs, a loss of schedule margin or launch dates, or loss of technical data to be collected?	What is the timeframe until this risk will be addressed? Short = 1-3 months Medium = 3-12 months Long = 12+ months	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]"	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle
1	C1	1.1 Project Management	Veronica Mok	Cost	Long Term	Global Economy Fluctuation	Given that the state of global economics is constantly changing, it is impossible to predict how much inflation or other changes to the cost of materials will be affected, as such, there is a possibility that subsystem manufacturing may be at risk of being delayed adversely impacting the completion to conduct proper testing, which can result in delays in delivering the completed aerobot	2	5	Moderate	Close (C)	→ Neutral →

Table 28: Thermal And Science Risks

Rank	Risk ID #	Risk Owner	Risk Title	Risk Statement	L	C	Rating	Approach	Trend
Prioritization order of risks to address	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	Which team member is assigned responsibility over seeing this risks closure	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]"	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle
7	P1	Karthik	Power Output Too Low	Given that the power system supplies a limited amount of energy, there is a possibility of insufficient power supplied to components adversely impacting Aerobot's operation, which can result in the failure of subsystems	2	5	Moderate	Watch (W)	↓ Decrease ↓
3	P2	Karthik	Power Reservoir Depletes Prematurely	Given that a limited amount of power onboard, there is a possibility of premature power loss adversely impacting the duration of the mission, which can result in an incomplete mission	3	3	Moderate	Watch (W)	↓ Decrease ↓
4	C1	Angy	Data Fails to Process Onboard	Given that many system architectures are onboard, there is a possibility of miscommunication adversely impacting onboard data and handling, which can result in missing or lost data or maneuvers	2	3	Moderate	Research (R)	↓ Decrease ↓
5	C2	Angy	Data Fails to Send to Earth	Given that the Aerobot will be operating on a distant planet, there is a possibility of a loss of communication adversely impacting the data relayed to Earth, which can result in lost data, or improper operation	2	5	Moderate	Research (R)	↓ Decrease ↓
9	T1	Eduardo	Components Exceed Operating Temperatures	Given that the Venusian atmosphere is very hot, there is a possibility of overheating adversely impacting the operation of subsystems, which can result in failure of the Aerobot	1	4	Low	Watch (W)	↓ Decrease ↓
9	T2	Eduardo	Thermal Management System Malfunctions	Given that the thermal system has a low TRL, there is a possibility of malfunctions adversely impacting thermal management, which can result in overheating components	1	2	Low	Research (R)	↓ Decrease ↓
8	S7	Madeline	Science Instruments Obscured	Given that the Venusian atmosphere blocks many wavelength of light, there is a possibility of surface features being obscured adversely impacting science package, which can result in failed science objectives	1	4	Low	Accept (A)	↓ Decrease ↓
6	S8	Madeline	Science Instruments are not provided all relevant data	Given that our knowledge of Venus is so low, there is a possibility of the selected location not possessing data adversely impacting what data can be gathered, which can result in failed science objectives	4	2	Moderate	Research (R)	↓ Decrease ↓

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans	Status of Mitigation Plans	Trigger Date	Trigger Event	Closure Date	Closure Event
Prioritization order of risks to address	Unique ID assigned to each risk, may have a precursor code to designate what level the risk is tracked at	What WBS element the risk affects the most	Which team member is assigned responsibility over seeing this risks closure	What effects will this risk cause? An increase in costs, a loss of schedule margin or launch dates, or loss of technical data to be collected?	What is the timeframe until this risk will be addressed? Short = 1-3 months Medium = 3-12 months Long = 12+ months	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]"	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LxC over its life cycle	Specific actions that will be implemented to mitigate the risk by either lowering the L, and/or the C, and/or a new line item and number (1,2,3, 10) from the risk register across the full life cycle plan, any mitigation plans are considered and coordinated	Updates on the mitigation plans throughout their life cycle	The date that a trigger event will be associated with the trigger date	A specific event that will be associated with the trigger date	The date that a trigger event will be associated with the trigger date	A specific event that will be associated with the closure date
2	M1	2.1.1.4	Adith	Cost/Schedule Scope	Short Term	Mechanisms Fail to Deploy	Given that the Aerobot will be supported by a pump balloon system, there is a possibility of improper operation adversely impacting the lift of the Aerobot, which can result in incomplete operation or total failure	3	4	Moderate	Mitigate (M)	↓ Decrease ↓	1. The analysis method will be utilized to verify the performance of the mechanisms in the intended conditions and circumstances. 2. The issues will be reviewed and comprehended. 3. The issues will be fixed. 4. The analysis method will be utilized to verify or validate the performance of the mechanisms in the intended conditions and circumstances. If any issues arise after step 1 through 4 will be used until the issue is resolved.	3 Complete	04/21/2025	PCR Submission	4/21/2025	PCR Submission
1	M2	2.1.1.4	Adith	Cost/Schedule Scope	Short Term	Structural Integrity Fails	Given that lightweight structures compose the Aerobot's frame, there is a possibility of structural failure adversely impacting the structures that compose it, which can result in total structural failure	2	5	Moderate	Mitigate (M)	↓ Decrease ↓	1. The analysis method will be utilized to verify the performance of the mechanisms in the intended conditions and circumstances. 2. The issues will be reviewed and comprehended. 3. The issues will be fixed. 4. The analysis method will be utilized to verify or validate the performance of the mechanisms in the intended conditions and circumstances. If any issues arise after step 1 through 4 will be used until the issue is resolved.	3 Complete	4/21/2025	PCR Submission	4/21/2025	PCR Submission

Table 29: Programmatic Risks

Science:

Rank	Risk ID #	WBS Element	Risk Owner	Category	Timeframe	Risk Title	Risk Statement	L	C	Rating	Approach	Trend	Mitigation Plans
Prioritize order of risks to address	Unique ID assigned to each risk, may have precursor code to designate what level the risk is tracked at	What WBS element the risk affects the most	Which team member is assigned responsibility over seeing the risks closure	What effects will this risk cause? An increase in costs, a loss of schedule margin or launch dates, or loss of technical data to be collected?	What is the timeframe and this risk will be addressed? Short = 1-3 months Medium = 3-12 months Long = 12+ months	A short descriptive title of the risk	A more thorough description of the risk following the appropriate format of "Given that [CONDITION], there is a possibility of [DEPARTURE] adversely impacting [ASSET], which can result in [CONSEQUENCE]"	Likelihood of the risk to occur, see the Resources sheet for more information	Consequences if the risk occurs, see the Resources sheet for more information	High level indicator of the criticality of the risk	How the risk will be handled or not	Trend of any changes to the risk's LoC over its life cycle	Specific actions that will be implemented to mitigate the risk by either lowering the L and/or the C Add or remove as many mitigation actions as needed and assign a new line and number (1,2,3...) to them. Carry this number across the next few rows where mitigation plans are monitored and controlled
2	S3	1.2 Science Traceability Matrix	Madeline Maldonado Gutierrez	Cost/Schedule/Scope	Long Term	Planetary Contamination	Given the highly acidic state of Venus' atmosphere, there is a possibility that Earth-based testing will not accurately confirm if the aerobot material will sustain the system for the entirety of the 30-hour data collection period, which can result in the atmosphere will unexpectedly disintegrate or damage the aerobot.	3	5	High	Mitigate (M)	Decrease	V&V methods will be utilized to prevent this. In the case of planetary contamination small scale tests can be utilized to collect data which can be uploaded to a software that analyzes the effects of the planetary contamination via the analysis method. This can be used in a design process to observe, collect data, create solutions, and repeat until success. This would just be a case of ensuring the material does not corrode which can be tested and analyzed.
5	S4	1.2 Science Traceability Matrix	Madeline Maldonado Gutierrez	Scope	Long Term	Concentrated Atmosphere Interactions	Given the varying high concentration of elements within the Venusian atmosphere from 100 km downwards, there is a possibility that the boundaries of the lower haze are heterogeneous around the entirety of Venus, such as potentially being close to the surface at the equator where Maat Mons is located, adversely impacting the visibility within the planned elevations for data collection outlined in the ConOps, which can result in photography data that may have lower resolution than expected and not fulfill the science objective.	1	3	Low	Mitigate (M)	Increase	V&V methods will be utilized to prevent this. The haze is unavoidable hence various heights from the ground in tandem with technology such as x-rays or data collected about weather via JMARS can be utilized when lowering the aerobot to collect more accurate data.
3	S5	1.2 Science Traceability Matrix	Laiba Mahmood	Cost/Schedule/Scope	Long Term	Unanticipated Wind Conditions	Given the highly variable wind speeds along altitudes in Venus' atmosphere, there is a possibility that the wind conditions will be stronger than anticipated, adversely impacting the location of the aerobot system during the CoT period, which can result in a failure to be situated over the Maat Mons region and a failure to fulfill all planned science objectives.	3	5	High	Research (R)	Decrease	The wind conditions can be unanticipated but there is control over the aerobot pump system. Hence, the aerobot pump system can be utilized to vary buoyancy in response to such weather.
6	S1	1.2 Science Traceability Matrix	Laiba Mahmood	Scope	Long Term	Science Instruments Obscured	Given the possibility of unexpected interactions with the Venus atmosphere or other events throughout the mission from launch, there is a possibility that the science instruments within the aerobot system will sustain damages that limit their capability to perform the expected physical characteristic collection, which would result in a failure to fulfill the science goals outlined by the customer.	1	4	Low	Accept (A)	Neutral	The various weather and wind conditions should not alter the condition of the instruments as they will be designed specifically for Venus but to provide further support removable and reusable protective casing can be placed over the instrument of choice when not in use.
4	S2	1.2 Science Traceability Matrix	Calin Stauffer	Cost/Scope	Long Term	Science Instruments are not provided all relevant data	Given that our knowledge of Venus' heterogeneous conditions across the planet are not well known, it is possible that the selected locations will not fulfill the science requirements and goals as they are theorized to, adversely impacting the useful data that will be gathered, which can result in failing to fulfill the science objectives.	4	2	Moderate	Research (R)	Decrease	This is something that can be altered over time. There will always be something that cannot be anticipated in terms of data collection. If there is something that cannot be analyzed the scope of the mission will only allow so much. To prevent this using current and past data in relation to analyzing similar planets with various telescopes to provide more data can help set precedence for data and how such data can be collected.
1	S6	1.2 Science Traceability Matrix	Calin Stauffer	Schedule/Scope	Long Term	Descope decreasing measurement time	Given that the mission descope due to unforeseen issues with balloon material strength, there is a possibility that the balloon will disintegrate after 30 hours instead of the planned 48 hours, adversely impacting the ability to collect all the anticipated science data in time, which can result in a failure to fulfill the science objectives and therefore the customer science goals.	5	3	High	Mitigate (M)	Decrease	The best way to deal with descope decreasing measurement time is to test ahead of time using the V&V techniques mainly testing. The material of the balloon strength can be improved through using the design process to research, develop, test, collect data, and repeat until a desirable balloon strength of material with tolerance is met.

Table 30: Science Risks

L I K E L I H O O D	5		C1 M2 P1 C2	S6		
	4	T1 S7	S2	M1		
	3		C1	P2		S3 S5 W1
	2	T2		R1 R2	S8	
	1			S4	S1	R3
		1	2	3	4	5
		CONSEQUENCES				

Table 31: Risk Table

L = Likelihood (1-5)

1 = Not likely
5 = Extremely likely
C = Consequence (1-5)
1 = Low consequence
5 = High consequence
LxC Trend
New - Added this month
↓ - Decreasing (improving)
↑ - Increasing (worsening)
→ - Unchanged
Closed - Not actively monitored
Approaches
Accept (A)
Watch (W)
Research (R)
Mitigate (M)
Elevate (E)
Close (C)

An important part of the risk analysis is also the decommission plans. This is because while the objective is to analyze Venus further, we do not want to affect Venus negatively as well. This is an important topic as we do not want to prevent any extraterrestrial life forms from any point of time being negatively affected by any human based contamination and to protect the planet in general. This is reinforced further with the use of decommission plans. The main items from the mission to contaminate Venus would be the material on the aerobot itself. This will be achieved with multiple methods. For sensors like the laser spectrometer, the design will be non-invasive to mitigate the risk of contamination. For other materials, cleaning and sterilizing with NASA's standards will be utilized to ensure that the materials eroding from Venus's ground and atmospheric conditions will not leave an invasive contaminates.

4.1.2. Failure Mode and Effect Analysis (FMEA)

In order to track and maintain a thorough record of current high level risks and the level of those risks, a Failure Mode and Effect Analysis table has been created and will be maintained. The priority of a given risk is to be tracked using a risk priority number. This number is a product of the severity of consequences, likelihood of occurrence, and ability to detect the failure prior to its occurrence. Items assigned the highest RPN are to be given priority to be handled.

The highest priority risks identified for the Aerobot system according to RPN are as follows; puncturing of the balloon, thermal management failure, communication failure, and a lack of data gathered to meet all science objectives. A balloon puncture ranks high due to the difficulty to discern if it has occurred. Modelling would need to be undertaken to be able to understand what expected descent velocities are and therefore allow for an error threshold to determine if a puncture has occurred. A thermal management failure is at a high RPN due to the extreme conditions the vehicle will be subject to, especially close to the end of its descent. Further, modelling of the venusian conditions are necessary to develop confidence in the capabilities of the proposed system. A similar situation exists for the communication system, in which the environment becomes progressively extreme, increasing the chances of failure. Maturing the CDH design will be key for mitigation of this risk. Lastly, due to the relatively low number of instruments and the relatively large amount of data needed to fulfill the demanding science objectives, a failure to achieve all the goals of the mission is relatively likely. Depending on the capabilities of the updated VIRTIS information, the specifications of a TLS system, and the mission path in the high winds, the mission stands to be relatively difficult to complete in totality.

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Balloon and Pump	Improper Inflation	Mission time drastically reduced	9	Malfunction of pump system	2	Test pump system multiple times	4	72	New pump system can be considered
	System Punctured	Expected lift more than actual	3	Puncture in manufacturing or deployment	5	Extensive testing of balloon to be used before installation	8	120	Pump can be adjusted to compensate
Thermal Management	Fails to cool the Aerobot to operating temperature	Aerobot systems overheat	8	Improper measurements fed to system/Insufficient cooling provided by system	3	Testing of thermal system in a simulated venusian atmosphere to validate performance	5	120	Redundant thermal systems/reduced operation of other systems
Power Supply	Provides insufficient maximum wattage	Subsystems must operate at a limited state	6	Calculated maximum draw is unavailable or greater than expected	5	Testing of max power draw from each subsystem and max output from power supply	2	60	Backup power maybe be added or a limited configuration must be achieved
	Power depletes early	Fails to allow for operation for 36 hours	5	Battery power insufficient for allotted mission time, or mean power draw is higher than expected	4	Battery lifetime must be verified and mean power draw modeled	3	60	Extra supply may be added or mission lifetime must be reduced
	Wiring to subsystems fails	Subsystems left unpowered	8	Faulty connections or cables installed or created through operation	2	Connections must be tested prior to final integration	3	48	New connection systems must be considered
Command and Data Handling	Antennas fail to communicate with Earth	Partial or complete loss of data and system failure	10	Failed verification of antennae performance in Venusian atmosphere	4	Planning of coms and mission orientation to ensure clear communication	4	160	Must be tested, mitigated, and accepted
	Internal data storage fails	Loss of scientific data and science objectives failed	8	Storage bank performance in conditions overestimated	3	Testing of storage devices in extreme conditions	3	72	Redundant storage devices may be installed
Structural System	Structure does not fit all components	Only critical components can be taken aboard	4	Improper dimensions provided to design team	5	Cross reference subsystems to structure design	1	20	Desclope Subsystems if they will not fit
	Structure exceeds 1x1x1m size constraint	Mission requirement exceeded	3	Improper design of structure	5	Measuring of structure dimensions	1	15	Consult customer about changing constraints
	Overload Failure	Spacecraft breaks up enroute or on mission	10	Stresses experienced greater than yield strength of material	2	Simulated and experimental testing of structure	2	40	Must be tested, mitigated, and accepted
Science System	Insufficient Data Gathered	Science objectives are not met	5	Data at target location may be insufficient for objectives	3	Further research and missions for information	6	90	Different location mission with more current data considered
	Target unavailable for data gathering	Science objectives are not met	5	High winds or insertion may blow the Aerobot away from mission area	6	Careful modeling of mission timeline and atmosphere	2	60	Consider possible targets nearby the target area
	Supplier instrument operates improperly	Instruments not ready by launch or fail on mission	9	Supplied instruments are faulty	1	Testing of instruments pre-integration	3	27	Consider different suppliers

Table 32: Failure Mode and Effective Analysis

4.1.3. Personnel Hazards and Mitigations

The significance of comprehending personnel hazards is critical in launch operations to make certain no injuries occur and keep the mission goal to run smoothly. In risky environments such as rocket launches, hazards can occur and escalate rapidly, possibly leading to fatal injuries or mission delays. To avoid serious personnel hazards, adhering to general lab safety rules is ideal. General lab safety is crucial to preventing incidents and making sure of a secure workplace environment. Personnel must always wear appropriate PPE, personal protective equipment, such as lab coats, safety glasses, and gloves to protect against any chemical spills and splashes. Proper handling and storage of chemicals are significant, for instance, substances need to be properly labeled and volatile liquids stored in ventilated areas. Without a doubt, eating and drinking are prohibited in any lab to avoid any contamination or possible ingestion of hazardous materials. Fire safety protocols are necessary, knowing the locations of extinguishers and emergency exits. Additionally, there must be immediate reporting once there is a spill, injury, or equipment malfunction to avoid worse conditions. Work areas must remain uncluttered as waste should be disposed of in designated containers. Training is crucial on safety procedures, emergency response, and hazard communication.

Making certain personnel are safe during the manufacturing, integration, and testing phases of the mission requires identifying and mitigating potential hazards that may threaten team members. Physical hazards, such as moving mechanical parts, sharp edges, and unforeseen equipment movement, may pose critical risks during assembly and testing. For instance, dangerous rotating machinery or components that are loose or unsecure may lead to crushing, entanglement, or fatal injuries. The best mitigation for these risks include designing to remove the hazard if possible, for instance eliminating sharp edges in workstations and walkways, as well as including rounded corners or protective coating. If elimination is not an option, guards, locks, and restraints should be implemented. Training and procedures should also be looked at, ensuring personnel are following lockout/tagout, LOTO, protocols or utilizing handles that are ergonomic based on manual material handling, MMH. This is emphasized on developing hardware with practical and comfortable gripping handles to prevent any injury from improper handling.

Electrical hazards also display concern such as exposure to live conductors or high-voltage circuits. Electrical hazards will be lowered by consistently inspecting cords for damage and avoiding shorting of circuits. There could also be exposure to high-voltage circuits and keeping note of damp conditions is also critical for outdoor launch preparations. Insulating wiring and grounding systems are crucial for mitigating the risk of shock. Serious injuries such as disruption in muscle control could occur when working with voltages exceeding 34 V DC under wet or damp conditions. To control this risk, the systems shall have a limit on voltage, utilize insulation and grounding, and include circuit breakers if possible. If higher-voltage is required, contact area and lab moisture should be checked frequently and maintained. In addition, training on electrical risks and wearing PPE overall reduces exposure and risk.

Coming up to and during launch operations pose great risk as well, especially extreme temperatures and noise. Temperatures from cryogenic propellants could lead to risk of burns, to avoid this, insulation and warning labels are necessary, as well as

safe zones during initial ignition. For instance, to avoid possible burns from hot surface temperatures, “The following references were utilized to determine the limits: Hot: Greene, L.C., et al. (1958) on human tolerance to heat pain showed that the pain threshold is reached at 43.7°C (110.7°F) skin temperature. Lloyd-Smith, D.L., and Mendelssohn, K. (1948) found the pain threshold to be 44.6°C (112.3°F). Defrin, et al. (2006), found the pain threshold to be between 43-46°C (109-115°F). Damage to porcine skin was determined to be at 49°C (120.2°F) (Moritz, et al., 1947). Hand dysfunction and the associated safety risk during occupational practices in the cold increases with decreasing skin temperature. Onset of cold pain has been reported to occur between 23°C (73.4°F) and 14°C (57.2°F) during cold contact (Havenith, et al., 1992). A marked deterioration in tactile discrimination occurs at finger skin temperatures <8°C (46.4°F) with numbness found in one-third of subjects at 7°C (44.6°F) (Morton and Provins, 1960). Risk of frostbite occurs at 0 °C (32°F) (Havenith, et al., 1992.).” (9.0 Hardware and Equipment Encyclopedia 2025, sec. 9.3) Excessive noise from testing or liftoff noise, over 180 decibels, may lead to permanent hearing damage. Noise cancelling headphones are constantly necessary to mitigate risks as well as remote monitoring when possible.

For any fluid and gases, potential hazards include pressurized leaks or release of toxic substances that require guidance, as pressurized leaks could lead to risk of explosion or injury and toxic substances may endanger health. To mitigate this risk includes designing well thought systems with sealed connected, and automated shutdowns to prevent unwanted release. Vacuum and fluid collections systems may be implemented and placed near maintenance points to take care of any spills. Extreme temperatures and excessive noise pose hazards and require insulation and proper labeling of protective zones to avoid burns or hearing damage. For example, if a surface exceeds the safe touching temperature, greater than 45 degrees celsius, it must be insulated or labeled with warning. By relying on NASAs Technical Standards, as well as prioritizing hazard elimination by design, engineering safeguards, and training, personnel hazards minimizes injuries and enhances operational efficiency.

5. Activity Plan

5.1. Project Management Approach

The Aerobot System mission will require a strong dedicated workforce of around forty five people beyond the Academy of student teams. Each will be assigned to subteams that correlate to specialized skills. This makes sure coverage is comprehensive across all technical, scientific, and business activities. Following the NASA programmatic guidelines a comprehensive and formal organizational structure that is established to promote efficient decision making, schedule maintenance and budget constraints.

The full mission team is to be split into two overarching categories, technical and business or programmatic support. The technical group will consist of five main subteams that will specialize in mechanical and structures, thermal management, command and data handling, power systems and verification and validation. Each of the subteams mentioned must contain technicians and engineers who will work together to develop, test and integrate the Aerobots hardware and software based solutions. A separate science and instrumentation based subteam will be responsible for payload selection and interpreting data. The business and

programmatic branch will compromise project controls and budgeting, contracts and supply chain, and outreach and communications. Each subteam will report to one project manager who will be responsible for resource distribution and alignment with system requirements. The project manager will collaborate with subteam leads on critical decisions but will retain ultimate authority over central reserves. This method of delegation offers subteam leads a limited degree of budget authority, enabling quick and crucial responses to routine needs without excessive administrative delay.

Subteam	Phase C (Jan 2028 – Sep 2028)	Phase D (Oct 2028 – Mar 2029)	Phase E (Apr 2029 – Dec 2030)
Mechanical & Structures	6	5	2
Thermal Management	4	4	2
Command & Data Handling	5	5	3
Power Systems	4	4	2
Verification & Validation	6	7	3
Science & Instrumentation	6	5	5
Project Controls & Budgeting	2	2	2
Contracts & Supply Chain	1	1	1
Outreach & Communications	1	1	1
Total FTEs	35	34	21

The mechanical and structures sub team will mainly focus on developing the Aerobot's structural frame, mechanisms, and overall features crucial for deployment. Mechanical engineers, materials specialists, and design professionals in this team must analyze system loads, environmental constraints, and size or mass limitations to ensure compliance with NASA's mission requirements. The environment on Venus demands strong and durable materials, this means this subteam must perform extensive trade studies to identify alloys, composites, and coatings that can withstand temperature extremes and corrosive atmospheric chemistry while staying in the 50-kilogram mass capacity. The mechanical and structures

subteam will work closely and efficiently with the thermal management subteam to confirm that all structural materials can accommodate mounting points for insulation, cooling lines, or heat pumps as required.

The thermal management sub team must be able to provide solutions and guidance for mitigating the heat fluxes that can be expected to be encountered in the harsh atmospheric conditions. Thermal engineers and technicians in this team will handle designing passive insulation layers and efficient active cooling methods, along with special coatings for protection from radiation. The sub team will also manage testing of the Aerobots ability to maintain internal temperatures that are near 300 K for an extended amount of time. Coordinating with the mechanical specialists is important as the thermal team depends on important and precise structural data to integrate hardware like the heat exchangers or change of materials during different phases. The project manager and the subteam lead must consult with each other regularly on cost estimates for such special materials, given the complexities and potential high expense of needed thermal solutions for an environment as of Venus.

Command and data handling (CDH) requires a separate subteam dedicated to on-board computing, software integration, and telecommunications. Personnel in this group will include software engineers, avionics specialists, and security professionals who will make sure that command pathways remain under normal contingency conditions. The CDH subteam defines communication protocols, on-board data storage, and fault protection routines. Extensive collaboration will occur with the science and instrumentation subteam, as well as the power team, to make sure and confirm that instruments receive timely commands and that power resources are managed effectively during critical events such as data downlink or high load instrument operations.

The power sub team will handle all of the components related to energy generation, storage, and distribution. Electrical engineers in this group focus on selecting appropriate energy sources and ensuring that the system is able to maintain stable voltage levels while in flight. Batteries like the salt based lithium based ones must operate reliably under the expected thermal environment. Adjusting the budget may be necessary with this sub team if new test data recommend changes to battery chemistry or packaging. Any shift in the approach has to be reviewed in coordination with the project manager and business group to avoid distribution to the overall schedule or cost margin.

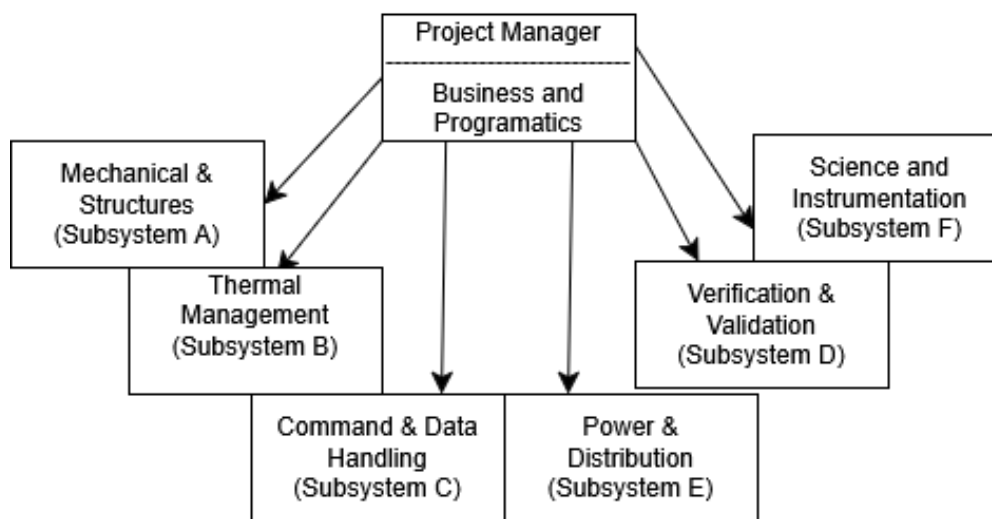
A dedicated sub team that conducts test campaigns for optimal performance and environmental tests will be formed. Test engineers and quality assurance specialists will plan procedures to altitude chambers, thermal vacuum tests, electromagnetic interference analysis and hardware simulations. This sub team is usually empowered to pause hardware progression, If significant design flaws or performance issues are noticed. This team will also documents all outcomes of the tests to provide NASA's review boards regarding the mission readiness as the project inches closer to launch.

Science and instrumentation activities will go under a specialized subteam, this will include planetary scientists and instrument electronics experts. Members of this group will refine payload definitions, specify scientific measurement requirements, and establish calibration standards for instruments measuring atmospheric chemistry and surface related variables. Coordination with the CDH group ensures adequate data throughput and on board storage capacity. In Phase E of the mission, the science team will oversee all data analysis and final

archiving, interacting with the outreach and communications unit to share what has been discovered with broader audiences and the mission stakeholders.

Programmatic and business functions are together under the project manager but consist of the different three sub teams with certain responsibilities. The project controls and budgeting subteam includes cost analysts, schedulers, and risk managers who track resource utilization across all technical groups. Outreach and communications personnel handle media relations, education, and mission press releases this will make sure external stakeholders can stay informed about project achievements and milestones the team is coming up on.

The central project manager will have oversight of these different sub teams and will make progress reports for major reviews such as the System Requirements Review and subsequent life cycle gates. Each sub team lead will coordinate activities and short term budget decisions under delegated authority daily. High or large risk cost changes, additions to the scope, and important schedule adjustments are escalated to the project manager and key mission executives.



The architecture positions envisioned above in the chart shows how each sub team will bear responsibilities. Fosters cross functional collaboration and establishes a straightforward escalation route for issues related to cost, schedule and technical issues. This design is to accommodate minor fluctuations in scope without a major reorganization being forced. Sub team leads will coordinate together and with the project manager to make sure that decisions, resource usage, and staffing levels remain on track. If critical issues were to arise, the project manager will consult with subteam leads and can reassign personnel or adjust the budget to preserve mission objectives and adhere to NASA's standards.

5.2. Mission Schedule

Schedule Basis of Estimate

The success of this mission hinges in the timing and planning of the team. As such, the mission will be subdivided into multiple stages, with the launch readiness date (LRD) being a crucial component in scheduling the mission timeline. In order for the mission to leave when Earth is properly aligned with Venus, the team should plan accordingly. Therefore, to ensure readiness, 10 weeks before the required launch date of October 1st, 2029, the majority of the design and mission operations should be verified and validated. This will allow the team to ensure a smooth transition from the planning stage to action, leaving enough time to plan for potential risks and unexpected mission descopes. Instead of relying on the calendar year, we will be using a fiscal year based on the ideal alignment of plants for the exploration of Venus. This section outlines the rationale behind the proposed timeline, including the Basis of Estimate.

The Schedule Basis of Estimate (BoE) is meant to be a framework to form the mission timeline. It includes rules that define the project's scope and estimates technical feasibility, constraints, and plans around the LRD. Ground rules, drivers, and assumptions are critical to the foundation of the mission's Schedule BoE. According to the NASA Space Flight Program and Project Management Handbook, ground rules and assumptions (GR&A) are derived from mission requirements to drive the initial projects and program development. The NASA Schedule Management Handbook, NASA Space Flight Program and Project Management Handbook, and SEH 3.0 NASA Program/Project Life Cycle outline the importance for implementation of GR&A's and drivers that were derived for the mission. The following are GR&A that define the Schedule BoE:

- Assumptions

- The schedule makes the assumption that key subsystems meet all technical requirements and are fully tested, validated, and verified meeting technical maturity on schedule.
- The schedule makes the assumption that budget and amount of staff remains stable throughout the mission. Having the expected number of resources ensures that the timeline is feasible.
- There should be enough time allocated for testing verification and validation in order to find any issues that must be addressed before the LRD.
- Contingency plans must be established throughout the mission by assigning margin time for major milestones to ensure the system is prepared for the LRD.
- The assumption is made that major milestones of the missions are defined by the NASA Mission Life Cycle and customer requirements ("SEH 3.0 NASA Program/Project Life Cycle - NASA" 2019).

- Schedule Drivers

Drivers are factors that significantly impact the cost and schedule estimates throughout the mission. The following are drivers that define the Schedule BoE:

- The schedule accounts for risk management, including buffer periods to address technical risks and integration challenges throughout the process.
- The mission descope due to unforeseen engineering setbacks of the aerobot system's balloon structure has driven the mission schedule. The science measurements must be collected within the shortened 36-hour window of cruising altitude time in Venus' lower atmosphere.
- Per the L'SPACE MCA Project Management Skill Module, a margin period of ~10% the task group duration is applied for each milestone. This allows for unexpected delays to not compound into potential delays to customer schedule requirements.
- Schedule Margin Policy. NASA Schedule Management Handbook recommends a ten percent time margin for Discovery class critical path tasks. Engineering build activities stay the full ten percent, low risk documentation tasks carry five percent. Missed Level-1 milestones will be escalated to the Project Manager to replan by five working days.

The timeline shall emphasize work after the completion of Pre-Phase A, Phase A, and Phase B, as these phases represent pre-sunk work, and thus will not be included in the Gantt chart. The focus shall begin at Phase C (Preliminary Design), leading through Phase D (Final Design and Fabrication), Phase E (Launch, Operations, and Data Analysis), and Phase F (Mission Closeout). The mission schedule shall align with the specific requirements for these phases. The launch readiness date shall be determined by the mission's optimal launch window, which, as mentioned before, is dependent on the alignment of Earth with Venus. This date is critical and non-negotiable. If the mission is not ready by the established LRD, it would lead to postponing the mission until the next available launch window, resulting in a delay of up to 19 months, that may jeopardize mission success and resource allocation. As such, the schedule shall include buffer periods for contingency in order to mitigate risks that may be due to unpredicted system integration problems, testing, or environmental changes.

5.2.2. Mission Schedule

The following Gantt Chart details the timeline of crucial tasks to be completed by the core mission personnel after the preliminary design and development that occurs throughout Pre-Phase A, Phase A, and Phase B of the NASA Mission Life Cycle

assuming there are no delays to the mission. Pre-Phase A, Phase A, and Phase B are not included since they are considered pre-sunk development phases. The Gantt Chart outlines the tasks within Phases C (final design and fabrication), D (system integration, assembly, and test, and launch), and E/F (operations and sustainment, and closeout) of the Life Cycle. The schedule estimates and tasks for completing each mission milestone within the Gantt Chart were derived from the SEH 3.0 NASA Program/Project Life Cycle, L'SPACE Mission Concept Academy timelines for LCR completion throughout Pre-Phase A to Phase B, NASA Procedural Requirements (NPR) 7123.1 C, and the foundational schedule created from the team's Schedule BoE. These sources provided the framework for defining the lifecycle reviews, key decision points, and the schedule's logic.

The schedule requirements outlined by the customer for system integration readiness by October 1st, 2028 and March 1st, 2029 for the LRD were included in our schedule BoE and were critical drivers for the scheduling of surrounding milestones. To account for uncertainties, a schedule margin of approximately 10% was applied to major task groups, following the L'Space Project Management Skill Module. These margins were incorporated into the Gantt Chart as buffer periods in order to maintain schedule integrity in the case of delays due to component deliveries, subsystem integration, or unexpected engineering descopes. For example, the balloon material degradation risk identified during Phase-B required a 112-hour reduction in science operations, reducing the cruise time from 48 to 36 hours. This change ended up also affecting Phase E timelines and caused changes for upstream design decisions and test windows to preserve the mission's timeline and viability despite the compressed science window.

The milestones for each task group were derived directly from the SEH 2.0 NASA Program/Project Life Cycle for Robotic Mission Projects. Milestones throughout the mission include Life Cycle Reviews (LCRs), Key Decision Points (KDPs), SRB (Standing Review Boards), and other major mission events such as launch date, mission closeout, and science data collection ("NASA Procedural Requirements NPR 7123.1C" 2020).

ID#	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	MARGIN	PHASE C												PHASE D																																								
								2025												2026												2027												2028																
								J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D																	
3 Validation Instruments Testing								0%	2/15/25	9/2/25	200	195																																																
3.1	Bench test instrument functionality	Science	Not complete	2/15/25	3/15/25	29																																																						
3.2	Conduct thermal and pressure testing to simulate Venus-like	Engineering, Science	Not complete	3/16/25	4/16/25	32																																																						
3.3	Verify the spacecraft avionics and power systems	Engineering, Science	Not complete	4/17/25	5/17/25	1																																																						
3.4	Compile test results into a formal test report	Programmatics	Not complete	5/18/25	7/18/25	62																																																						
3.5	Prepare launch site checkout and post launch activation	All	Not complete	7/19/25	9/2/25	46																																																						
3.6	Schedule Margin			2/20/25	9/2/25	195																																																						
3.7	Validation Instruments Testing		Not complete	9/2/25	9/2/25	1																																																						
2 Critical Design Review (CDR)								0%	9/3/25	9/30/25	26	25																																																
2.1	Review and update system requirements in previous phases	Engineering, Science	Not complete	9/3/25	9/12/25	8																																																						
2.2	Finalize structural design with mission constraints	Engineering, Science	Not complete	9/13/25	9/22/25	8																																																						
2.3	Update and validate Science Traceability Matrix (STM) and Trade Study	Science	Not complete	9/23/25	9/30/25	6																																																						
2.5	Schedule Margin			9/4/25	9/30/25	25																																																						
2.6	◆ Critical Design Review (CDR)		Not complete	9/30/25	9/30/25	1																																																						
4 System Integration Review (SIR)								0%	10/1/25	2/16/26	139	5																																																
4.1	Finalize operational procedures	Programmatics	Not complete	10/1/25	10/31/25	31																																																						
4.2	Conduct training and mission launch simulation for all personnel	All	Not complete	11/1/25	12/30/25	60																																																						
4.3	Verify ground system performance	Engineering	Not complete	12/31/25	1/10/26	11																																																						
4.4	Review and finalize mission operation procedures and documentation	Programmatics	Not complete	1/11/26	2/10/26	31																																																						
4.5	Schedule Margin			2/11/26	2/15/26	5																																																						
4.6	◆ Operational Readiness Review (SRB)		Not complete	2/16/26	2/16/26	1																																																						
1 Key Decision Point D								0%	1/23/26	2/17/26	21	4																																																
1.4	Draft final bill of materials (BOM)	All	Not complete	1/28/26	2/10/26	14																																																						
1.5	Conduct Preliminary Design Review Meeting	All	Not complete	2/11/26	2/14/26	4																																																						
1.6	Schedule Margin			2/14/26	2/17/26	4																																																						
1.7	◆ Key Decision Point D		Not complete	2/17/26	2/17/26	1																																																						
5 Launch Vehicle Readiness Review								0%	2/17/26	6/12/27	481	62																																																
5.1	Confirm compatibility between Aerobot and launch machinery	Engineering, Science	Not complete	2/17/26	4/10/27	418																																																						
5.2	Assessments in electrical, mechanical, and system components	Engineering, Science	Not complete	2/18/26	4/18/26	60																																																						
5.3	Cycles of testing, modifications, re-testing (below)	Engineering	Not complete	4/19/26	4/10/27	357																																																						
	Conduct launch equipment performance review	Engineering	Not complete	4/20/26	4/11/27																																																							
	Verify launch vehicle tolerances and constraints	Engineering	Not complete	4/21/26	4/12/27																																																							
	Identify and resolve launch machinery risks and abnormalities	Engineering, Programmatics	Not complete	4/22/26	4/13/27																																																							
5.4	Schedule Margin			4/11/27	6/11/27	62																																																						
5.5	◆ Launch Vehicle Readiness Review		Not complete	6/12/27	6/12/27	1																																																						
6 Mission Readiness Review								0%	6/13/27	7/5/28	389	7																																																
6.1	Verify Aerobot's system performance and health	Engineering, Science	Not complete	6/13/27	8/13/27	62																																																						
6.2	Validate ground system readiness for operation	Engineering, Programmatics	Not complete	8/14/27	10/19/27	67																																																						
6.3	Confirm mission procedures and emergency protocols	All	Not complete	10/20/27	12/30/27	72																																																						
6.4	Conduct integration system test between Aerobot and ground control	Engineering, Science	Not complete	12/31/27	2/19/28	51																																																						
6.5	Identify and resolve machinery risks and abnormalities	All	Not complete	2/20/28	4/20/28	61																																																						
6.6	Perform risk analyses and mitigation planning	All	Not complete	4/21/28	6/28/28	69																																																						
6.7	Schedule Margin			6/28/28	7/4/28	7																																																						
6.8	◆ Mission Readiness Review		Not complete	7/5/28	7/5/28	1																																																						
7 Safety and Mission Success Review								0%	7/6/28	8/29/28	55	11																																																
7.1	Review and finalize documentation	Engineering, Programmatics	Not complete	7/6/28	7/10/28	5																																																						
7.2	Validate risk procedures and potential mission anomalies	Engineering, Science	Not complete	7/11/28	7/23/28	13																																																						
7.3	Ensure compliance with NASA mission and safety protocols	Programmatics	Not complete	7/24/28	7/31/28	8																																																						
7.4	Conduct final risk assessment	All	Not complete	8/1/28	8/7/28	7																																																						
7.5	Validate risk procedures and potential mission anomalies	All	Not complete	8/8/28	8/17/28	10																																																						
7.6	Schedule Margin			8/18/28	8/28/28	11																																																						
7.7	◆ Safety and Mission Success Review		Not complete	8/29/28	8/29/28	1																																																						
8 Mission Readiness Board Review								0%	8/30/28	10/1/28	33	5																																																
8.1	Compile and analyze Aerobot performance results	Engineering, Programmatics	Not complete	8/30/28	9/4/28	6																																																						
8.2	Conduct final system evaluation	Engineering	Not complete	9/5/28	9/25/28	21																																																						
8.3	Schedule Margin			9/26/28	9/30/28	5																																																						
8.4	◆ System Integration Date Constraint		Not complete	10/1/28	10/1/28	1																																																						
9 Mission Readiness Board Review								0%	10/2/28	11/7/28	37	3																																																
9.1	Review and finalize launch risk assessment	All	Not complete	10/2/28	10/29/28	28																																																						
9.2	Confirm mission operation readiness	Programmatics	Not complete	10/29/28	11/3/28	6																																																						
9.3	Schedule Margin			11/4/28	11/6/28	3																																																						
9.4	◆ Mission Readiness Board Review		Not complete	11/7/28	11/7/28	1																																																						

[illegible]

Table 9: Gantt Chart showing the tasks for each phase, respectively

5.3. Budget

5.3.1. Budget Basis of Estimate

The Budget Basis of Estimate delivers a design framework for the Aerobot System for the L'SPACE Mission Concept Review. The intention of this document is to provide the ground rules, cost estimates, assumptions, and drivers guiding the financial planning for this part of the mission phase. The estimate is assembled to guarantee that the project abides to NASA's cost management principles while still sustaining value within the \$200 million budget restriction. Phases A and B are considered sunk costs and are not included. Cost estimates presented in this document take in consideration for development, testing, integration, launch preparation (which is covered by a contractor and excludes the rocket launch vehicle), mission operations, and contingency reserves, as well as outreach, travel, personnel, and direct costs. When developing this mission budget, the four required cost categories are personnel costs, direct costs, travel costs, and outreach costs. The budget allocated was determined through NASA's Cost Estimating Handbook (CEH). This ensures the budget is realistic and aligns with past missions such as DAVINCI+ and VERTIAS (Historical Venus Mission Data).

Ground Rules, Assumptions, and Cost Drivers

The budget estimate applies to Phases C-F of the mission life cycle. The budget for the Aerobot System mission is capped at \$200 million, launch vehicle, EDI system, and descent system costs are excluded as it is covered by external contractors. Cost sharing agreements must be documented properly to maintain integrity. A contingency reserve of 10% is applied to categories that mitigate unforeseen costs due to high risk. This reserved percentage is aligned with NASA's recommended contingency range of 10-30%, which is stated in the NASA Cost Estimating Handbook (CEH). This reserve percentage was based on cost overruns observed in similar planetary exploration missions such as DAVINCI+ and VERITAS (Historical Venus Mission Data).

Few assumptions impact the cost structure of this mission and are accounted for in this budget estimate. A 2.5% annual inflation rate is assumed and aligns with NASA's New Start Inflation Index (NNSI) (CEH). This rate will affect long term projections of cost, multi-year personnel contracts and procurement of specialized materials where price adjustments overtime will need to be considered. Stating this rate is an example of how we address inflation as an assumption in our basis of estimate.

The Aerobot System is predicted to start at TRL 4-5, this will require extra maturation investment. The cost is factored into the system development margin (39%) ensuring that additional testing and design processing do not exceed the allocated

funding (CEH). Meeting TRL requirements and scientific objectives also drives the schedule, influencing manufacturing and testing timelines.

The workforce crucial to this mission includes scientists, engineers, technicians, administrative staff and project managers. Salaries are estimated from the data provided in the Stakeholder Requirements. Scientists and engineers salaries are projected at \$80,000 per year, technicians and administrative support at \$60,000 per year and project managers at \$120,000 per year. These costs directly affect the personnel budget category and are accordingly incorporated.

The lifespan of the Aerobot System is expected to be mission defined with the main data collection lasting a period that is aligned with mission goals, mission control and data processing will go on beyond the main operations (Stakeholder Requirements). This will impact long-term operational and personnel costs. The Aerobot system must be ready for integration by October 1, 2028, in preparation for the launch on March 1, 2029. This timeline will affect testing and manufacturing schedules which will increase costs related to component procurement and staffing (Stakeholder Requirements). Radioisotope Thermoelectric Generators will not be used. This will require an alternate power solution such as battery storage systems or solar based and will increase research and development costs (Historical Venus Mission Data).

Venus's high temperatures of 475°C and corrosive sulfuric acid clouds require advanced thermal and material engineering, this will increase the costs (Historical Venus Mission Data). Scientific instruments have to be lightweight, durable and able to function in extreme conditions. This will drive up the development and testing cost (Stakeholder Requirements). Precise environmental and vacuum chamber testing will be required to confirm system performance under conditions that are similar to Venus' (CEH).

Real time communication will require autonomous power. This will increase software and AI development costs (PMH). The Aerobot System has to interface with the Descent system and EDI system, this will require extensive validation (Stakeholder Requirements).

The budget is aligned with NASA's cost estimation practices, making sure that each category reflects planetary mission expenditures. System development will account for the largest portion of the budget at 39% this reflects on the complication of designing a spacecraft able to withstand Venus' harsh conditions. Testing and integration is allocated at 14% due to the need for long term testing. Launch preparation is allocated at 10%, this will cover transportation and logistical expenses. Mission operations are allocated at 12%, this will fund ground control, data processing and communication infrastructure. Personnel costs account for 15% taken from the workforce salary estimates. Outreach and travel are included at 5% covering team related travel and public engagement. A 10% contingency reserve has been set aside as a risk mitigation measure which aligns with NASA's recommendations.

Cost Category	Estimated Cost (\$Million)	Margin (%)	Notes
System Development	\$78 Million	39%	Spacecraft Design, subsystems and fabrication
Testing and Integration	\$28 Million	14%	Environmental testing, Integration and validation
Launch preparation	\$20 Million	10%	Preflight handling, Logistics and transport
Mission Operations	\$24 Million	12%	Ground control, data processing and communication infrastructure
Personnel Costs	\$30 Million	15%	Salaries from the mission task document
Reserves	\$20 Million	10%	Unexpected costs
Total	\$200 Million	100%	Mission constraints

Table 33: Cost Estimation and Margins

Total | \$200 Million | 100% | Mission constraints

The budget is aligned with NASA's cost estimation practices, making sure that each category reflects planetary mission expenditures. System development will account for the largest portion of the budget at 39%—this reflects on the complication of designing a spacecraft able to withstand Venus' harsh conditions. Testing and integration is allocated at 14% due to the need for long term testing. Launch preparation (10%) covers transportation and logistical expenses only, since the actual launch vehicle is contracted out. Mission operations are allocated at 12%, this will fund ground control, data processing and communication infrastructure. Personnel costs account for 15%, taken from the workforce salary estimates. Outreach and travel are included at 5% covering team related travel and public engagement. A 10% contingency reserve has been set aside as risk mitigation, aligning with NASA's recommendations (CEH).

5.3.2. Total Mission Cost

The following table dictates the total cost of this mission:

Additional Information							
	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6	
Science Personnel:	3	3	3	3	6	6	
Engineering Personnel:	10	10	10	10	12	12	
Technicians:	15	15	15	15	10	10	
Administration Personnel:	7	7	7	7	7	7	
Management Personnel:	4	4	4	4	4	4	
NASA L'SPACE Mission Concept Academy Budget - <i>Venus Aerobot</i>							
Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
PERSONNEL							
Science Personnel	\$ 240,000	\$ 246,240	\$ 252,480	\$ 258,720	\$ 529,920	\$ 542,400	\$ 2,069,760
Engineering Personnel	\$ 800,000	\$ 820,800	\$ 841,600	\$ 862,400	\$ 1,059,840	\$ 1,084,800	\$ 5,469,440
Technicians	\$ 900,000	\$ 923,400	\$ 946,800	\$ 970,200	\$ 662,400	\$ 678,000	\$ 5,080,800
Administration Personnel	\$ 420,000	\$ 430,920	\$ 441,840	\$ 452,760	\$ 463,680	\$ 474,600	\$ 2,683,800
Project Management	\$ 480,000	\$ 492,480	\$ 504,960	\$ 517,440	\$ 529,920	\$ 542,400	\$ 3,067,200
Total Salaries	\$ 2,840,000	\$ 2,913,840	\$ 2,987,680	\$ 3,061,520	\$ 3,245,760	\$ 3,322,200	\$ 18,371,000

Total F&A	\$ 1,828,333	\$ 1,828,333	\$ 1,828,333	\$ 1,828,333	\$ 1,828,333	\$ 1,828,333	\$ 10,970,000
Total Projected Cost	\$ 24,668,157	\$ 24,924,761	\$ 25,189,445	\$ 25,474,740	\$ 25,945,546	\$ 26,247,658	\$ 152,450,308
Total Cost Margin	\$ 777,719	\$ 817,905	\$ 859,116	\$ 900,811	\$ 1,019,319	\$ 1,067,314	\$ 5,442,184
	3.2%	3.3%	3.4%	3.5%	3.9%	4.1%	
Total Project Cost	\$ 24,668,157	\$ 24,924,761	\$ 25,189,445	\$ 25,474,740	\$ 25,945,546	\$ 26,247,658	\$ 152,450,308
F&A %	10%	10%	10%	10%	10%	10%	
ERE - Staff	28%	28%	28%	28%	28%	28%	
Inflation Rate	0.0%	2.6%	5.2%	7.8%	10.4%	13.0%	
Science Personnel Salary	\$ 80,000						
Engineering Personnel Salary	\$ 80,000						
Technicians Salary	\$ 60,000						
Administration Personnel Salary	\$ 60,000						
Project Management Salary	\$ 120,000						

Direct Costs

<i>Subsystem</i>	<i>Estimated Cost Range</i>
Power	\$354,000
Thermal	\$210,000 - \$300,000
Mechanical	\$29,000,000
Payload	\$60,000,000 – \$80,000,000

Command and Data Handling	TBA
<i>Total</i>	\$89,600,000 - \$109,700,000

5.3.3. Personnel Budget

	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6	
Science Personnel:	3	3	3	3	6	6	
Engineering Personnel:	10	10	10	10	12	12	
Technicians:	15	15	15	15	10	10	
Administration Personnel:	7	7	7	7	7	7	
Management Personnel:	4	4	4	4	4	4	
NASA L'SPACE Mission Concept Academy Budget - Venus Visionaries							
Mission Phase	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Cumulative Total
PERSONNEL							
Science Personnel	\$ 240,000	\$ 246,240	\$ 252,480	\$ 258,720	\$ 529,920	\$ 542,400	\$ 2,069,760
Engineering Personnel	\$ 800,000	\$ 820,800	\$ 841,600	\$ 862,400	\$ 1,059,840	\$ 1,084,800	\$ 5,469,440
Technicians	\$ 900,000	\$ 923,400	\$ 946,800	\$ 970,200	\$ 662,400	\$ 678,000	\$ 5,080,800
Administration Personnel	\$ 420,000	\$ 430,920	\$ 441,840	\$ 452,760	\$ 463,680	\$ 474,600	\$ 2,683,800
Project Management	\$ 480,000	\$ 492,480	\$ 504,960	\$ 517,440	\$ 529,920	\$ 542,400	\$ 3,067,200
Total Salaries	\$ 2,840,000	\$ 2,913,840	\$ 2,987,680	\$ 3,061,520	\$ 3,245,760	\$ 3,322,200	\$ 18,371,000

	0	0	0	0	0	0	
	\$	\$	\$	\$	\$	\$	
Total ERE	792,644	813,253	833,861	854,470	905,892	927,226	\$ 5,127,346
Personnel	\$	\$	\$	\$	\$	\$	
Margin	757,227	797,114	838,026	879,961	1,003,988	1,051,834	\$ 5,328,150
TOTAL	\$	\$	\$	\$	\$	\$	
PERSONNEL	4,389,871	4,641,836	4,901,865	5,170,035	5,691,826	5,990,424	\$ 30,785,858

Table 34: Personnel Cost

As seen above, the personnel budget closely allocates the costs necessary for the individuals that are working on the mission directly. This includes Scientists, Engineers, Technicians, Administration, and Management. In the table, only Phases C, D, E, and F are listed, however, these values were derived from the initial values in Phase A and Phase B. Meaning, these phases started with 5 scientists, 8 engineers, 5 technicians, 7 admins, and 4 managers. The current values of Phase A and Phase B were extrapolated from the current standing and positions of Venus Visionaries' team structure. Venus Visionaries has been able to complete all tasks and deliverables in a timely manner leading it to believe that it has the suitable organizational structure.

According to the mission constraints, it is recommended that the project operates amongst 30 to 50 people. However, the number of people does not stay the same as it changes depending on the phase of the mission. For instance, in Phases C and D there is a decrease in scientists, but increase in engineers and technicians. Scientists will decrease from 5 to 3 due to the science objectives being established in the earlier fiscal years. Phases C and D will primarily focus on the development of the mission itself including creation of the final designs, manufacturing the Aerobot System, and running a series of tests and launches. Given this reasoning, engineers will increase from 8 to 10 and technicians will increase from 5 to 15. If these two groups increase, this will provide a more seamless transition between the phases and preparation of the Aerobot System.

Furthermore, Phase E focuses on operations and sustainment, more specifically "conduct the prime mission to meet the initially identified need and to maintain support for that need" while Phase F is the closeout stage where data analysis occurs and confirmation of system functionality is in good shape. During these phases, there will be an increase in scientists and engineers of 6 and 12 because these roles are responsible

for conducting analysis on the gathered data and ensuring that the systems and instruments are performing the intended purpose. There will be a decrease in technicians because the Aerobot system has been launched meaning there is no hands-on work to be conducted.

Overall, throughout the phases there are changes in the number of each subteam, however, there will be roughly 39 people working on the mission. In addition, it will be assumed that all personnel involved in the mission are full-time employees (FTE).

In the table, it is found that the salary for each role is listed and these are predetermined from the mission constraints. As stated, scientists and engineers earn \$80,000 per year, technicians and admin earn \$60,000 per year, and managers earn \$120,000 per year. Over the span of 6 fiscal years, there will be an arbitrarily determined 20% personnel margin per year of the total personnel cost which is suggested to be approximately \$30,785,858.

5.3.4. Travel Budget

	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6
Science Personnel:	3	3	3	3	6	6
Engineering Personnel:	10	10	10	10	12	12
Technicians:	15	15	15	15	10	10
Administration Personnel:	7	7	7	7	7	7
Management Personnel:	4	4	4	4	4	4

Total Flights Cost	\$ 29,200	\$ 29,200	\$ 29,200	\$ 27,924	\$ 8,586	\$ 8,586	\$ 132,696
Total Hotel Cost	\$ 20,325	\$ 20,325	\$ 20,325	\$ 19,305	\$ 10,368	\$ 10,368	\$ 101,016
Total Transportation Cost	\$ 5,250	\$ 5,250	\$ 5,250	\$ 15,600	\$ 4,860	\$ 4,860	\$ 41,070
Total Per Diem Cost	\$ 6,686	\$ 6,686	\$ 6,686	\$ 10,725	\$ 4,860	\$ 4,860	\$ 40,502
Travel Margin	\$ 11,492	\$ 11,791	\$ 12,090	\$ 11,850	\$ 6,331	\$ 6,480	\$ 60,034

Total Travel Costs	\$ 72,953	\$ 75,156	\$ 77,375	\$ 92,066	\$ 38,646	\$ 39,724	\$ 395,919

Travel costs related to the mission for reasons such as launch attendance, inspection of manufacturing facilities, and meeting with administration were determined in the table above. The basis of assumption for these travel costs were from the primary city of Charlotte, North Carolina. In the first three fiscal years encompassing Phase C, trips were made to Pasadena, California and Rome, Italy to meet with representatives from the Jet Propulsion Laboratory (JPL) and Leonardo Systems respectively to discuss and inspect the manufacturing of parts for the Aerobot System.

The flight costs from Charlotte to Los Angeles and Charlotte to Rome are \$424 and \$800. The duration of the stay was determined to be 3 days per visit per person alongside hotel costs of \$191 and \$200 per night. The Per Diem cost is taken from the U.S. General Services Administration and the Department of State were determined to be \$64.50 and \$61.60. For transportation costs, the average costs of an Uber roundtrip for Los Angeles and Rome are \$40 and \$75.

The flight costs from Charlotte to Orlando are approximately \$358 round trip. The duration of the stay was determined to be 5 days per visit per person alongside hotel costs of \$143 per night. The Per Diem cost was taken from the U.S. General Services Administration and was determined to be \$55. For transportation costs, the average costs of an Uber roundtrip for Orlando is \$80.

The flight costs from Charlotte to Houston was \$159. The duration of the stay was determined to be 3 days per visit per person alongside hotel costs of \$128 per night. The Per Diem cost was taken from the U.S. General Services Administration and turned out to be \$60. For transportation costs, the average cost of an Uber roundtrip in Houston is \$60.

All these costs added together plus a travel cost margin of an arbitrarily determined 20% sums to \$375,879.

5.3.5. Outreach Budget

The overall outreach budget will be split up into five categories for programs described in the Outreach Plan. The budget was estimated with standard costs of industry-standard items, pricing from similar NASA outreach efforts, and commercial market rates.

The Outreach Training is a 10-week program with \$5,000 set aside for training materials and supplies. These supplies can include printed handouts, digital resources, etc. Facility rentals have a \$3000 budget assuming sessions will be hosted weekly at a mid-range venue. \$2000 are for guest speakers.

For the “On the Way to Venus” Tour and University Outreach, a tour bus rental would cost \$60,000. Chartered tour buses cost around \$1,000 a day according to Bus.com and rates can be negotiated. A bus could cover 15,000 miles and typically have a fuel efficiency of 6 mpg. At \$4.00 a gallon, this would be \$10,000 per bus and we’re looking to take 2 buses. Lodging and meals for the outreach team throughout the entire trip would have a \$180,000 budget. \$50,000 is set aside for aerobot kits to be distributed to 10,000 students, aiming to keep the cost per kit at \$5. Promotional materials will cost an additional \$30,000.

For conferences and public engagement, booth rentals at six major conferences will cost \$30,000 and an additional \$30,000 will cover 10 people’s travel and lodging. \$15,000 shall go towards hosting speaker events.

Advertising and media outreach will have \$30,000 for billboards in urban areas, newspapers and ads will cost around \$15,000, and \$50,000 will be for social media and digital ad campaigns. More money will be focused in the digital sector due to how prominent we rely and consume digital media on a day to day, aiming to be a more effective advertising technique. Additionally, a website will be created for this mission and all these efforts combined shall be designed to maintain public interest and engagement leading up to the launch date. Website development and maintenance has a budget of \$20,000.

The following table outlines the Outreach Budget:

Category	Estimated Cost (\$)	Description
Outreach Team Training	10,000	Facility rentals
K-12 and University Tour	340,000	Bus rental, fuel, lodging, STEM kits
Conferences and Public Engagements	75,000	Rentals, travel, and speaker events
Advertising and Media Outreach	115,000	Billboard, digital ads, newspapers, website
Misc./Contingency	54,000	Unforeseen circumstances

Total Est. Budget	594,000	
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Table 35: Outreach Budget Outline

In the best case scenario, the budget is designed with the intention of being more than enough to cover the necessities of hosting and conducting such events. In the worst case scenario, around 10% of the total estimated budget is set aside for miscellaneous and contingency costs to account for cost fluctuations and to provide financial flexibility. When considering where costs may be reduced, we can potentially save ~\$150,000 through the following conditions:

- If we reduce the number of K-12 and university visits to 30 per leg, that would save approximately \$100,000.
- Scaling down the amount of billboards and newspaper ads to only the most densely populated metro areas would save \$20,000.
- Combining outreach conferences and events with ones hosted/sponsored by NASA would save \$25,000 on standalone booths.
- Conducting a portion of the outreach training virtually has the potential of saving \$5,000.

OUTREACH							
Total Outreach Materials	\$ 5,833	\$ 5,833	\$ 5,833	\$ 5,833	\$ 5,833	\$ 5,833	\$ 35,000
Total Outreach Venue Costs	\$ 500	\$ 500	\$ 500	\$ 500	\$ 500	\$ 500	\$ 3,000
Total Outreach Travel Costs	\$ 46,667	\$ 46,667	\$ 46,667	\$ 46,667	\$ 46,667	\$ 46,667	\$ 280,000
Total Outreach Services Costs	\$ 19,167	\$ 19,167	\$ 19,167	\$ 19,167	\$ 19,167	\$ 19,167	\$ 115,000
Total Outreach Personnel Costs	\$ 12,500	\$ 12,500	\$ 12,500	\$ 12,500	\$ 12,500	\$ 12,500	\$ 75,000
Outreach Margin	\$ 9,000	\$ 9,000	\$ 9,000	\$ 9,000	\$ 9,000	\$ 9,000	\$ 54,000
Total Outreach Costs	\$ 93,667	\$ 96,102	\$ 98,537	\$ 100,973	\$ 103,408	\$ 105,843	\$ 598,530

5.3.6. Direct Costs

Direct Costs

This section provides a breakdown of the estimated direct costs associated with each primary system of the Aerobot. This section includes descriptions about the Power, Thermal, Mechanical, Payload, and Command and Data Handling subsystems. Each estimate will also reflect research about comparable missions, historical data, and hardware costs including manufacturing, testing, and integration considerations.

Power Subsystem

The Power subsystem encompasses energy generation, management, storage, and distribution to other systems as well. For this mission, the solution combines advanced battery storage with a PMAD (Power Management and Distribution) architecture, adapted from CubeSat heritage systems (MarCo from 2018).

- **Battery Storage:** For battery storage, the cost is estimated at \$4,000 for a 4kWh capacity (\$1,000/kWh). This depends on the selected lithium-ion batteries based on contemporary costs for space-rated and mission dependent energy storage criteria.
- **PMAD System:** For the PMAD system, the associated costs are estimated at approximately \$300,000. This estimate is derived from the MarCO CubeSat mission in 2018. This mission utilized a similarly scaled PMAD system and as a result, the price point reflects necessary enhancements in fault-tolerance and redundancy.
- **Testing and Assembly:** For testing, the radiation hardening and validation testing for the batteries is estimated at around \$10,000. For final assembly labor and QA/QC, the expected amount to be spent is around \$40,000. Cost elements for battery strings and the PMAD board were estimated with the lithium-ion and regulated-bus cost estimating relationships in the NASA Cost Estimating Handbook (CEH v4).

All in all, the total estimated direct costs associated with the majority of the aspects of the Power Subsystem are slated to be \$354,000.

Thermal Subsystem

The Thermal Subsystem is a crucial part of the Aerobot as it must regulate internal temperatures against Venus' extreme environmental conditions where temperatures reach almost 475°C. The system includes insulation, heat pumps, thermostats, and phase change materials, and is based on costs adapted for aerospace applications. Among other costs, the primary costs accounted for from this subsystem come from the Phase Change Material (PCM), space-rated fans, heat pump assembly, insulation, thermostats, and testing and integration.

- **Phase Change Material (PCM):** For the PCM, we selected $\text{LiBO}_2 \cdot 8\text{H}_2\text{O}$. This material is based on thermophysical properties from Purdue research and has a material cost of up to \$32.2/kg. For this application however, this cost is nearly negligible as only 0.176kg of this material is needed, bringing the cost to less than \$6. However, \$2,000 is allocated for custom containment development.
- **Space-rated Fans:** The space-rated fans are crucial for circulating internal air and dissipating hot spots in the Aerobot. Each fan is estimated at around \$5,000 each and it is planned that 4 of these fans will be used. This brings the total cost for fans to \$20,000.
- **Heat Pump Assembly:** Due to the nature of this mission, it has been decided that the heat and cooling pump assembly will be entirely custom built. As this is a custom job, it is difficult to accurately predict the expenditure but it is likely to cost around \$100,000+. This estimate also includes modeling, material sourcing, and thermal testing infrastructure for developing the assembly.
- **Insulation:** Insulation is crucial to the Aerobot to maintain system health of other subsystems and to ensure mission success. For this application, foam insulation adapted for spacecraft has been chosen. This has a 100x cost multiplier vs. consumer-grade insulation and is priced at \$450/ft². For the mission, approximately 6 m² of coverage is needed which brings the insulation cost to around \$30,000.
- **Thermostats:** For this part of the Aerobot, aerospace-graded Honeywell-style thermostats have been chosen to be used. Each is priced at \$4,500 (it has been scaled from \$45 retail units) and 12 units are projected to be used. This brings the cost to around \$54,000.
- **Testing & Integration:** For testing and integration for all parts of the Aerobot, rigorous validation will be required due to the criticality of thermal performance (as it plays a major role in prolonging the mission timeline). As this is difficult to exactly estimate, a placeholder budget of around \$100,000–\$150,000 has been

set aside for vacuum chamber testing, simulation, and thermal vacuum cycles among other tests.

All in all, the total estimated direct costs associated with the majority of the aspects of the Thermal Subsystem are slated to be between \$210,000–\$300,000.

Mechanical Subsystem

The Mechanical subsystem is the backbone of the Aerobot and plays a major role in the structural design, deployment systems, and mobility elements as well as ensuring integration between all engineering subsystems and science systems is efficient and effective. As it is a very large subsystem, it is difficult to directly pinpoint costs like in previous sections so more board estimates are used. Component masses and material classes match the ADVENTS longerons and skin panels within a give or take 8 percent, satisfying the analog selection requirements in the Space Flight Program & Project Management Handbook

- **NASA's ADVENTS Aerobot Study:** Based on NASA's ADVENTS Aerobot study conducted in 2023, it was seen that similar mechanical systems tailored for Venus which also included structural design, aerobot fabrication, and subsystem integration cost around \$29 million.

All in all, the total estimated direct costs associated with the majority of the aspects of the Mechanical Subsystem are slated to be around \$29 million.

Payload Subsystem

The Payload Subsystem is a crucial system that houses all scientific instruments focusing on atmospheric analysis and surface imaging onboard as well as ensures they remain safe and protected from external forces and circumstances (to an extent to endure they are able to function properly without being damaged). This system encompasses the VIRTIS and IRLTS/TLS systems among other instruments.

- **VIRTIS:** The VIRTIS is estimated to cost around \$30 million. This is based on numbers from similar instruments and the total costs of previous missions it was

on.

- **IRLTS/TLS:** The Infrared Tunable Laser Spectrometer (IRLS/TLS) is considered a heritage instrument as it was used in the Curiosity mission to Mars. As a result, the research used to develop this system can be restructured to fit a Venus mission profile as well, helping save time and costs. The associated cost, including possible refurbishment of lab spares, calibration, and integration, is projected to be between \$30–\$50 million.

All in all, the total estimated direct costs associated with the majority of the aspects of the Payload Subsystem are slated to be between \$60–\$80 million.

Command and Data Handling

The CDH subsystem is still under active development and system selection. As of this stage in the mission concept, the final architecture and components for CDH have not been finalized. Therefore, a precise cost estimate is not yet available.

The team is currently evaluating multiple CDH solutions based on prior CubeSat missions and Venus-adapted avionics packages. These options vary significantly in terms of complexity, autonomy, and environmental tolerance, which will directly influence the final cost. Once a system is selected, cost estimates will be derived using parametric models and CERs from missions with similar mission profiles and operating conditions.

Applying digital avionics CER-A5 and FPGA processor CER-A8 from the NASA Cost Estimating Handbook gives a median CDH hardware cost of around \$ 3.9 million. An additional 25 percent equivalent of \$0.98 million covers the EMI/EMC screening and environmental testing, this gives the CDH direct cost of around \$ 4.9 million. Cost-Growth Margin (CEH v4). Schedule and cost margins follows the CEH guidance closely, direct hardware lines carry a 50 percent cost growth allocation (CEH v4). \$ 52 million is placed in management reserve, which keeps the mission total below the \$ 200 million restriction.

<i>Subsystem</i>	<i>Estimated Cost Range</i>
Power	\$354,000
Thermal	\$210,000 - \$300,000
Mechanical	\$29,000,000
Payload	\$60,000,000 – \$80,000,000
Command and Data Handling	TBA
<i>Total</i>	\$89,600,000 - \$109,700,000

5.4. Scope Management

5.4.1. Change Control Management

Making proper note of changes is crucial to the success of the team since all members must be aware that a change has occurred and other systems can be revised accordingly. Throughout the formation of the mission's MDR, decisions and changes are being tracked through a document which can be accessed and edited by every team member. If a team member would like to request changes, there must be research and explanation that will support the new change. If adequately provided, the next step will be to follow up with the sub-team leader and if approved, must also go through a democratic voting process with the rest of the team to make sure the provided research and explanations will benefit the mission. Below are images of the live changes document Venus Visionaries has implemented. The first image outlines how to enter the process of suggesting a change. The second image shows an example of Programmatics and the overall mission's subteams proposing changes. The Engineering and Science subteams have identical tables to the ones seen in the second image.

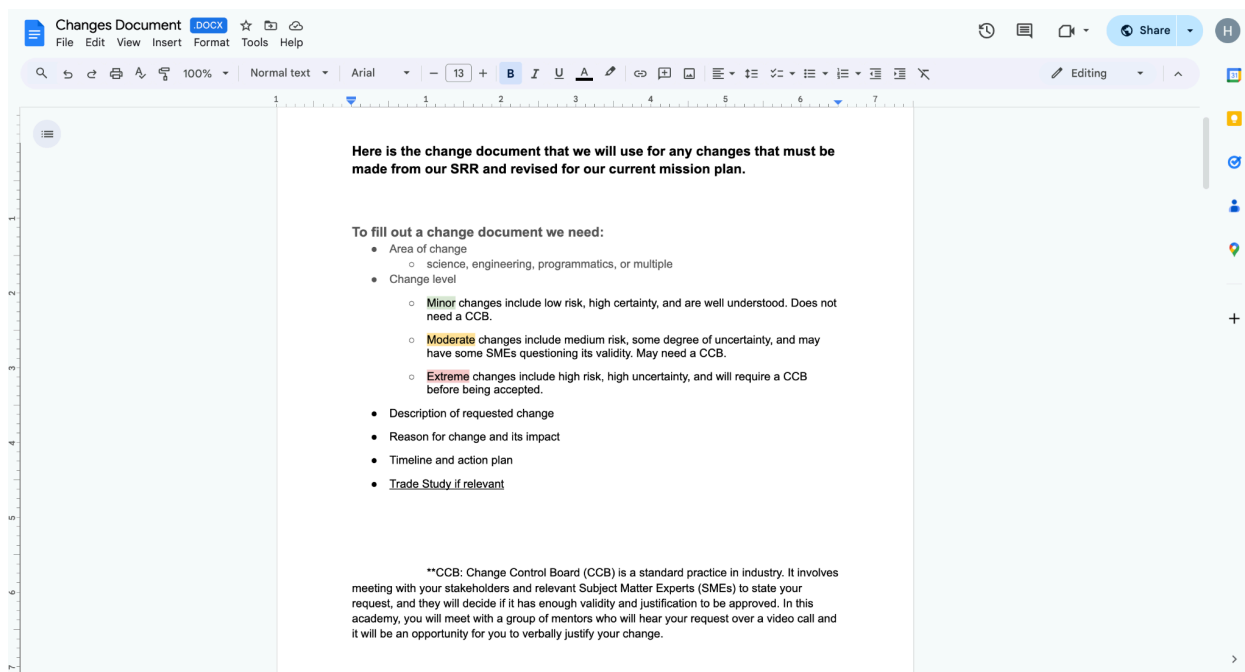


Figure 13: Introduction of Changes Document

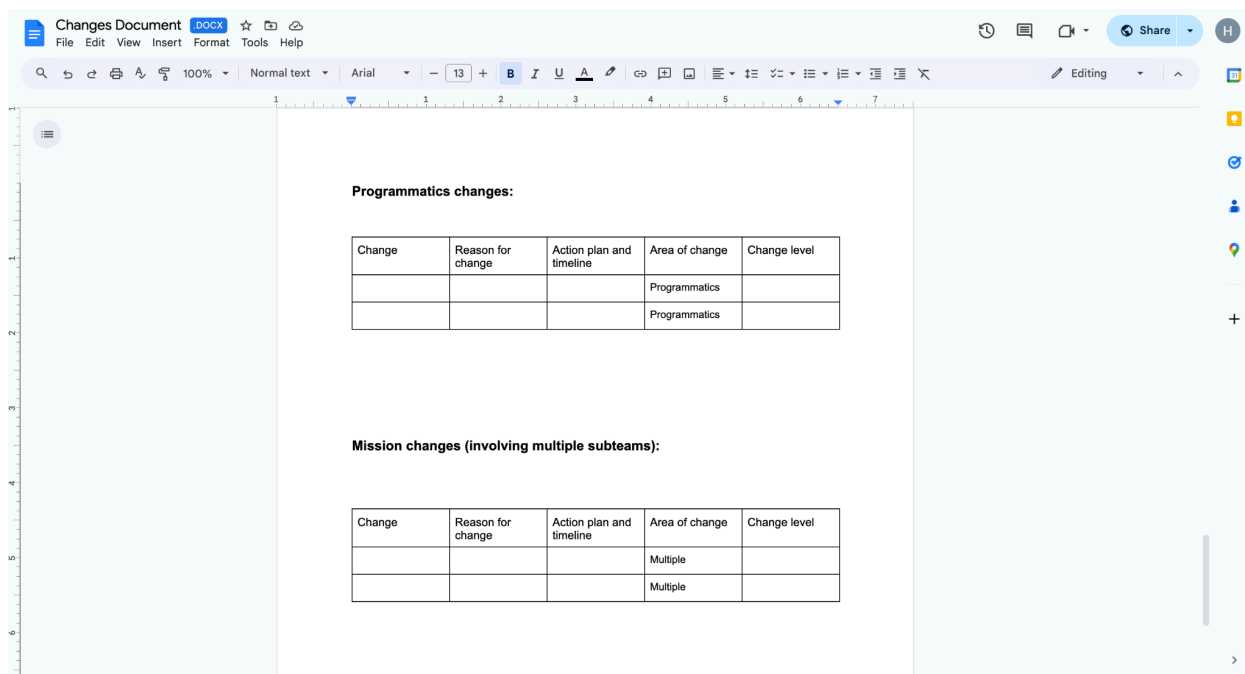


Figure 14: Changes Tables per Subteam

To be precise, after the voting takes place, the process in addressing internal changes is more detailed and several steps must be taken. After evaluating how the potential change can be beneficial to the overall mission, the team member(s) will need to discuss with the subteam lead to go to the Change Control Board (CCB) upon completion of the Change Request Form (CRF). The process of the CCB must be communicated through Discord clearly to the rest of Venus Visionaries to ensure a smooth flow of the mission's development. When changes are potentially being made, communication within the subteams must occur every few days with updates/questions/comments/concerns as this will allow all team members to be on the same page. Team leads may choose to set a brief meeting about any specific changes for utmost clarity amongst all subteam-specific members.

Additionally, the team's overall goal is to learn a bit more from each deliverable and build upon it in future assignments. That being said, Venus Visionaries' weekly meetings will consist of discussing the RFAs and ADVs that were given in the feedback report. This will be crucial for the team's learning as it not only discusses how to strengthen certain aspects of the project, but also allow the team to view the deliverables with a fresh perspective. RFAs must be taken seriously as they will put the deliverables on the proper track towards success. ADVs must be carefully thought through because they can heavily clarify certain components and increase readability.

Furthermore, usage of Discord for discussion purposes will be heavily requested as this is the main form of communication. Specifically, when implementing new changes, it is crucial that each team member thinks outside the box and develops new strategies to improve the score of the prior deliverable. After brainstorming ideas, it will be important to conduct research and find valuable explanations as to why this change will be beneficial for the overall team. Additionally, to make sure everyone on the team agrees, there will be a voting system where the majority vote means the new idea will be added into the deliverable. In a time sensitive scenario, team leads and the Project Manager will utilize Discord to discuss potentiality to any new idea and go forward with it immediately. The tasks will be delegated based on present availability.

To ensure organization, any requested changes or new implementations be recorded in a Google Document. Within the Google Document, it must be neatly formatted with the team member's name, idea, research, and reasoning as to why it will be best for the team to consider this change. In the future, this document may be shifted into a Google Spreadsheet as this can precisely show the thought process for a possible solution to reach success.

Lastly, decision making is a crucial component of Venus Visionaries' current task, so many factors will need to be considered when working towards the final goal. For example, Venus Visionaries will heavily prioritize budgets and risks for the entirety of the mission. This is because there is a strict \$200 million budget that cannot be overstepped. There are several moving parts, so it is important that the team has proper communication and allocation regarding the budget to remain below the constraint. In addition, the budgets and risks work in tandem due to only having so many resources. It will be key to make responsible decisions by clearly thinking through the benefits of a possible change that will benefit the Aerobot System as an end result.

5.4.2. Scope Control Management

In the case that a scope change is necessary, we as a team must have an outlined plan of action to account for such scope changes. Our goal is to implement strategies that ensure we meet all the technical and operational requirements for this mission while staying within budget, schedule, and technical constraints.

To start off, we establish a baseline and a change control process. This mission began with defining clear missions objectives, deliverables, and performance metrics. As such, any potential scope changes must go through a formal review process that involves the engineering, science, and programmatic team as a whole to discuss the best course of action. Any proposed changes shall be assessed based on technical feasibility, impact on our cost budget, schedule implications, and how it will affect any mission goals before a scope change can be approved.

A strategy we implement includes a comprehensive risk assessment in order to determine how changes will impact mission objectives. Every scope change decision must include a risk assessment matrix to quantify the likelihood and impact, such that mitigation efforts can be prioritized effectively. In the case that budget constraints get compromised, we shall explore alternative suppliers, purchase standardized components, or take on more of the labor in-house to reduce costs without compromising performance. If the schedule is at risk and delays occur, we shall reevaluate if we are placing enough and/or the proper personnel for the right tasks. Additional personnel may be allocated to subsystems that are in a more critical phase that may be blocking other subsystems. As a last resort subsystem modifications and instrument removals will be considered and extensively studied and discussed with technical experts to see if the overall aerobot will still meet all system objectives. Non-critical subsystems may be simplified or modified with lower-cost materials to allocate more resources to more critical instruments. This descope plan emphasizes

risk-based decision making. By ensuring that we continuously monitor any potential risks, we can make educated scope decisions early on and have it affect the timeline as little as possible.

In the case that more resources become available for the mission to use, we shall prioritize improvements to the system that will provide the highest return on investment. These will include adding redundancy to systems we are not as confident in, improving less critical subsystems to ensure broader data collection, improving frequency of data collection for more accurate research, and extending the mission timeline to allot time for more extensive testing. Any scope increase will be thought out as carefully as descope considerations. A trade study shall be conducted on any changes made due to a scope increase to determine if we are allocating any additional resources in the most efficient way possible.

As the timeline progresses, it is important that we analyze historical data from similar Venus exploration missions to gauge realistic budget, schedule, and performance expectations. It is also important that thorough review is conducted throughout all phases of the mission and early-stage testing allows us to validate and verify system design choices and identify any risks or changes to be made before committing to a full-scale implementation.

5.5. Outreach Summary

A crucial step to take throughout mission development is generating an outreach plan that will not only notify the public of the Aerobot mission to Venus but also make the people excited and intrigued in how revolutionary the collected data will be. It will also be necessary to gain the support of the public given that NASA missions are funded by United States tax dollars. In order to increase public awareness about the mission at hand, an outreach team of about 50 individuals will be formed. Approximately 10 of the individuals will be incharge of outreach with K-12 schools, 10 in charge of outreach with universities and colleges, 10 reaching out to the general public, 10 reaching out to underrepresented communities, and 10 that will handle designs, logos, and outreach progression. Each team will have a different approach based on the audience but the message and intended goal of making United States citizens excited about the developing mission will be the same.

Before any outreach can occur it is vital to make sure the official outreach team has a proper understanding of the mission to Venus and how it will not only impact NASA's research but also how it will impact society. To help the team have the ultimate understanding of the mission, a ten-week preparation program will be held. During the first weeks of the program, the outreach team will be "shadowing" each of the subteams

to properly understand each component of the mission and how they play into each other. The first week will be for shadowing the science team since it is where the mission's roots lie. Week two will be to shadow the engineering team and see how the objectives decided by the science team will impact decisions made by the engineering team. Week three will be to shadow the programmatic team in order to be aware of how important it is to meet mission requirements and constraints and how the team adjusts according to decisions made by the science and engineering team. Weeks four and five are for the outreach team to brainstorm and create designs, logos, and outreach activities based on each specific group where outreach is being done. Weeks six through ten will be a reiteration of the first five weeks which will allow the outreach team to go through another "cycle" of seeing how objectives are thought of, engineered, and made realistically. Once the ten-week preparation is complete, the team will be tested on their knowledge and finally prepare for official outreach.

Since part of the outreach goal is to connect with the future generations of NASA, the outreach department is to host an "On the way to Venus" tour that will consist of two legs. The first leg will involve traveling to about 50 major K-12 schools all across the United States. The team of about 10 individuals will be provided a tour bus that can comfortably sleep each one and store the supplies that will be needed during each stop. Starting in Florida, the team will stop at schools containing a large population of students to maximize the number of individuals reached. The tour will travel throughout the U.S. and ultimately arrive back at the initial starting point. During each stop, the team will spend part of the school day briefly learning about NASA and then more specifically the mission. A brief informational presentation will be shown at the beginning of the session to provide relevant background. Students in grades K-12 generally learn best through visuals and hands-on learning so each student will then be provided with miniature aerobot kits that can be built and can be personalized by drawing designs on the aerobot (Begin learning 2021). This will also allow students to visually see what kind of systems NASA has the ability to create and create an excited attitude (NASA n.d.).

The second leg of the "On the way to Venus" tour will be tasked with visiting about 50 major universities and colleges all across the United States. The team of about 10 individuals will also be provided a tour bus that can comfortably sleep each one and store the supplies that will be needed during each stop. Starting again in Florida, the team will stop at schools containing a large population of students to maximize the number of individuals reached. The tour will travel throughout the U.S. and ultimately arrive back at the initial starting point. College students may learn in slightly different ways when compared with K-12 students and they tend to appreciate rewards. During each stop, the team will post up in the "center" of each college/university and provide giveaways that include free NASA and mission gear such as t-shirts, pins, and hats

once the students have visited each “station” of the mission lifecycle. There will be about 5 stations and one team member will stand at each station to briefly explain an important component of the mission. Once a student reaches the end they can spin the wheel for a prize and will also receive a card on how to stay up to date with the latest mission updates by opting in to receiving text messages, a webpage that will provide the same information, or social media pages that promote the mission.



Figure 15: Aerobot Mission to Venus Logo



Figure 16: Aerobot Mission to Venus T-Shirt design

The general public will have outreach opportunities that are different from those of students since it consists of the people who are funding the mission so it is very important to formulate an outreach event that will intrigue and gain the support of the people. During the time prior to March 1, 2029, when the mission is set to launch, 10 members of the outreach team will be attending any conferences that are relevant to the space industry or the mission. While attending these conferences, the team will host a booth to provide a sufficient amount of information about the mission that individuals can come up and interact with. Depending on the conference, the team may have the opportunity to host a speaker event Q&A session where the team will go more in depth about the processes and current progress of the mission and attendees can ask questions or even express any concerns.

It must be kept in mind that not all citizens live in very populated areas and a large chunk live more towards the suburbs. Through this, it is still extremely important to also reach out to those individuals about the mission. To help with this type of outreach, the team plans to purchase billboard ads around the surrounding areas which will have an enticing graphic about the mission and ways to learn more about it. The team will also reach out to local newspaper companies and request that a section be dedicated to information about the mission, upcoming plans, and how to get involved with the progression of the mission.

Although different age groups usually need different approaches for outreach, cell phones are a tool that are extremely favorable when it comes to reaching out to people of all ages. Another way that the outreach team can inform people about the upcoming mission is by creating and posting ads that can be seen on social media. Applications such as Instagram, Facebook, and YouTube are used by all people regardless of age and posting ads on these platforms will greatly expand the number of people that can be reached. The ads can include an attention-grabbing graphic or short promotional video that should hook the viewer. A link should also be provided to view the NASA webpage associated with the progression details of the mission or the number to opt into text message updates. To be even more involved with the social media side of outreach, accounts can be created on any large platforms that would likely increase the amount of exposure the mission gets.

In order for all of the necessary outreach to occur before the official launch date of the Aerobot mission, a timeline of events must be identified. The teams in charge of school and university level outreach will likely require that one week be dedicated to each school. This one week time slot will account for the travel time, verification that there will be enough things such as materials for crafts and merchandise, and the actually outreach events.

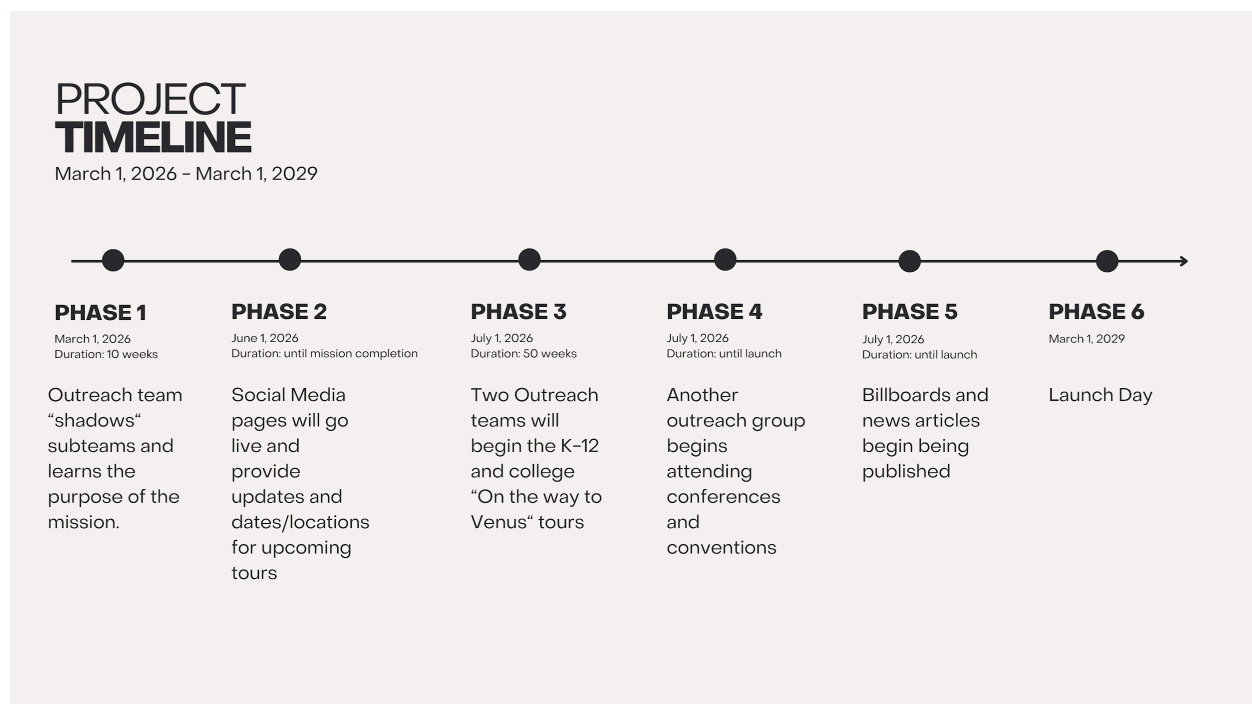


Figure 17: Timeline of planned outreach events and activities.

The biggest part of creating outreach events is not only to inform the public of the missions that NASA is working on, it is also to gain the support of the people and provide reassurance that tax dollars are being used as efficiently as possible. By hosting events, the people have the opportunity to speak up about questions and concerns that might be had and can be answered to gain as much support as possible.

6. Conclusion

This mission aims to obtain surficial and atmospheric data to enhance the understanding of the Venusian environment. Specifically, the scientific objectives for this mission will be to understand the role that volcanic outgassing plays in atmospheric concentration, the mechanisms of chemical weathering on surficial rocks, and paleoatmospheric conditions. With scientific and engineering system research, the team has selected the Virtual and Infrared Thermal Imaging Spectrometry (VIRTIS) and the IR Tunable Laser Spectroscopy (IRTLS) as relevant and realistic data collection instruments. Following all requirements, the proposed aerobot system is intended to maintain steady-state operation at reasonable internal temperatures for the science instruments. The aerobot must do this while cruising along the Venusian winds and obtaining data to fulfill the mission's goals.

In order to provide the highest quality content for the PDR, the team greatly utilized the feedback received from the MDR scoresheet document. The sections noted with RFA's were adjusted accordingly and the sections noted for ADV were enhanced in order to provide a better quality deliverable. If more time were allocated to the team to complete the Preliminary Design Review, more evaluating would have been performed by the sub teams to enhance the completeness/consistency of the planning, technical, cost, and schedule baselines developed for the Aerobot mission. For the Critical Design Review, the team plans on providing increased assurance that the proposed design, and the planned manufacturing and test methods and procedures, will result in an acceptable mission, with minimal project risk (Bell 1999). The team will also make it evident that the mission is ready to begin implementation. Given that mitigation #1 was selected for the descope, the constraints still include a total mass limit of 50 kilograms, stored dimensions remaining 1 m x 1 m x 1 m, and the capped cost of \$200M. But, 5kg of the mass limit will now be redistributed to the balloon's new wall thickness, and the mission must now be completed within 36 hours rather than the initially given 48 hours. The team shall ready the aerobot system by October 1st, 2026, for a launch at Cape Canaveral, Florida, by January 1st, 2028.

During the preparation of this work, the team used Canva Image Generator in order to create the Aerobot component in the Concept of Operations graphic. After using this tool/service, the team reviewed and edited the content as needed and takes full responsibility for the content of the deliverable.

During the preparation of this work, the team used Canva Image Generator in order to create the Rocket component in the Outreach Summary Logo graphic. After using this tool/service, the team reviewed and edited the content as needed and takes full responsibility for the content of the deliverable.

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Appendix

TBD #	Plans and Timeline for Resolution
[1]	
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Table 36: Table of TBD/TBR Resolution for the Mission Definition Review

SELECTING MISSION SITE LOCATION FROM THREE PROPOSED SITES						
Criteria	Explanation	Grade	Weight	Site 1 (Ozza and Maat Mons)	Site 2 (Haastse-baad Tessera)	Site 3 (Ishtar Terra)
Identifying volcanism and atmospheric composition	The site provides access to ongoing or records of volcanic activity, potential mantle plume regions, and weathering, and atmospheric (CO ₂ , H ₂ O, and SO ₂) chemical reactions with the composition and physical properties of the Venusian	10 = high, 5 = medium 1 = low 0 = Fail	25%	10	8	8

	surface.					
Entry and descent feasibility	The atmospheric entry should minimize risks (high wind shear, turbulence, and extreme heating) while allowing controlled descent. Ideally, the site should not expose the prototype to excessive thermal, pressure, or wind (mean zonal) stress that could potentially damage it.	10 = high, 5 = medium 1 = low 0 = Fail	30%	9	9	9
Tectonic activity and ancient surface exposure	The site has regions of crustal thickening, rifting, fault activity, and deformation exposing ancient rocks for C, H, O, and N measurement.	10 = high, 5 = medium 1 = low 0 = Fail	10%	7	10	8
Site location data can be collected at 50-70 km altitude	Region is large enough for at least some data to be collected in the likely chance that lower altitude measurements cannot be taken/large surface area for larger chance of seeing features	10 = high, 5 = medium 1 = low 0 = Fail	35%	10	10	10
		TOTALS:	100%	94.00%	92.00%	90.00%

Table 37: Trade study conducted to determine the mission site location from three proposed sites given valuable criteria